Radio Listening Guide

30 sheets to understand, listen to and identify signals, with a UV-K1, its open-source firmwares and an SDR dongle.

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9	UV-K1 and overlay applications
10	Antennas to build

Values marked "from memory" must be verified before use; each sheet states what was confirmed online.

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How to use this guide

> **Listening is free; transmitting is not.** This guide covers reception. Transmitting requires, depending on the service, an amateur radio licence and callsign, approved equipment and compliance with local regulations. Never transmit on a distress frequency (121.5, 156.8, 243, 406 MHz; 2182 kHz) and never replay a command you have captured. Private conversations you overhear must not be disclosed.

How to read a sheet

- Each chapter has its own colour, repeated in the strip at the top of the page: the current chapter is shown larger.
- The badge at the top right shows reliability: "checked online", "from memory, to be verified" or "partly checked".
- Cross-references ("sheet 12") avoid repeating information: antenna sizes are in sheet 1, tools in sheet 2b, emergencies and DSC in sheets 12, 15 and 16.

Two tools for the whole guide

- **UV-K1** and its open-source firmwares (Fusion, Labs, Apex, Falcon, GoguFW): VHF/UHF listening, HF with the SI4732, on-board applications. See sheet 2b.
- **RTL-SDR dongle**: everything that needs decoding (ADS-B, ACARS, AIS, radiosondes, rtl_433, satellites). See sheet 2b.

Where to start

I want to...	Sheets
Understand the basics and choose an antenna	1, 2, 2b
Work out what I am hearing	3, then the chapter for the service
Track aircraft	7 to 10
Track ships	11, 12
Follow a weather balloon or sensors	13, 14, 14b
Know what to do if I hear a distress call	15, 16
Decode digital modes or images	17 to 24 (including 20b)
Find the origin of a signal	25, 26

Index by subject

ACARS	sheet 9	Meshtastic, MeshCore, LoRa	sheet 14b, 6
ADS-B, squawk 7700	sheet 8, 15	METAR, aeronautical weather	sheet 7
AIS, AIS-SART	sheet 11, 16	MMSI, MID	sheet 12
Amateur radio bands	sheet 4	Morse, phonetic alphabet	sheet 27
Amateur radio callsigns	sheet 27	NAVTEX	sheet 12
Antennas to build, plans (dipole, J-pole, EFHW)	sheet 29	Overlay applications (Labs)	sheet 28
Antennas: sizes to use	sheet 1	Propagation	sheet 1
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CW	sheet 3, 25, 27	Satellites (NOAA, Meteor, Iridium, Inmarsat)	sheet 24
DSC	sheet 12	Selcall	sheet 19
Emergencies: frequencies	sheet 15	Settings (mode, bandwidth, squelch)	sheet 2
Firmwares (Fusion, Labs, Apex, Falcon, GoguFW)	sheet 2b	SigID	sheet 25
FT8, FT4, PSK31	sheet 17	SSTV	sheet 21
ILS	sheet 10	UV Studio, CHIRP	sheet 2b
Images in the spectrum	sheet 23	UV-K1 keys by firmware	sheet 28
KISS, TNC	sheet 20b	Weather fax (radiofax)	sheet 22

Waves, propagation and antennas

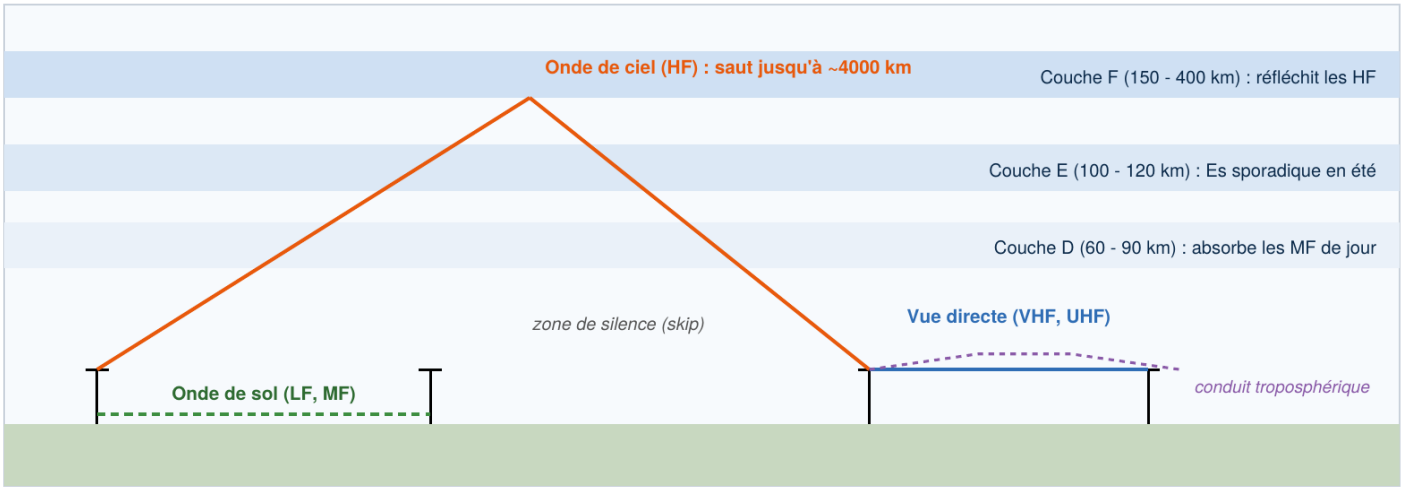
Principles, calculation formulas and sizes to use

1 · Start

Radio waves in brief

- A radio wave is an electromagnetic wave: an electric field and a magnetic field that travel together at the speed of light, **c = 300,000 km/s** (299,792 km/s in a vacuum).
- Frequency **f** (in Hz: oscillations per second) and wavelength **λ** (distance travelled during one oscillation) are related by **λ = c / f**. The higher the frequency, the shorter the wavelength, and so the smaller the antennas.
- **Polarisation**: direction of the electric field, vertical or horizontal depending on the antenna. Two cross-polarised antennas lose about 20 dB. On VHF/UHF FM, polarisation is vertical; for weak-signal SSB/CW, it is horizontal.
- An antenna transmits and receives in the same way (**reciprocity**): a good transmitting antenna is also a good listening antenna.

How waves propagate



Basic diagram: ground wave, sky wave reflected by the ionosphere, line of sight.

- **Ground wave** (below ~3 MHz): follows the curvature of the Earth, better over the sea. By day, the D layer absorbs MF: only the ground wave carries (a few hundred km). At night, the D layer disappears and the sky wave brings in distant stations on MW.
- **Sky wave** (HF, 3 - 30 MHz): the ionosphere (E and F layers, ionised by the Sun) reflects the wave. One hop covers 2000 to 4000 km, and several hops go around the world. Between the end of the ground wave and the point where the sky wave returns, there is the **skip zone**.
- **MUF** (maximum usable frequency): above it, the wave passes through the ionosphere. It rises with solar activity (cycle of about 11 years) and by day. The high bands (15, 12, 10 m) open only when it is high; the low ones (80, 160 m) work mostly at night.
- **Line of sight** (VHF, UHF and above): the wave does not follow the Earth. Range is limited to the **radio horizon**, slightly beyond the optical horizon because of refraction.
- **Special propagation**: sporadic E (50 - 144 MHz, mostly in summer, 1000 to 2000 km), tropospheric ducting (temperature inversion: a few hundred km on VHF/UHF), meteor scatter (50 - 144 MHz), moonbounce (EME), satellites. Above 10 GHz, rain strongly attenuates the signal.

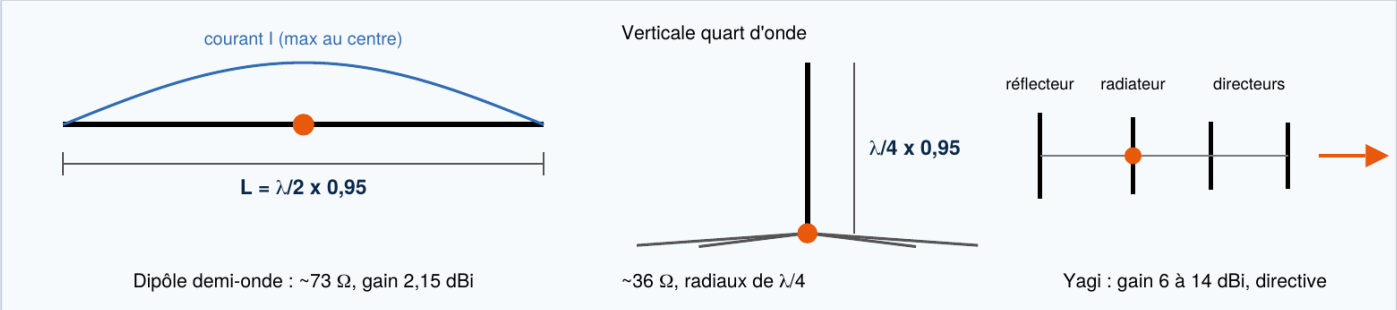
Bands and dominant propagation

Band (ITU)	Frequencies	Wavelength	Dominant propagation	Typical range
LF	30 - 300 kHz	10 km - 1 km	Ground wave	Several hundred to thousands of km
MF	300 kHz - 3 MHz	1 km - 100 m	Ground wave by day, sky wave at night	~300 km by day, very far at night
HF	3 - 30 MHz	100 m - 10 m	Sky wave (ionosphere)	Regional to worldwide, depending on time of day and the Sun
VHF	30 - 300 MHz	10 m - 1 m	Line of sight; Es, tropo	Radio horizon; exceptionally 1000 km and more
UHF	300 MHz - 3 GHz	1 m - 10 cm	Line of sight; tropo	Radio horizon
SHF	3 - 30 GHz	10 cm - 1 cm	Strict line of sight	Point-to-point links, satellites

How an antenna works

Waves, propagation and antennas

Principles, calculation formulas and sizes to use



Half-wave dipole, quarter-wave vertical, Yagi.

- An alternating current in a conductor radiates a wave. The antenna is efficient when it is **resonant**: its electrical length is a multiple of λ/4, the current is then at its maximum at the feed point and the impedance is purely resistive, so little energy is sent back to the transmitter.
- **Half-wave dipole**: two λ/4 elements, fed at the centre. Impedance of about 73 Ω in free space (close to 50 Ω as an inverted V).
- **Quarter-wave vertical**: a λ/4 element above a ground plane (or λ/4 radials). The ground "reflects" the other half. Impedance of about 36 Ω. Radiates in all horizontal directions, with a low angle, useful for DX.
- **Gain**: an antenna does not add power, it concentrates the radiation. It is expressed in dBi (relative to an isotropic source) or in dBd (relative to a dipole). +3 dB = power × 2; +6 dB = range × 2 in free space.
- **Height**: a horizontal antenna more than λ/2 above the ground has a low take-off angle (DX). Lower (λ/8 to λ/4), it sends the energy upwards: regional NVIS propagation on 80 and 40 m (0 - 500 km).
- **Balanced and unbalanced**: a dipole is balanced, a coax is unbalanced. A **balun** (1:1 or 4:1) stops current from flowing back up the cable (stray radiation, noise). End-fed half-wave antennas (EFHW) use a **49:1 unun**.

Calculation formulas

Calculation	Formula	Example
Wavelength	λ (m) = 300 / f (MHz)	145 MHz: 300 / 145 = 2.07 m
Half-wave dipole	L (m) = 142.5 / f (MHz)	7.1 MHz: 142.5 / 7.1 = 20.1 m (2 × 10.0 m)
Quarter-wave vertical	L (m) = 71.25 / f (MHz)	145 MHz: 71.25 / 145 = 49 cm
5/8-wave vertical	L (m) = 187.5 / f (MHz)	145 MHz: 187.5 / 145 = 1.29 m
Full-wave loop	L (m) = 306 / f (MHz)	14.2 MHz: 306 / 14.2 = 21.5 m perimeter
Half-wave cable	L (m) = 150 × VF / f (MHz)	145 MHz, VF 0.66: 150 × 0.66 / 145 = 68 cm
Yagi (rules of thumb)	Driven element 0.47 λ; reflector 0.49 λ; directors 0.43 to 0.45 λ	Spacing 0.15 to 0.25 λ
Discone	Cone slant length (m) = 71.25 / fmin (MHz)	fmin 100 MHz: 71 cm
Gain in dB	G (dB) = 10 log (P2 / P1); dBi = dBd + 2.15	× 10 in power: +10 dB
Reflection coefficient	Γ = (ZL - Z0) / (ZL + Z0)	100 Ω on 50 Ω: Γ = 1/3
SWR	SWR = (1 + |Γ|) / (1 - |Γ|)	Γ = 1/3: SWR = 2.0
Reflected power	Preflected / P = Γ^2	SWR 2: 11 %; SWR 3: 25 %
EIRP / ERP	EIRP (dBm) = P (dBm) - cable loss (dB) + G (dBi); ERP = EIRP - 2.15 dB	5 W, 1 dB loss, 2.15 dBi: 5 W × 0.79 × 1.64 = 6.5 W EIRP
Radio horizon	d (km) = 4.12 × (√h1 + √h2), h in metres	Aircraft at 10,000 m, ground at 10 m: 4.12 × (100 + 3.2) = 425 km
Free-space path loss	A (dB) = 32.44 + 20 log f (MHz) + 20 log d (km)	145 MHz, 100 km: 32.4 + 43.2 + 40 = 115.7 dB
Link budget	Preceived = Ptransmitted + Gt + Gr - A - losses	Each dB of loss costs 1 dB of signal
Dish gain	G = 10 log (η (π D / λ)^2), η ≈ 0.55	D 0.6 m at 12 GHz: about 35 dBi

Velocity factor (VF): solid coax 0.66; foam-dielectric coax 0.80; twin lead 0.90 to 0.95. For a length of wire in air, the 0.95 "shortening" coefficient is already included in the 142.5 and 71.25 lengths above.

Waves, propagation and antennas

Principles, calculation formulas and sizes to use

Sizes to use (theoretical values)

Band / use	Frequency (MHz)	λ	Dipole λ/2	Vertical λ/4
160 m	1.85	162.16 m	77.03 m	38.51 m
80 m	3.65	82.19 m	39.04 m	19.52 m
40 m	7.1	42.25 m	20.07 m	10.04 m
30 m	10.12	29.64 m	14.08 m	7.04 m
20 m	14.2	21.13 m	10.04 m	5.02 m
15 m	21.2	14.15 m	6.72 m	3.36 m
CB (channel 19)	27.185	11.04 m	5.24 m	2.62 m
10 m	28.5	10.53 m	5.00 m	2.50 m
6 m	50.15	5.98 m	2.84 m	1.42 m
ILS: localizer	110	2.73 m	1.30 m	64.8 cm
Aviation distress	121.5	2.47 m	1.17 m	58.6 cm
Aviation (band centre)	125	2.40 m	1.14 m	57.0 cm
ACARS	131.725	2.28 m	1.08 m	54.1 cm
APT reception 137 MHz	137.5	2.18 m	1.04 m	51.8 cm
2 m / APRS	145	2.07 m	98.3 cm	49.1 cm
Marine VHF	156.8	1.91 m	90.9 cm	45.4 cm
AIS	162	1.85 m	88.0 cm	44.0 cm
ILS: glide slope (UHF)	332	90.4 cm	42.9 cm	21.5 cm
Radiosondes	403	74.4 cm	35.4 cm	17.7 cm
70 cm	433	69.3 cm	32.9 cm	16.5 cm
PMR446	446	67.3 cm	32.0 cm	16.0 cm
868 MHz sensors	868.3	34.6 cm	16.4 cm	8.2 cm
ADS-B	1090	27.5 cm	13.1 cm	6.5 cm
23 cm	1296	23.1 cm	11.0 cm	5.5 cm

Single reference table. All the reception sheets refer back here for antenna size: do not recalculate elsewhere. Lengths are for wire elements in air (0.95 coefficient included); for the ILS, the dipole is horizontal, with two equal elements, each half the length given. For 137 MHz APT, use a V dipole: two elements of about 52 cm forming an angle of 120°, horizontal. Always measure: the wall, the mast and the height shift the resonance. Cut 3 to 5 % long, then trim with an analyser.

Which antenna for what?

Antenna	Principle and size	Gain / impedance	Use
Half-wave dipole	2 λ/4 elements, centre-fed, horizontal or inverted V	2.15 dBi, 50 to 73 Ω	Basis of any HF antenna, good for listening on 40 and 80 m
λ/4 vertical	Vertical element + at least 4 λ/4 radials	~0 dBi, 36 Ω	DX, VHF/UHF FM, CB, car antenna
Long wire / random wire	10 to 20 m of wire, with a 9:1 unun or a tuner	Variable	Multi-band HF and SW listening, discreet
EFHW (end-fed)	λ/2 wire on the low band, 49:1 unun, multi-band (harmonics)	~2 dBi, ~2500 Ω at the feed point	Portable, 80/40/20/15/10 m
Full-wave loop	Perimeter of 306 / f (MHz)	~3 dBi, 100 to 120 Ω	Less sensitive to noise, good for reception
Yagi	Driven element + reflector + directors, boom of 0.5 to 3 λ	3 el.: 7 dBi; 5 el.: 10 dBi; 10 el.: 13 dBi	Directional: DX, satellites, VHF/UHF
Discone / wideband	Disc + cone, covers ~25 to 1300 MHz	0 to 2 dBi, vertical	Scanner, aviation, marine and PMR listening
Collinear / 5/8 λ	Several stacked sections; 5/8 λ = 187.5 / f	3 to 6 dBi	VHF/UHF repeaters, ADS-B, base stations
Magnetic loop / active	Small tuned loop, or short sensor + amplifier	Low efficiency, narrow band	HF and LF in a restricted space, listening
Dish	Reflector + feed at the focus	30 to 40 dBi	Satellites, microwaves

Practical tips

- **Height and clearance** matter more than gain: on VHF/UHF, every extra metre increases the radio horizon. Keep the antenna away from metal walls and cables.
- **Cable**: losses increase with frequency and length (rule of thumb: RG58 ~15 dB / 100 m at 145 MHz, ~30 dB / 100 m at 435 MHz; RG213 about half that). Keep the cable short on UHF.
- **SWR**: below 1.5 is very good, 2 is acceptable. When receiving, a high SWR mostly reveals a fault (contact, length, cable).
- **Safety**: no antenna near power lines; earthing and a surge arrester outdoors.
- **UV-K1**: the original whip antenna is a compromise. A quarter-wave tuned to the band you are monitoring (table above) on SMA, or an outdoor antenna, clearly improves reception.

Listening settings

Modes, bandwidth, squelch, gain and audio connection

1 · Start

The settings that make the difference

Setting	Usual reference points
Modulation	**FM**: marine, VHF/UHF amateur radio, PMR, radiosondes (narrow FM). **AM**: civil and military aviation, broadcasting, CB. **LSB / USB**: HF amateur radio and utilities (LSB below 10 MHz, USB above), all digital modes in USB. **CW**: Morse.
Bandwidth	SSB: 2.4 to 3 kHz. Aviation AM: 8 to 10 kHz. Narrow FM: 12.5 kHz; wide FM (FM broadcast): about 150 kHz. CW: 300 to 500 Hz. Narrower = less noise, but the signal may be clipped.
Frequency step	12.5 or 25 kHz on VHF/UHF; 8.33 kHz on some aviation channels (set to the nearest step, 6.25 kHz); 9 kHz (Europe) or 10 kHz for MW; 5 kHz on shortwave.
Squelch	Cuts the noise when there is no signal. Open to listen to a weak signal or a digital burst; closed for standby monitoring. CTCSS / DCS sub-tones are masked by the squelch: disable it to hear everything.
Gain and attenuation	Too much gain overloads (false lines, ghost signals); too little hides weak signals. Attenuate near a strong transmitter. A band-pass or notch filter helps in crowded areas (FM, DAB).
Drift correction (ppm)	For an SDR without a TCXO, set it on a known signal. Essential for narrow modes (radiosondes, FT8).

Which mode for what?

I want to listen to...	Mode and setting	Sheet
Aviation (towers, approaches)	AM, 118 - 137 MHz, 25 kHz step, low squelch	7
Marine VHF	FM, 156 - 162 MHz, channel 16 on standby	15
HF amateur radio	USB / LSB, 2.4 kHz bandwidth	4
DATA modes (FT8, RTTY, SSTV...)	USB, audio sent to a decoder on a PC or smartphone	17
Live aircraft, boats, sondes, sensors	SDR with the matching decoder	8, 11, 13, 14

From receiver to decoder

- **Audio cable**: headphone or speaker output to the computer's line or mic input, at a medium volume. A signal that is too loud distorts and makes decoding fail.
- **Via the microphone**: with no cable, the smartphone microphone placed near the speaker is enough for slow modes, in a quiet room.
- **Local noise**: switch-mode power supplies, chargers, screens and LEDs radiate noise. Move the antenna and receiver away, add ferrites on the cables.
- **Antenna**: the biggest improvement almost always comes from the antenna, outdoors and in the clear (sheet 1).
- **Equipment**: choice of receiver, UV-K1 firmwares and SDR dongle: sheet 2b.

Quick troubleshooting

Symptom	Things to check
No sound	Squelch too high, wrong modulation, frequency off by one step, antenna disconnected
Distorted or overloaded sound	Gain or volume too high, signal too strong: attenuate, move the antenna away from the transmitter
Muffled or distorted voice on SSB	Wrong sideband (USB / LSB) or frequency offset: fine-tune
Constant noise	Switch-mode power supply, charger, nearby screen: move away, add ferrites, try on battery
Nothing decodes	Audio level, sideband, dial frequency, clock (for FT8), SDR drift correction. These causes are more common than a decoder fault.
Ghost signals	Receiver overload: reduce the gain, add a filter, change location

UV-K1, firmwares and SDR dongle

The two tools of the guide: what to choose, what to flash, for which sheet

1 · Start

In brief. The guide rests on two tools. The **UV-K1** (PY32F071 microcontroller, BK4829 radio chip) listens to VHF/UHF, and to HF with its SI4732 when the firmware supports it. An open-source firmware adds applications, a spectrum analyser and tuning tools. The **SDR dongle** takes over for everything that gets decoded: ADS-B, ACARS, AIS, radiosondes, sensors. The following sheets do not repeat these basics: they refer back here.

Choosing your receiver

Receiver	What it listens to	Limits
UV-K1 (VHF/UHF)	FM and AM over the range allowed by the firmware: aviation, marine, amateur radio, PMR, beacons	No built-in digital decoding (apart from Labs applications); sensitive to strong signals
UV-K1 with SI4732 (HF)	0.15 to 30 MHz in AM, and in LSB/USB with the SSB patch: broadcasting, utilities, amateurs	Modest sensitivity and filters; audio output to be sent to an external decoder
RTL-SDR dongle	About 24 to 1700 MHz: ADS-B (1090 MHz), AIS, radiosondes, 433 / 868 MHz sensors, ACARS, APRS, fax and weather satellites. HF needs "direct sampling" mode or a converter	Instantaneous bandwidth of about 2 MHz; drift correction (ppm) matters, especially around 400 MHz
Wideband SDR (Airspy, SDRplay, HackRF...)	Wider coverage and larger instantaneous bandwidth: the whole radiosonde band or several channels at once	More expensive, more demanding on the computer

The firmwares in the guide

Firmware	What it is	Key points	Checked
Fusion (F4HWN)	Reference edition of the project by armel and muzkr for UV-K1 and UV-K5 V3. Brings the former K5 V1 editions together in a single firmware.	UV Studio (flash, calibration, Live Viewer), CHIRP driver shipped with each release, Fox Hunt, spectrum, APRS. Hidden menu 76/77 SetNav: set UP/DOWN or LEFT/RIGHT depending on the model.	Online
Labs	Experimental F4HWN edition with installable "overlay" applications.	Applications installed with UV Studio, launched with F + 7: full list of applications: sheet 28 (ATIS RX, Sig Finder, APRS, SSTV, sondes...). An application is compiled for a specific firmware version.	Guide
GOGUFW	Derived from F4HWN (base 6.2.0 cited), stable version 3.2.0 seen online. UV-K1 and UV-K5 V3 only.	Messenger: FSK text messages between GOGUFW radios. Also HEARD, Range Check, CALLTX and memory spectrum. Since 3.0.0: 16 application slots. Dedicated CHIRP module, "multiboot" image.	Online
Apex (NR7Y CW doc.)	CW (Morse) firmware by NR7Y, based on Fusion (F4HWN) and on egzumer's firmware. UV-K1, UV-K5 V3 and UV-K5 V1. The documentation provided does not mention the name "Apex": match to be confirmed.	Automatic iambic keyer (modes A, B, Ultimatic, Bug), paddle on PTT / Side1 or external, stored macros, Code Practice mode, RIT / XIT, decoded CW displayed. Long press 0: FM, AM, USB, CW; long press 5: practice oscillator; long press 7: break-in.	Online doc.
Sonic (V71 manual)	Fork of F4HWN with code from NTOIVOLA, egzumer and Dual Tachyon. Sonic 1K and 4K versions. UV-K1 (USB-C) and RS232 versions.	Spectrum at ~300 channels/s with 50 scan lists and 50 custom bands (CHIRP), ULTRA WATCH ~600 channels/s, history of 200 detections, close call, no hidden menu. No applications; logo and calibration to be handled with F4HWN.	Manual
Falcon	Falcon 4K v2.3.0-OA (overlay applications) and v1.3.0-SE (all-in-one). UV-K5 V3 and UV-K1.	Spectrum and Sentinel (LIST, BAND, RANGE), history of 200 detections, dual, triple, quad or multi-list standby, discreet Falcon mode, FM broadcast radio. CHIRP driver included in the archive.	Leaflet

Falcon: installing and listening

Version	Contents
v1.3.0-SE (standard)	All-in-one, without F4HWN overlay applications; own CHIRP driver; flash memory 99.25 % full
v2.3.0-OA (overlay)	Compatible with the F4HWN 6.2.0 / 6.3.0 overlay applications installed with UV Studio; 10 application modules to upload after flashing (menu, FM radio, editors, scan); flash 94.18 % full. Three Falcon applications included: system info, LED torch and Morse tone, Snake

Installing v2.3.0-OA (the Falcon .bin and the modules must come from the same archive): 1) delete the old modules from slots 39 to 48 with UV Studio; 2) boot into F4HWN and delete the old Falcon multiboot slot; 3) install Falcon in a free slot; 4) boot into Falcon; 5) upload the modules to slots 39 to 48; 6) power off and on again. Do not use these slots for F4HWN 6.2.0 applications, otherwise the Falcon modules are erased. Falcon has only been tested with the memory layout of F4HWN 6.3.0.

To listen	Keys (VFO)
LIST / BAND / RANGE spectrum	F + 7 / F + 8 / F + 9; Sentinel: F + 6
Scan the channel's list / all memories (fast)	Long press on * / F + *

To listen	Keys (VFO)
Scan between two free frequencies (VFO A and B)	Long press on * when A and B are free
Monitor a frequency	F + 4 (up to 50 frequencies)
Triple, quad or multi-list standby	Menu RxMode = FULL; priority channels PRICH1 and PRICH2
Squelch threshold in dBm	Menus SQLSCN and SQLHYS, or F + left / right
FM broadcast radio	F + 0; nine favourites, long press 1 to 9 to save
Falcon mode (discreet, backlight off)	Long press on 7 in SINGLE or DUAL

In the spectrum and Sentinel, 9 opens the history, SK1 locks or opens the audio, SK2 blacklists the frequency. Source: Falcon v2.3.0 user guide provided.

Apex / NR7Y CW (UV-K1, K5 V3): binary nr7y.k1-k5v3.v1.4.bin (v1.4 seen online) with its CHIRP module nr7y.k1-k5v3.chirp.v1.4a.py. To flash: switch the radio on while holding PTT (LED on, screen black) with UVTools2 or n7six's Multi-firmware tool. Back up the calibration first. In CHIRP-next: developer mode, File > Load Module, then model "UV-K1 & UV-K5 V3 (F4HWN Fusion)", NR7Y variant, and reload the module each session. Documentation: briand.github.io/cw-firmware-docs. For CW, see also sheets 3 and 27.

The Fusion version seen on the releases page was 5.9.0, whereas GOGUFW cites an F4HWN base of 6.2.0: check the releases page for the latest version before flashing.

Flashing and backing up

- **UV Studio** (armel.github.io/uvstudio), in Chrome, Edge or Chromium on a computer: flash in DFU mode, calibration backup, Live Viewer (radio screen on the computer), Apps page for the applications.
- **Back up the calibration right after the first flash.** It is specific to your radio and is restored before going back to the original firmware.
- **CHIRP**: use the driver shipped with the flashed version (GOGUFW driver for GOGUFW). A driver from another version reads the settings wrongly.
- **quansheng_toolkit** (open source, Apache 2.0): reading and writing channels, calibration and DFU flashing with a list of compatible firmwares.
- The wrong firmware can make the radio unusable. GOGUFW must not be installed on a UV-K5 V1 or V2.

Which tool for which sheet

I want to...	UV-K1 (firmware)	SDR dongle (software)	Sheet
Listen to aircraft, marine, amateur radio, PMR	Direct: AM/FM depending on the band, low squelch, memories	Optional: wide spectrum with SDR++ or GQRX	2, 4, 6, 7
See aircraft and their messages	Voice listening only	readsb or dump1090 (ADS-B), acarsdec (ACARS), dumpvdl2	8, 9
Track boats	ATIS RX (Labs) for inland waterways	AIS-catcher or rtl_ais, OpenCPN	11
Track weather balloons	A crackling carrier	radiosonde_auto_rx, SondeHub	13
Read wireless sensors	Detection of a burst only	rtl_433	14
Look for a beacon or a source	Sig Finder (Labs), AM on 121.5 or 243 MHz	Decoding of the 406 MHz frame with dec406	16, 26
Identify a tone signal	SigID, then decoding on a smartphone (Labs applications: sheet 28)	SDR + waterfall to see the shape	3, 25
Decode FT8, RTTY, SSTV, fax, APRS	SI4732 in USB, audio to PC or smartphone	Decoder on the computer (fldigi, WSJT-X, Direwolf)	17 to 22
Write to another radio without Internet	GOGUFW Messenger, between GOGUFW radios	Not applicable	this sheet

Division of roles

- The radio picks up, filters and delivers the signal or the audio. It does not decode the heavy digital modes (FT8, AIS, ADS-B).
- The SDR dongle digitises a wide band and decodes, but has no practical squelch and no battery: it stays at home or in the bag with a computer.
- The UV Studio Live Viewer mirrors the radio's screen on the computer: handy for capturing a Sig Finder reading.

> **Good to know.** GOGUFW Messenger messages are transmissions: they require a callsign or authorised use on the chosen frequency. Information on Fusion and GOGUFW checked online; Sonic from its V71 manual; Falcon from its user guide; Labs applications taken from the guide. Apex from the NR7Y CW documentation provided (name to be confirmed).

Recognising a signal by ear

From the sound heard to the service and the sheet to consult

A four-question method

- **On which frequency?** The band already tells you the service: 118 - 137 MHz aviation, 156 - 162 MHz marine, 144 MHz amateur radio...
- **In which modulation does the signal become clear?** Clear voice in AM or FM: speech. Whistles or crackles: data.
- **How many tones, and what rhythm?** A chopped tone (CW), two alternating tones (RTTY, NAVTEX, DSC), several tones in sequence (Selcall, FT8).
- **When and at what rate?** A single burst after a voice, repetition every second, 15 s cycle, continuous loop...

What I hear on VHF, UHF and above

What I hear	What is it?	Where?	Sheet
Voice in a loop: "Information Alpha, wind..."	Aeronautical ATIS	VHF 118 - 137 MHz, AM	7
Short "bzzz" burst every few minutes	ACARS, aircraft-to-ground text messages	VHF about 131.5 to 131.9 MHz, AM	9
A few Morse letters, fixed tone, in a loop	Identification of an ILS or navigation beacon	VHF 108 - 112 MHz (ILS), AM	10
Nothing audible but pulses on the SDR	ADS-B, aircraft positions	1090 MHz, SDR	8
Data warbling, two channels	AIS, ship positions	161.975 and 162.025 MHz, FM	11
Short digital burst at the end of a sentence	Inland waterways ATIS	Marine working channel, FM	11
Marine voice then a fast, high-pitched "beep-beep"	DSC, digital selective calling	156.525 MHz (channel 70), 2187.5 kHz	12
Two fast tones, continuous	NAVTEX, notices to mariners	518, 490, 4209.5 kHz	12
Continuous data, narrow signal, no silence	Radiosonde	400 - 406 MHz, narrow FM	13
Short burst, no voice, once or twice a minute	Wireless sensor (weather, probe)	433.92 and 868 MHz, SDR	14
Tone sweeping downwards, repeated	Homing on a distress beacon	121.5 and 243 MHz, AM	16
Whistling in packets of 1 to 2 seconds	APRS, amateur radio positions	144.800 MHz, FM	20

On HF and in USB

What I hear	What is it?	Where?	Sheet
Fixed tone chopped into dots and dashes	CW, Morse	Amateur bands, bottom of the band	27
Two tones alternating without a pause, regular warble	RTTY	HF, amateur and utility bands	18
Sequence of tones in 15 s cycles	FT8	Reference frequencies: sheet 4	17
Burst of 4 to 6 short notes in half a second	CCIR Selcall	VHF / HF, network channels	19
Screeching that changes pitch, over a minute long	SSTV, slow-scan images	14.230 MHz, 28.680 MHz...	21
Machine noise paced line by line, low and high tones	Weather fax	HF, weather stations	22

Reference frequencies and emergency frequencies are grouped in one place only: distress and emergencies in sheet 15, amateur radio bands in sheet 4. To classify a tone signal automatically, use the SigID application (sheet 25); to find the direction of a transmitter, Sig Finder (sheet 26). For an unknown signal, record a few seconds of audio and look at it in a spectrogram: the shape it draws helps you decide.

This table gives clues, not certainties: several services use similar sounds, and frequencies vary from country to country.

Amateur radio bands

Allocations (IARU Region 1 / France), sub-bands, modes and reference frequencies

Band	Range	Usual segments	Reference frequencies	Mode / remarks
LF, MF and HF				
2200 m	135.7 - 137.8 kHz	CW and slow modes (WSPR, QRSS)	136.000 kHz (WSPR)	Secondary, limited power
630 m	472 - 479 kHz	CW and slow modes	475.5 kHz (WSPR)	Secondary, limited power
160 m	1810 - 2000 kHz	CW 1810-1838 · digital 1838-1843 · phone 1843-2000	—	LSB. Night band, atmospheric noise
80 m	3500 - 3800 kHz	CW 3500-3570 · digital 3570-3600 · phone 3600-3800	3580 (RTTY) · 3730 (SSTV)	LSB. Night and regional
60 m	5351.5 - 5366.5 kHz	CW, digital and phone within the narrow range	—	USB. Secondary, limited power
40 m	7000 - 7200 kHz	CW 7000-7040 · digital 7040-7060 · phone 7060-7200	7110 (emergency) · 7171 (SSTV)	LSB. Day and night
30 m	10 100 - 10 150 kHz	CW 10 100-10 130 · digital 10 130-10 150	—	No phone. Reduced power
20 m	14 000 - 14 350 kHz	CW 14 000-14 070 · digital 14 070-14 099 · IBP beacons 14 100 · phone 14 101-14 350	14 080 (RTTY) · 14 230 (SSTV) · 14 300 (emergency)	USB. The most active band for DX
17 m	18 068 - 18 168 kHz	CW 18 068-18 095 · digital 18 095-18 109 · phone 18 111-18 168	—	USB. WARC band
15 m	21 000 - 21 450 kHz	CW 21 000-21 070 · digital 21 070-21 149 · phone 21 151-21 450	21 340 (SSTV)	USB. Depends on the solar cycle
12 m	24 890 - 24 990 kHz	CW 24 890-24 915 · digital 24 915-24 929 · phone 24 931-24 990	—	USB. WARC band
10 m	28.000 - 29.700 MHz	CW 28.000-28.070 · digital 28.070-28.190 · beacons 28.190-28.225 · phone 28.225-29.200 · FM 29.100-29.300 · repeaters 29.520-29.700	28.680 (SSTV) · 29.600 (FM calling)	USB; FM above 29.1 MHz
VHF, UHF and microwaves				
6 m	50 - 52 MHz (France: 50.2 - 51.2 MHz, to be verified)	CW 50.0-50.1 · SSB 50.1-50.5 · beacons 50.4-50.5 · digital 50.3-50.4	50.150 (SSB calling) · 51.510 (FM calling)	Opened by sporadic-E propagation in summer
2 m	144 - 146 MHz	CW 144.000-144.150 · SSB 144.150-144.400 · beacons 144.400-144.491 · FM 145.200-145.5875 · repeaters 145.600-145.7875 · satellites 145.800-146.000	144.300 (SSB calling) · 145.500 (FM calling) · 144.800 (APRS)	FM, offset -600 kHz. ISS: 145.800 (voice) and 145.825 (APRS)
70 cm	430 - 440 MHz	SSB/CW 432.000-432.100 · beacons 432.400-432.500 · FM 433.400-433.575 · repeaters and digital 430-431.9 and 438-440 · satellites 435-438	432.200 (SSB calling) · 433.500 (FM calling) · 433.775 (LoRa APRS)	FM, offset +/-1.6 MHz. Also DMR, D-STAR, Fusion
23 cm	1240 - 1300 MHz	Repeaters, ATV, satellites, EME	1296.200 MHz (SSB calling)	FM, digital, amateur TV
13 cm	2300 - 2450 MHz	Links, ATV, EME	2320.200 MHz (SSB calling)	Shared with 2.4 GHz Wi-Fi
5 cm	5650 - 5850 MHz	Links, ATV, satellites	5760.200 MHz (SSB calling)	Shared with 5 GHz Wi-Fi
3 cm	10 - 10.5 GHz	Links, EME, beacons	10 368.200 MHz (SSB calling)	The most popular microwave band
24 GHz and above	24 - 24.25 · 47 - 47.2 · 76 - 81 · 122.25 - 123 GHz	Experimental links	24 048.200 MHz (SSB calling)	Specialised equipment

Reference frequencies for digital modes (kHz, USB unless stated)

Mode	160 m	80 m	40 m	30 m	20 m	17 m	15 m	12 m	10 m	6 m
FT8	1840	3573	7074	10 136	14 074	18 100	21 074	24 915	28 074	50 313
FT4	1842	3575	7047.5	10 140	14 080	18 104	21 140	24 919	28 180	50 318
WSPR	1836.6	3568.6	7038.6	10 138.7	14 095.6	18 104.6	21 094.6	24 924.6	28 124.6	50 293.0
PSK31	1838	3580	7040	10 142	14 070	18 100	21 070	24 920	28 120	50 305

Frequencies are USB dial settings, usual reference points (not mandatory channels). The PSK31 and FT4 rows are the least certain. WSPR: dial frequency; the signal is offset by 1400 to 1600 Hz. In FT8, most stations sit between 200 and 3000 Hz above the dial. For 144.174 MHz (FT8 on 2 m),

the same logic applies.

Usage, calls and conventions

- **Sideband**: LSB below 10 MHz (160, 80, 40 m), USB from 10 MHz upwards. Exception: 60 m uses USB.
- **Calls**: "CQ" is a general call; "QRZ?" means who is calling me; emergency nets on 3760 and 7110 kHz (Region 1) and 14 300 kHz (worldwide).
- **Callsigns, RST reports, CW abbreviations and Q code**: sheet 27. Digital modes: sheet 17.
- **Power**: depends on licence class and band, with lower limits on secondary bands (2200 m, 630 m, 60 m). To be verified in the ARCEP decision.
- **Transmitting**: requires a callsign (ANFR exam). For listening, the UV-K1 covers VHF/UHF in FM and HF with its SI4732 (sheet 2b).

Broadcast bands

Radio and television: frequencies, modulation and listening (Region 1 / Europe)

AM radio (HF and below)

Band	Range	Modulation / format	Step	Remarks
Long wave (LW)	148.5 - 283.5 kHz (Europe: 153 - 279 kHz)	AM, DRM	9 kHz	Range of several hundred km. Many transmitters have been shut down.
Medium wave (MW)	526.5 - 1606.5 kHz (Region 1: 531 - 1602 kHz)	AM, DRM, HD Radio (Americas)	9 kHz (10 kHz in the Americas)	Ground wave by day, sky wave at night.
Short wave (SW)	2.3 - 26.1 MHz (14 bands below)	AM, DRM, SSB for utility stations	5 kHz	International broadcasting, depends on propagation.

Short wave: the broadcast bands (kHz)

Band	Range	Format	Best period	Remarks
120 m	2300 - 2495	AM	Tropical	Mainly tropical regions
90 m	3200 - 3400	AM	Tropical	Mainly tropical regions
75 m	3900 - 4000	AM	Night	Europe, night and regional
60 m	4750 - 5060	AM	Tropical / night	Regional stations
49 m	5900 - 6200	AM, DRM	Night	The most active band in the evening
41 m	7200 - 7450	AM, DRM	Day and night	Shared with the 40 m amateur band (7000-7200)
31 m	9400 - 9900	AM, DRM	Day and night	The most used band
25 m	11 600 - 12 100	AM, DRM	Day	Good range in daytime
22 m	13 570 - 13 870	AM, DRM	Day	
19 m	15 100 - 15 800	AM, DRM	Day	Depends on the solar cycle
16 m	17 480 - 17 900	AM, DRM	Day	Depends on the solar cycle
15 m	18 900 - 19 020	AM, DRM	Day	Narrow band
13 m	21 450 - 21 850	AM, DRM	Day	Depends on the solar cycle
11 m	25 670 - 26 100	AM, DRM	Day	Little used

FM radio, digital and TV

Band	Range	Modulation / format	Step / channel	Remarks
FM (band II)	87.5 - 108 MHz	Wide FM (WFM), +/-75 kHz	200 kHz (100 kHz in France)	Stereo: 19 kHz pilot; RDS: 57 kHz subcarrier; 50 µs pre-emphasis.
VOR / ILS (aviation)	108 - 118 MHz	AM (VOR: 30 Hz)	50 kHz	Radionavigation, not public broadcast radio (sheet 10).
DAB / DAB+ (band III)	174 - 240 MHz (blocks 5A to 13F)	OFDM, COFDM	1.536 MHz per block	5A: 174.928 MHz; 13F: 239.200 MHz.
DTT (UHF)	470 - 694 MHz (France)	DVB-T / DVB-T2 depending on the multiplex, MPEG-4 (H.264)	8 MHz per channel	700 MHz band freed in 2019 for mobile telephony.
Legacy VHF TV	47 - 68 · 174 - 230 MHz	Analogue (switched off in France in 2011)	7 / 8 MHz	The bands have now been reallocated.
Satellite TV (Ku)	10.7 - 12.75 GHz	DVB-S2, QPSK / 8PSK	27.5 to 36 MHz per transponder	Local oscillator 9.75 / 10.6 GHz; Astra 19.2 E, Hotbird 13 E.
Satellite TV (C)	3.4 - 4.2 GHz	DVB-S, analogue FM	36 MHz	Larger dishes, better rain resistance.

Time signals (indicative)

Station	Frequency	Modulation / format	Remarks
WWV / WWVH	2.5 · 5 · 10 · 15 · 20 MHz	AM, voice announcements	UTC time, seconds and service tones.
DCF77 (Germany)	77.5 kHz	AM, BCD time code	Used by radio-controlled clocks in Europe.
MSF (United Kingdom)	60 kHz	AM, time code	Also WWVB (United States) on 60 kHz.

Broadcast bands

Radio and television: frequencies, modulation and listening (Region 1 / Europe)

Listening with the UV-K1

- **FM (87.5 - 108 MHz)**: received by the SI4732 as well as by the VHF section, in wide FM. RDS (station name, title) is only readable if the firmware decodes it.
- **LW, MW, SW**: the SI4732 covers 0.15 - 30 MHz in AM (bandwidth 2 to 6 kHz) and in SSB with a patch. SSB is for utility and amateur stations, AM for broadcasting.
- **Tuning step**: on MW, channels are spaced 9 kHz apart in Europe; on SW, 5 kHz. Set the firmware step accordingly.
- **DRM, DAB and DTT** are digital: the UV-K1 only gets noise from them; you need an SDR or a dedicated receiver.
- **Tips**: listen to MW at night and SW according to the time of day (49 m in the evening, 25 and 19 m by day). Broadcasters' schedules and frequencies change with the seasons (A and B): check the HFCC schedules.

Comparison

Service	Frequencies	Mode	Power	Range
PMR446	446.00625 - 446.19375 MHz (16 channels)	Narrow FM, or digital (dPMR / DMR Tier 1)	0.5 W	1 to 3 km
CB 27 MHz	26.965 - 27.405 MHz (40 channels)	AM, FM, SSB	4 W (AM/FM), 12 W (SSB)	A few km, more with propagation
LPD433	433.075 - 434.775 MHz (69 channels)	FM	10 mW	Under 1 km
LoRa	863 - 870 MHz (and 433 MHz)	Spread spectrum (chirp)	~25 mW, 1% duty cycle	Several km to tens of km

PMR446, CB, LPD433 and LoRa are licence-free in France: no licence or fee, provided the equipment complies with the regulations (approved devices).

PMR446: the 16 channels (narrow FM, 12.5 kHz step)

Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7	Channel 8
446.00625	446.01875	446.03125	446.04375	446.05625	446.06875	446.08125	446.09375
Channel 9	Channel 10	Channel 11	Channel 12	Channel 13	Channel 14	Channel 15	Channel 16
446.10625	446.11875	446.13125	446.14375	446.15625	446.16875	446.18125	446.19375

- Users add CTCSS or DCS codes (sub-tones): the squelch hides them, so disable them to hear everything (sheet 2).
- Digital (dPMR, DMR) is not decoded by the UV-K1: you hear a "buzz" or nothing.

CB 27 MHz: reference channels

Channel	Frequency	Usual use
1	26.965 MHz	Start of band
9	27.065 MHz	Emergency channel by convention (sheet 15). Monitoring not guaranteed.
19	27.185 MHz	Truckers and traffic info
40	27.405 MHz	End of band

SSB (USB/LSB) is used on some of the upper channels of the band, with long-distance contacts when propagation opens (summer, daytime).

LPD433 and the 433 MHz ISM band

- 69 channels of 25 kHz from 433.075 to 434.775 MHz, FM, 10 mW. Channel 1: 433.075 MHz.
- The 433 MHz band also hosts remote controls, weather stations, doorbells and sensors: 433.920 MHz is the most common frequency. They transmit in OOK or FSK, in short bursts (sheet 14).
- The band overlaps the 70 cm amateur band (secondary use).

LoRa: usual frequencies

Use	Frequency	Typical parameters
LoRaWAN EU868	868.1 / 868.3 / 868.5 MHz	BW 125 kHz, SF7 to SF12
Meshtastic (Europe 868)	869.525 MHz	LongFast: BW 250 kHz, SF11
LoRaWAN EU433	433.175 / 433.375 / 433.575 MHz	BW 125 kHz
LoRa APRS	433.775 MHz	BW 125 kHz, SF12 (amateur callsign required to transmit)

- LoRa modulates the frequency in a fast sweep ("chirp"): SF7 is fast with short range, SF12 is slow with long range.
- **Detection only**: at best the UV-K1 hears a brief hiss. To decode, you need an SDR or a LoRa module (SX127x) configured with the same settings (SF, BW).

Listening with the UV-K1

- **PMR446 and LPD433**: narrow FM, 12.5 kHz step (PMR) or 25 kHz (LPD). Store the channels in memory and scan.
- **CB**: 26.965 - 27.405 MHz in AM or SSB with the HF receiver (SI4732). CB FM is only received if the firmware tunes low enough in VHF.
- PMR446 and CB require approved devices for transmitting.

Aviation and aeronautical weather

Listening to the air: bands, traffic, ATIS, VOLMET, METAR, military, balloons and radiosondes

The bands to know

Range	Mode	Use
118.000 - 136.975 MHz	AM	Civil aviation: towers, approaches, centres, ATIS
138 - 144 MHz	AM / FM	Military in several countries (varies by state)
225.000 - 399.975 MHz	AM	Military aviation (NATO)
400.15 - 406.0 MHz	Narrow FM (GFSK)	Weather radiosondes (sheet 13)
406.0 - 406.1 MHz	BPSK	Distress beacons (sheet 16)
1090 MHz	Pulses	ADS-B: aircraft positions, SDR required (sheet 8)

Aviation emergencies: frequencies 121.500 and 243.000 MHz, search and rescue: sheet 15. The transponder codes **7700** (emergency), **7600** (radio failure) and **7500** (hijack) are visible in ADS-B (sheet 8).

Civil traffic (indicative breakdown, to be verified in the AIP)

Range	Typical use
118.0 - 121.4 MHz	Control towers and approaches
121.6 - 121.975 MHz	Ground: taxiing, departure clearances
122.0 - 123.65 MHz	Self-information, gliders, microlights, air-to-air. 123.450 MHz: unofficial air-to-air, often used by crews
124 - 136 MHz	Approaches and en-route centres (Reims, Brest, Paris, Bordeaux, Marseille): aircraft in cruise
131.5 - 131.9 MHz	ACARS (sheet 9)
136.7 - 136.975 MHz	VDL Mode 2, data (sheet 9)

- **Callsigns**: AFR (Air France), RYR (Ryanair), EZY (easyJet), DLH (Lufthansa). Languages: English and French.
- **8.33 kHz step**: since 2018, many channels use this step. The channel name is not the exact frequency: set the UV-K1 to the nearest one (6.25 kHz step), method below.
- **Range**: line of sight only. An aircraft at 10,000 m can be heard up to ~400 km; on the ground, only nearby airports.

Weather by ear: ATIS, VOLMET and tower

Service	Content	Where
ATIS	Weather and information for one aerodrome on a loop: runway in use, wind, visibility, clouds, temperature, QNH. Identified by a letter (Alpha, Bravo...).	VHF 118 - 137 MHz, AM, one frequency per aerodrome
VOLMET	METAR messages (observations) and sometimes TAF (forecasts) for a list of airports, read continuously.	HF in USB (long range) or VHF depending on the country
Tower / approach	Weather information given to crews (wind, QNH).	VHF AM, local

HF VOLMET (kHz, USB)	Frequencies	Area
Shannon VOLMET (Ireland)	3413 · 5505 · 8957 · 13264	Europe, North Atlantic
Gander VOLMET (Canada)	3485 · 6604 · 10051 · 13270	North America, North Atlantic

VOLMET frequencies from memory, to be verified. The low ones carry mostly at night, the high ones by day. For ATIS and VHF VOLMET in France, see the AIP (SIA).

Reading a METAR (fictional example)

```
METAR LFPG 081930Z 24012KT 9999 FEW040 BKN090 18/12 Q1015 NOSIG
```

Item	Meaning
LFPG	Airport, ICAO code (Paris-Charles de Gaulle)
081930Z	Day 08, 19:30 UTC (Z = UTC time)
24012KT	Wind from 240°, 12 knots (G25 = gusting 25 kt; VRB = variable)
9999	Visibility of 10 km or more (in metres). CAVOK = good visibility, no low cloud
FEW040 BKN090	Few clouds at 4,000 ft, broken at 9,000 ft (height in hundreds of feet)
18/12	Temperature 18 °C, dew point 12 °C
Q1015	QNH pressure 1015 hPa
NOSIG	No significant change expected within 2 h

Clouds	Coverage
FEW	1 to 2 oktas
SCT	3 to 4 oktas
BKN	5 to 7 oktas
OVC	8 oktas (overcast)

Code	Phenomenon
RA / DZ	Rain / drizzle
SN	Snow
FG / BR	Fog / mist
TS / SH	Thunderstorm / shower

Intensity: "-" light, "+" heavy (e.g. +RA = heavy rain). TAF: forecast for 9 to 30 h, with BECMG (becoming), TEMPO (temporary), PROB30 / PROB40 (probability). Aeronautical Q codes: QNH, QFE, QFU, QDM, QDR (sheet 27).

Military

- UHF band 225 - 400 MHz in AM: exercises, patrols, air-to-air refuelling, air bases. Frequencies are published in the military AIP and in listening hobbyists' databases.
- **Partial reception**: many exchanges use frequency hopping (HAVE QUICK) or encryption: you hear bursts, a hiss or nothing.
- 243.000 MHz remains the military emergency guard frequency: program it into memory together with 121.500 MHz.

Hot-air balloons, balloons and radiosondes

Type	Frequency / band	Note
Hot-air balloon (ground crew)	PMR446: 446.0 - 446.2 MHz (FM); CB 27 MHz	Common, unofficial use: listen to the recovery crew (sheet 6).
Hot-air balloon (in flight)	Aviation VHF 118 - 137 MHz, AM	When the craft carries a radio, on the local frequency.
Weather radiosondes	400.15 - 406.0 MHz	Reception, models and tracking: sheet 13.
Amateur stratospheric balloons	434 MHz (RTTY, LoRa); APRS 144.800 MHz; WSPR 14.0956 MHz	Tracking on habhub / SondeHub (sheets 18 and 20).

The 8.33 kHz step: from channel name to tuned frequency

Since 2018, a 25 kHz channel is split into three 8.33 kHz channels. Controllers and charts give the **channel name** (6 digits, like 124.835), which is not the actual transmitted frequency. A radio that only tunes in steps of 25, 12.5 or 6.25 kHz cannot display this name: it has to be converted.

Name ending (mod 25 kHz)	Channel type	Actual frequency (carrier)	Examples
00, 25, 50, 75	25 kHz	the name itself	118.000 → 118.000; 121.500 → 121.500
05, 30, 55, 80	8.33 kHz	25 kHz block, + 0 kHz	118.005 → 118.000; 124.830 → 124.825
10, 35, 60, 85	8.33 kHz	25 kHz block, + 8.333 kHz	118.010 → 118.0083; 124.835 → 124.8333
15, 40, 65, 90	8.33 kHz	25 kHz block, + 16.667 kHz	118.015 → 118.0167; 124.840 → 124.8417

Method: 1) take the last two digits of the name; 2) subtract the remainder of the division by 25 to find the start of the block (35 gives 25 + 10); 3) add 0, 8.333 or 16.667 kHz depending on the remainder (5, 10 or 15); 4) round to your radio's step.

> **Example: your aerodrome's tower is announced on 124.835** The remainder is 10: block start 124.825, plus 8.333 kHz = **124.8333 MHz** actual. Your radio does not offer this value. With a 6.25 kHz step: you can tune **124.83125** (offset 2.1 kHz, good reception in AM). With a 12.5 kHz step: **124.8375** (offset 4.2 kHz, borderline but audible). With a 25 kHz step only: 124.825 (offset 8.3 kHz), the sound is weak or distorted. Preferably program one memory per channel with the name as its label.

Channel name	Remainder	Actual (MHz)	6.25 kHz step	12.5 kHz step
118.005	5	118.0000	118.00000	118.0000
118.010	10	118.0083	118.00625	118.0125
118.015	15	118.0167	118.01875	118.0125
127.440	15	127.4417	127.44375	127.4375

Reception only. A stronger neighbouring channel can bleed in (wide AM receiver). If your firmware offers an 8.33 kHz step (to be verified), enter the actual value.

UV-K1 settings and tips

- **Aviation**: AM mode (AIR), low squelch, 25 kHz step. Program into memory: 121.500 / 243.000 MHz, your nearby airport (tower, ATIS), an en-route centre. Scan 118 - 137 MHz: note the frequency, time and callsign of each exchange.
- **ATIS and VOLMET**: ATIS in AM on the aerodrome frequency (line-of-sight range). HF VOLMET in USB with the SI4732; the voice is read quickly, in English: note the airport, UTC time, wind and QNH.
- **ADS-B and ACARS**: cannot be decoded on the UV-K1, see sheets 8 and 9 (SDR).
- **Never transmit** on aeronautical frequencies: it endangers crews. To prepare a flight, use official sources (Météo-France Aéro, aviationweather.gov): radio listening is not reliable for navigation.

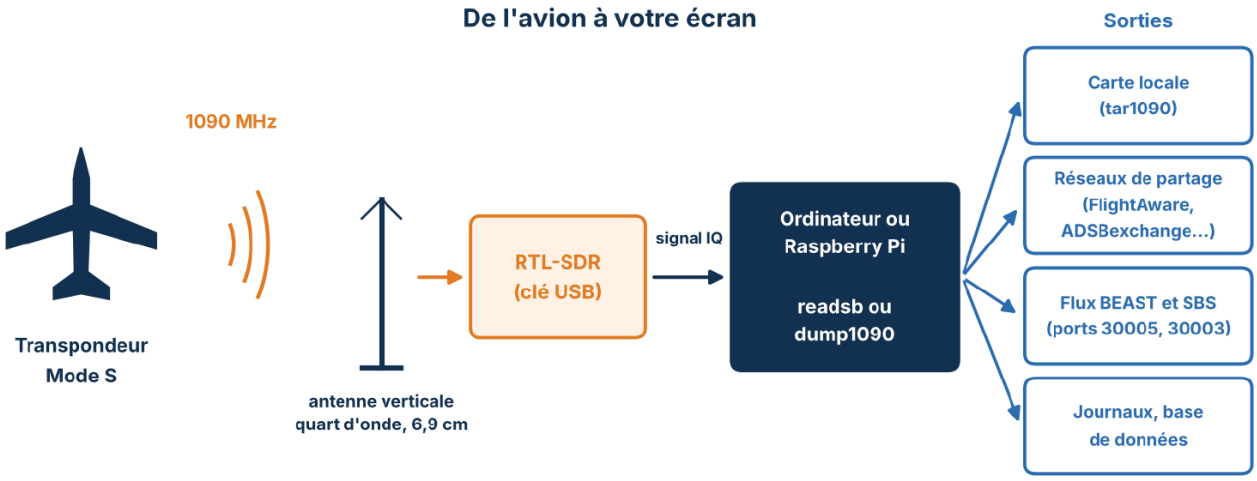
ADS-B

Aircraft positions on 1090 MHz: pulses, frame, decoding, reception

In brief. ADS-B (Automatic Dependent Surveillance-Broadcast) is a system in which every equipped aircraft automatically broadcasts, in clear, its identity, GNSS position, altitude and speed, about once a second or more. It mainly uses 1090 MHz ("1090ES", an extension of the Mode S transponder). In the United States, general aviation flying below 18,000 ft may also use 978 MHz (UAT). The equipment is mandatory in many airspaces: since 1 January 2020 in the United States, and for some aircraft since 2017 in Europe. Messages are neither encrypted nor authenticated: any receiver in range can read them.

1090 MHz	1 Mbit/s	112 bits	24 bits
ADS-B frequency (1090ES)	data rate, pulse position modulation	length of an ADS-B frame	aircraft ICAO address

How it flows

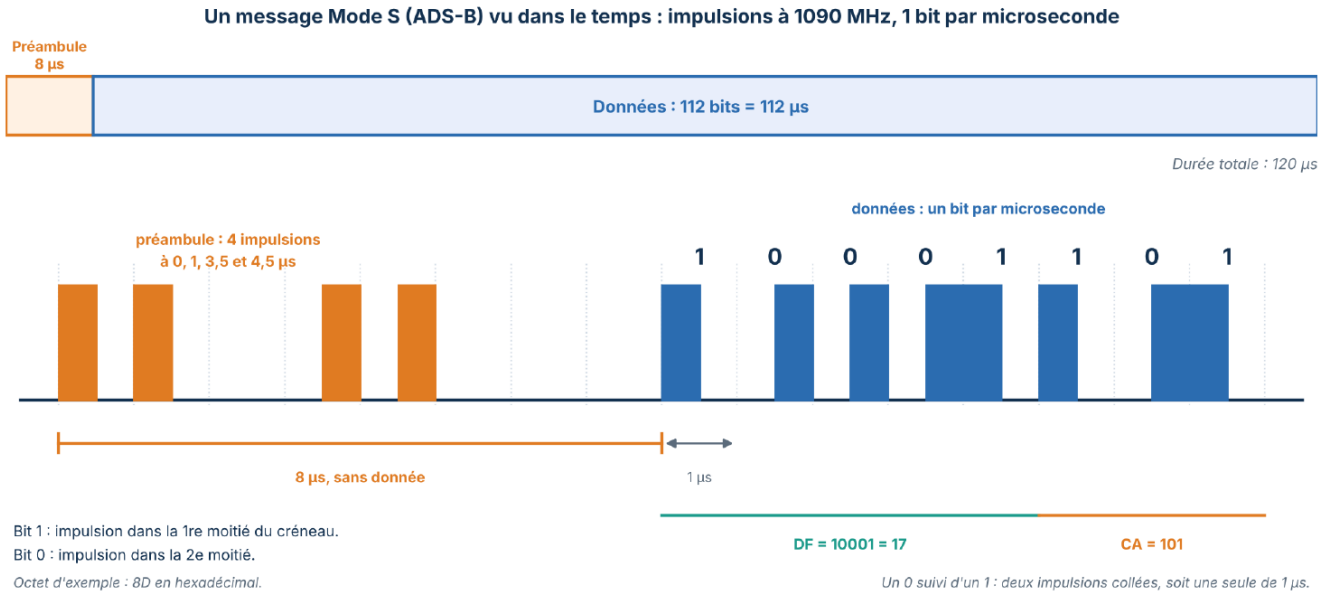


The receiver digitises the band around 1090 MHz, detects preambles, demodulates the pulses and keeps only frames with a valid CRC. The software then groups messages by aircraft (ICAO address) and publishes them on a map or over the network.

Setting up and running

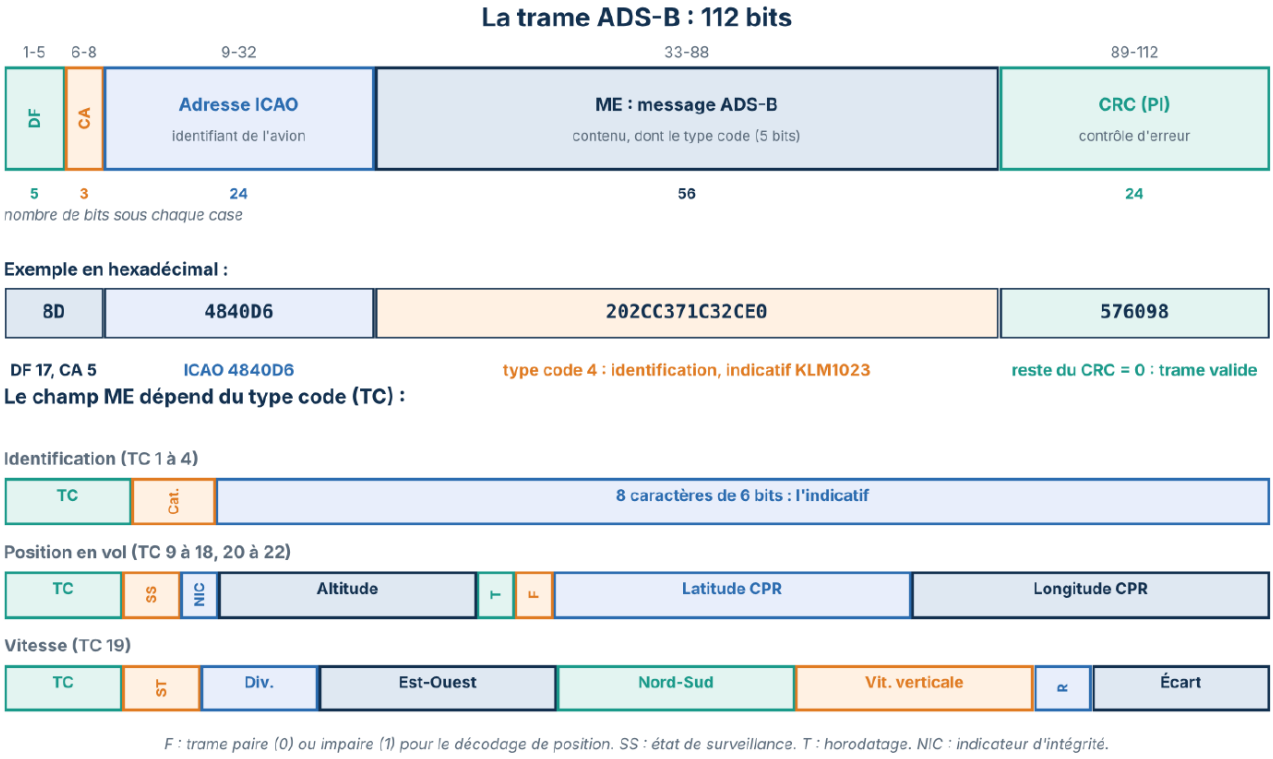
Item	Detail
Receiver	RTL-SDR dongle (sheet 2b). A 1090 MHz filter and a low-noise amplifier near the antenna improve range (optional)
Antenna	Quarter-wave vertical (size: sheet 1, ADS-B row). Place it high, in the clear, with a short cable
Software	readsb or dump1090 (including the dump1090-fa variant) to decode, tar1090 for the web map
First test	dump1090 --interactive --net: table of received aircraft. rtl_adsb shows raw frames in hexadecimal
Sharing	FlightAware, FlightRadar24, ADS-B Exchange, OpenSky or airplanes.live accept feeds from personal receivers

The pulses



- The **preamble** lasts 8 microseconds: four pulses of 0.5 microsecond, at 0, 1, 3.5 and 4.5 microseconds. It lets you detect a message and synchronise.
- Each **bit** lasts 1 microsecond: a 1 is a pulse in the first half-microsecond, a 0 in the second (pulse position modulation).
- ADS-B frames are long: 112 bits, or 120 microseconds with the preamble. Short Mode S replies are 56 bits.

The frame



For a DF 17 frame, the 24-bit field at the end of the frame is a CRC (generator polynomial 0xFFF409): when dividing the 112 bits, the remainder must be zero. A damaged frame is rejected. DF 17 denotes ADS-B sent by a transponder, DF 18 that from equipment without a transponder or relayed by TIS-B.

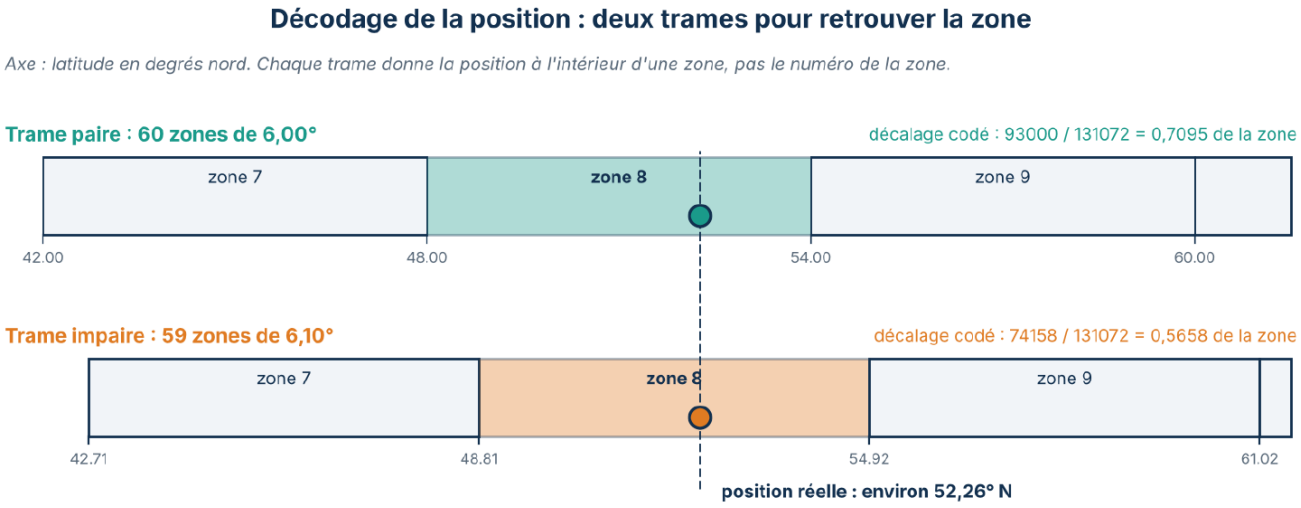
ADS-B

Aircraft positions on 1090 MHz: pulses, frame, decoding, reception

Message types (type code)

Type code	Content	Detail
1 to 4	Identification	Callsign (8 characters) and aircraft category
5 to 8	Surface position	Position, speed and track on the runway or taxiways
9 to 18	Airborne position	Barometric altitude, latitude and longitude in CPR
19	Airborne velocity	Ground speed, track, vertical rate
20 to 22	Airborne position	GNSS height, latitude and longitude in CPR
28, 29, 31	Aircraft status, target state and selected values, operational status	Emergency, selected values, equipment capabilities

Position: CPR decoding



Un seul numéro de zone est compatible avec les deux découpages : on obtient la latitude, ici 52,2572° N, puis de même la longitude, ici 3,9194° E.

- Latitude and longitude each fit in 17 bits: compact, but ambiguous. **CPR** (Compact Position Reporting) encodes the position within a zone.
- The aircraft alternates even and odd frames, which use 60 and 59 latitude zones. By combining one frame of each, the receiver finds the zone.
- With a nearby reference position (the receiver or a previous fix), a single frame may be enough.

A decoded example, step by step

Frame	Decoded content
8D4840D6202CC371C32CE0576098	Identification: ICAO 4840D6, type code 4, callsign KLM1023, CRC valid
8D40621D58C382D690C8AC2863A7	Position, even frame: ICAO 40621D, altitude 38,000 ft, CPR latitude 93000, CPR longitude 51372
8D40621D58C386435CC412692AD6	Position, odd frame: same aircraft, CPR latitude 74158, CPR longitude 50194
even + odd	Position: 52.2572° N, 3.9194° E (most recent frame is the even one)
8D485020994409940838175B284F	Velocity: ICAO 485020, type code 19: 159 kt, track 182.9°, vertical rate -832 ft/min

Altitude: 12-bit code; if the Q bit is 1, altitude = N × 25 - 1000 ft. Velocity (subtype 1): East-West and North-South components, giving ground speed and track. Values checked by a full decode of these frames.

ADS-B

Aircraft positions on 1090 MHz: pulses, frame, decoding, reception

The other Mode S messages

DF	Message	Note
17	ADS-B (extended squitter)	Broadcast spontaneously by the transponder, 112 bits
18	ADS-B from non-transponder devices, TIS-B	Equipment that does not answer interrogations
11	All-call reply	Announces the aircraft's presence to radars and collision avoidance
4, 5	Altitude and identity replies	Replies to a radar interrogation, 56 bits
20, 21	Comm-B replies	Onboard data returned on radar interrogation
0, 16	Collision avoidance (ACAS)	Dialogue between nearby aircraft

Formats other than DF 17 and 18 depend on radar interrogations: a passive receiver mostly picks up the replies, not always their context.

Range and limits

- **Optical range**: about 200 to 250 nautical miles for an aircraft in cruise with a good, unobstructed antenna. It drops sharply behind hills and buildings.
- **Invisible aircraft**: aircraft without ADS-B Out (older ones, some military) do not appear; some networks locate them by multilateration (MLAT).
- **Congestion**: 1090 MHz is shared with secondary radars and collision avoidance; in dense areas, messages overlap and are lost.

Troubleshooting

Symptom	Things to try
No aircraft	Antenna outdoors and vertical, short cable, gain on automatic, check that the dongle is detected
Short range	Raise the antenna, add a 1090 MHz filter and a low-noise amplifier near the antenna
Many rejected frames	Gain too high (saturation): reduce it. Move the dongle away from the computer with a USB extension
Wrong or late positions	Only one position frame received: wait for the pair, or provide a reference position

Going further

- **mode-s.org**: online book on Mode S and ADS-B (frame formats, detailed decoding).
- **pyModeS**: Python library to decode Mode S and ADS-B frames.
- **tar1090** and **graphs1090**: web map and reception statistics.

Good to know. ADS-B messages are public, but they remain sensitive: privacy programmes exist (LADD in the United States, temporary ICAO address PIA) and some countries restrict or monitor ADS-B receivers. Find out about local legislation. **Never transmit on 1090 MHz**: this frequency is used for air safety (radars, collision avoidance), and broadcasting false positions is illegal and dangerous.

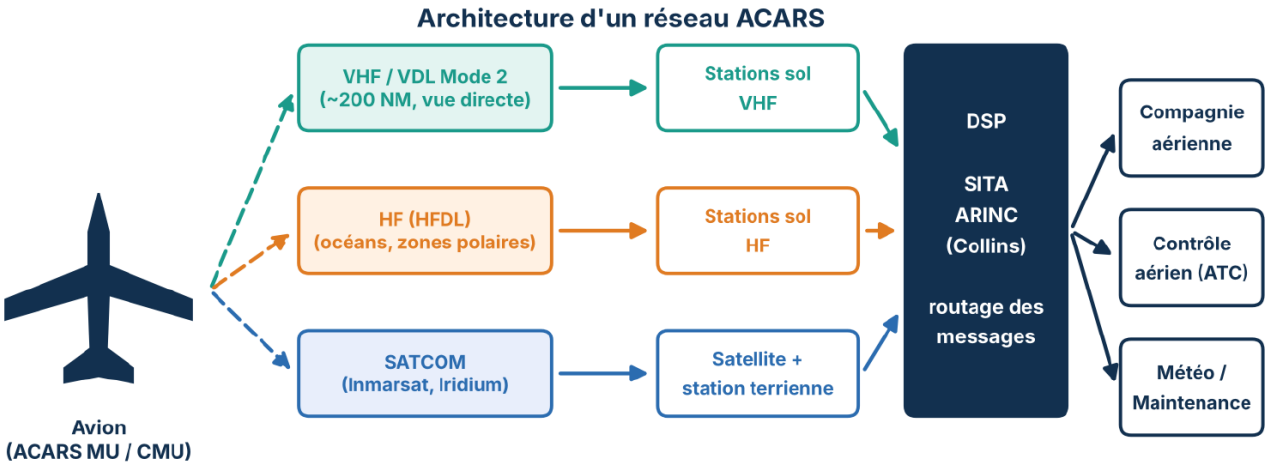
ACARS

Aircraft-to-ground text messages on VHF: links, modulation, frame, reception

In brief. ACARS is a digital text messaging system between aircraft and the ground, created in 1978 by ARINC. It is mainly used by airlines (operations, maintenance, weather) and, via dedicated applications, by control services. It is not pilot/ATC voice: these are short automatic messages, or ones typed by the crew, carried over several types of radio links.

1978	2400 bit/s	131.550 MHz	~220 chars
year of creation	data rate of classic VHF ACARS	worldwide VHF frequency	max. text per block

How it flows



Les messages circulent dans les deux sens : liaison montante (sol vers avion) et descendante (avion vers sol).

The onboard equipment (ACARS management unit) chooses the available link. Ground stations or the satellite pass messages to the data service provider (DSP, e.g. SITA or ARINC), which routes the message to the airline or organisation concerned.

The links used

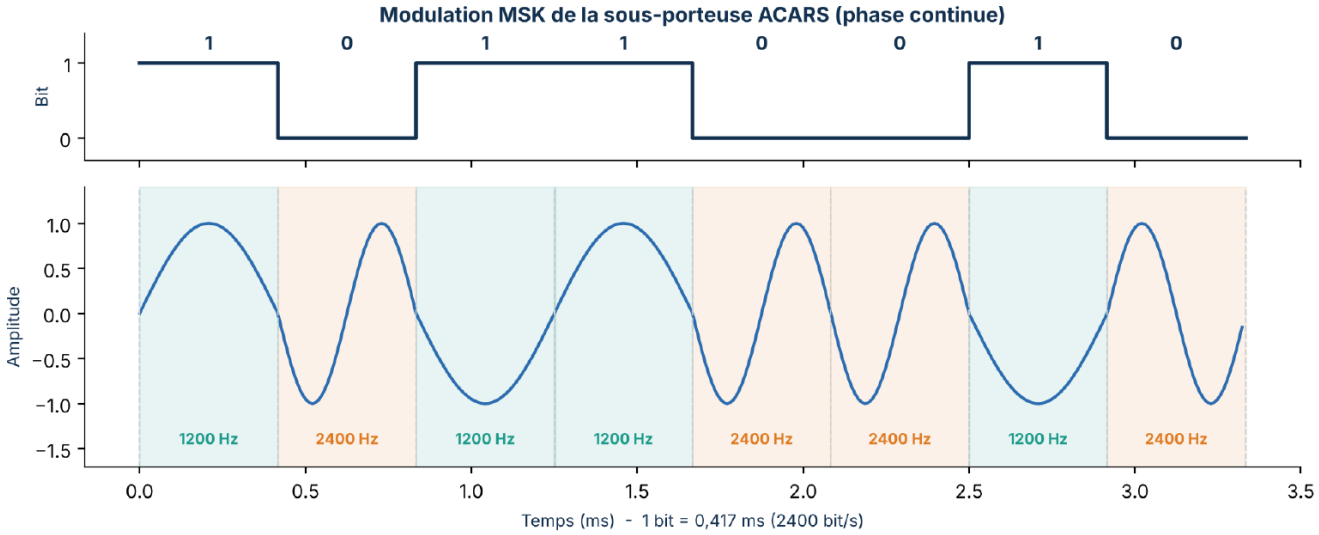
Link	Band	Data rate	Use
VHF (classic ACARS)	129 - 132 MHz	2400 bit/s (AM, MSK)	Continental areas, the most widespread
VDL Mode 2	136.7 - 136.975 MHz	31.5 kbit/s (D8PSK)	Gradually replacing classic VHF
HFDL (HF)	2 - 22 MHz	300 to 1800 bit/s	Oceans and polar regions, beyond VHF range
Satellite	Inmarsat ~1545 MHz; Iridium ~1616 - 1626 MHz	variable	Worldwide coverage

Classic VHF frequencies

Area	Frequencies
Worldwide (common)	131.550 MHz
Europe	131.725 / 131.525 / 131.825 MHz
United States	129.125 / 130.025 / 130.425 / 130.450 MHz

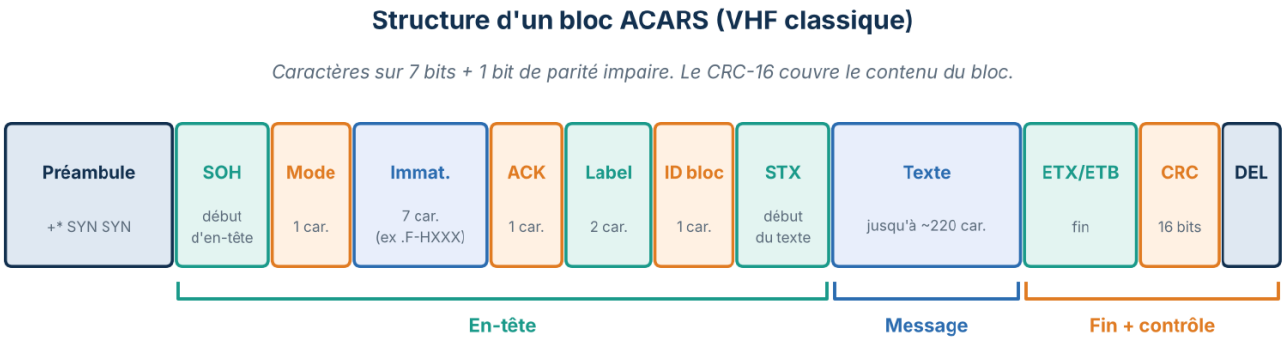
25 kHz channels, amplitude modulation (AM). Frequencies vary by region and change over time: check local allocations before programming a receiver.

Modulation: MSK on a subcarrier



- The audio signal is a subcarrier at 1200 Hz and 2400 Hz, then transmitted in AM on the VHF carrier.
- A 1 bit corresponds to 1200 Hz (half a cycle per bit), a 0 bit to 2400 Hz (one full cycle).
- The phase stays continuous at bit changes: this is what defines MSK and limits spectral spreading.

The frame



Un bloc fait au plus ~220 caractères de texte : les messages longs sont découpés (ETB) puis réassemblés.

The preamble is used for receiver synchronisation. The registration identifies the aircraft, the label (2 characters) indicates the message type, and the CRC-16 detects transmission errors.

What the messages contain

The **OOOI** events punctuate a flight and are sent automatically: OUT (leaving the gate), OFF (take-off), ON (landing), IN (arrival at the gate).

- Weather (requests and reports), positions and estimated times of arrival (ETA)
- Departure clearances (PDC) and air traffic control messages
- Engine and maintenance data, fault reports
- Free text between the crew and the airline

Label	Usual meaning
Q0	Link test
SQ	Ground station identification (squitter)
H1	Data messages (reports, ATC applications depending on context)
5Z	Airline-defined message

Non-exhaustive list: usage varies between airlines.

Decoding them

Need	Detail
Receiver	RTL-SDR dongle (or another SDR covering 130 MHz), sheet 2b
Antenna	VHF quarter-wave (size: sheet 1, ACARS row)
acarsdec	Classic VHF ACARS decoder (command line)
dumpvdl2	VDL Mode 2 decoder
JAERO	Satellite ACARS decoder (Inmarsat)
acarshub	Web interface to view and archive received messages
airframes.io	Community site that aggregates messages received by volunteers

Legality. Simply receiving these signals is generally tolerated, but disclosing the content of messages may be regulated (confidentiality of correspondence). Rules vary by country: find out about local legislation before publishing or sharing what you decode.

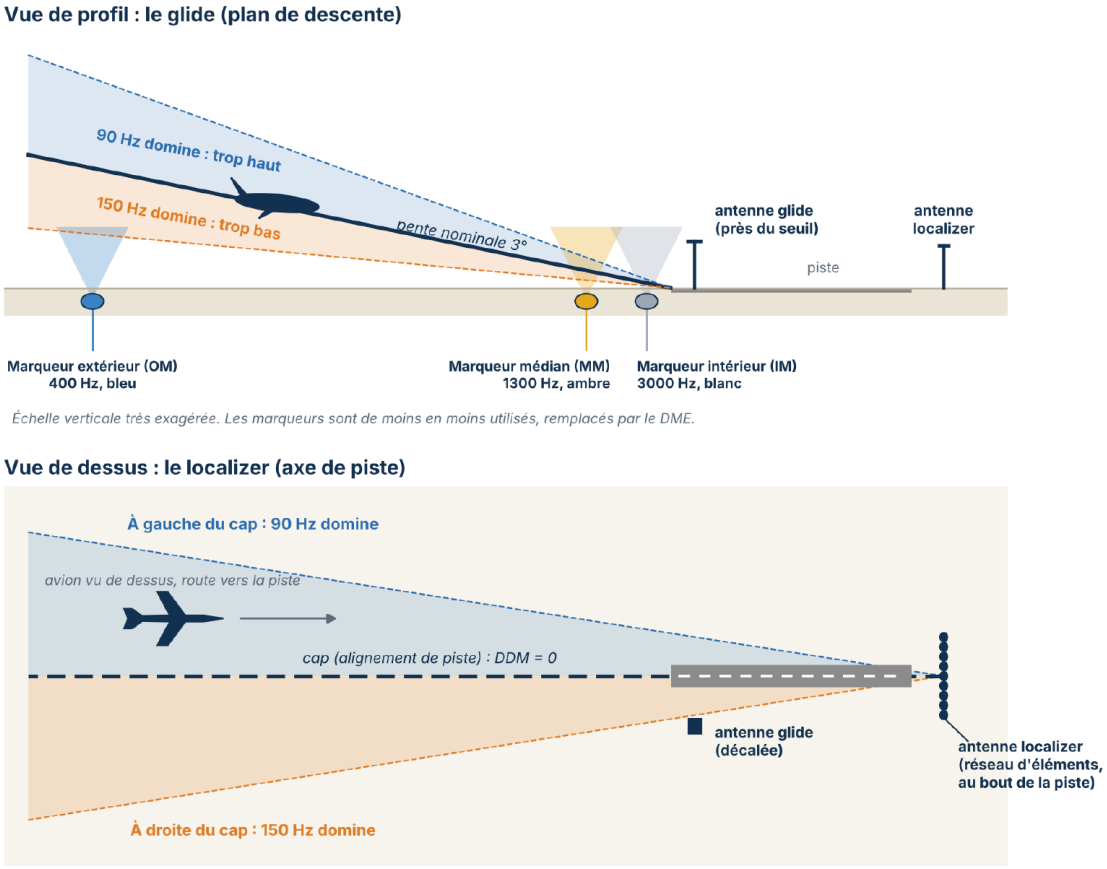
ILS

Instrument landing system: localizer, glide, DDM, markers

In brief. The ILS is an instrument landing system: it guides the aircraft precisely, laterally (localizer) and vertically (glide), down to the runway, even without visibility. The principle relies on two radio beams amplitude-modulated by two tones, 90 Hz and 150 Hz. The aircraft measures the difference in depth of modulation (DDM) between these two tones: it is zero on the centreline and changes sign on either side. The pilot, or the autopilot, corrects to bring the needle back to the centre. The system is standardised by ICAO (Annex 10).

108 - 112 MHz	329 - 335 MHz	90 / 150 Hz	3°
localizer (VHF)	glide (UHF)	the two modulation tones	nominal slope

Overview



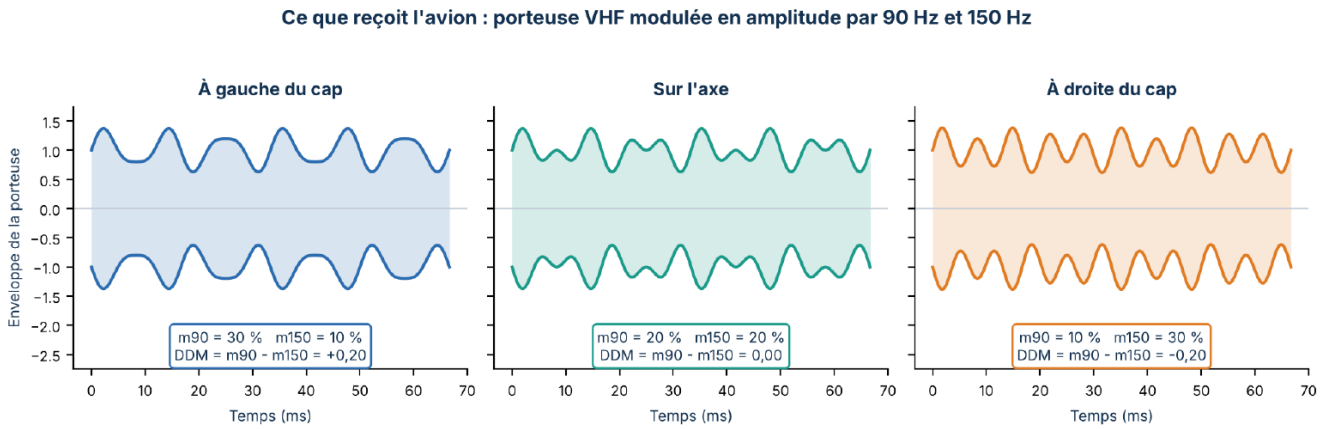
The two beams are radiated from the ground. The localizer marks the runway centreline, the glide marks the descent slope. Seen from an aircraft on approach, 90 Hz dominates on the left (or above), 150 Hz on the right (or below).

The components

Item	Role	Key points
Localizer (LOC)	Lateral guidance, runway centreline	VHF, 108.10 to 111.95 MHz. Antenna at the end of the runway
Glide (GS)	Vertical guidance, slope	UHF, 329.15 to 335.00 MHz. Antenna near the threshold, offset from the centreline
Markers	Distance marks to the threshold	75 MHz. Increasingly rarely used
DME	Distance to the runway	Frequency paired with the LOC's, often replaces the markers
Approach lighting	Transition to visual guidance	Lights aligned on the centreline, before the threshold
Monitors	Continuous supervision	Switch to the standby transmitter or shut down the emission if the signal goes out of tolerance

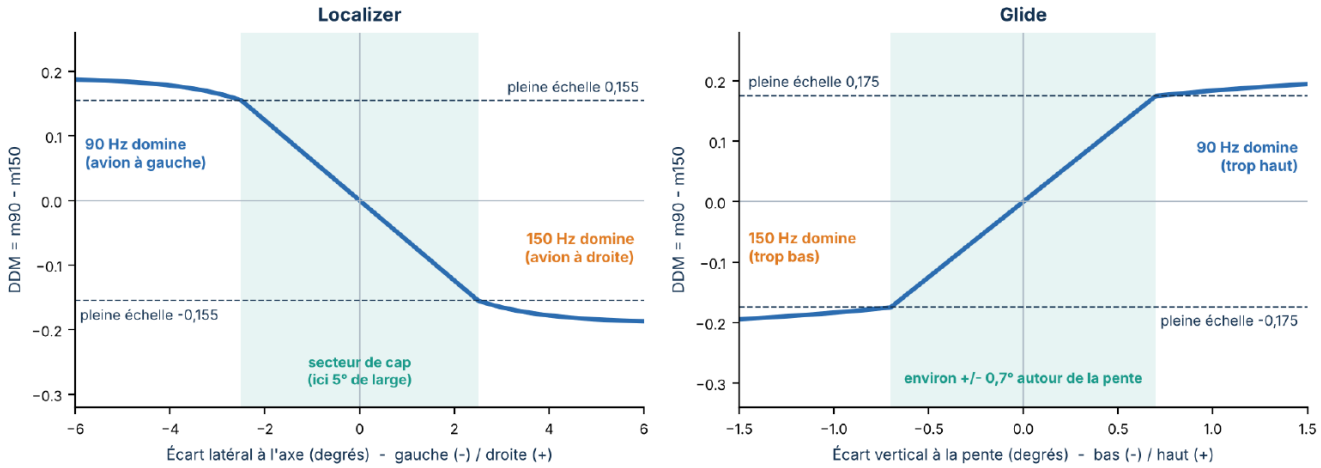
The principle: two tones

- The modulation depth of each tone varies with the aircraft's position in the beam.
- On the centreline, both tones have the same depth: the DDM is zero. To one side, one of the tones takes over.
- The receiver separates the two tones by filtering, measures their amplitudes and deduces the deviation.



What the aircraft receives: VHF carrier amplitude-modulated by 90 Hz and 150 Hz.

DDM as a function of offset



Near the centreline, the DDM grows linearly with the offset: this gives proportional guidance. Beyond full scale, it keeps increasing slowly, which preserves the direction of correction even at large offsets.

Reading it in the cockpit

Aircraft situation	Dominant tone	Needle	Correction
Left of the centreline	90 Hz	Course needle to the right	Correct to the right
Right of the centreline	150 Hz	Course needle to the left	Correct to the left
Above the slope	90 Hz	Slope needle down	Descend
Below the slope	150 Hz	Slope needle up	Climb or reduce the descent

The needle shows where the path to follow is: it "goes towards the centreline". Flying towards the needle brings the aircraft back onto the beam.

ILS

Instrument landing system: localizer, glide, DDM, markers

The radio in numbers

	Localizer	Glide
Band	108.10 to 111.95 MHz (VHF)	329.15 to 335.00 MHz (UHF)
Channels	40 channels. The 100 kHz digit is odd (108.10, 108.15, 108.30...)	Frequency paired with each localizer frequency
Modulation	AM, 90 Hz and 150 Hz, 20 % each on the centreline	AM, 90 Hz and 150 Hz, 40 % each on the centreline
Dominant tone	90 Hz on the left, 150 Hz on the right (seen from the approach)	90 Hz above, 150 Hz below
Full scale	DDM 0.155 (course sector of about 5°, adjustable from 3° to 6°)	DDM 0.175 (about ±0.7° around the slope)
Coverage	±10° up to 25 NM, ±35° up to 17 NM	±8° up to 10 NM, from 0.45 to 1.75 times the slope angle
Identification	Morse code of 3 to 4 letters starting with I, 1020 Hz tone	No identification of its own
Polarisation	Horizontal	Horizontal

Frequency pairs: 108.10 MHz with 334.70 MHz, 108.15 with 334.55, 110.30 with 335.00, 110.50 with 329.60, 111.90 with 331.10, 111.95 with 330.95.

The markers

Marker	Tone	Light	Typical position
Outer (OM)	400 Hz, dashes	Blue	About 7 km from the threshold
Middle (MM)	1300 Hz, alternating dots and dashes	Amber	About 1 km from the threshold
Inner (IM)	3000 Hz, dots	White	A few hundred metres from the threshold

Positions vary by runway and are published on approach charts.

Approach categories

Category	Decision height	RVR (visibility along the runway)
CAT I	60 m (200 ft) or more	550 m or more (or visibility of 800 m)
CAT II	30 to 60 m (100 to 200 ft)	300 m or more
CAT IIIA	Less than 30 m, or none	About 200 m or more
CAT IIIB	Less than 15 m, or none	About 50 to 200 m
CAT IIIC	None	None (no routine operational use)

Indicative values (ICAO and EASA); the exact rules depend on the country and the operator. The category actually usable depends on the ground installation, the aircraft's equipment and the crew's qualification.

Pitfalls and limits

- **False glides**: the glide also radiates spurious lobes above the true slope (around 6° and 9°). The glide is therefore intercepted from below.
- **Back beam**: the localizer also radiates backwards, with reversed indications if followed without care.
- **Critical areas**: vehicles and aircraft must stay behind the holding points, because they reflect the beams and distort them.
- **Terrain and obstacles**: reflections off the ground or buildings create ripples in the centreline or slope.
- **A single path**: the ILS imposes a straight-line approach. GLS (GBAS) or LPV (SBAS) approaches are satellite-based alternatives.

Receiving and observing them

Need	Detail
Receiver	RTL-SDR dongle, in AM mode (sheet 2b)
Antenna	Horizontal dipole (horizontal polarisation): sizes in sheet 1, ILS rows (two equal elements, each half the given length)
Software	SDR++, SDR# or GQRX to listen, Audacity or a Python script (FFT) to measure
What you hear	The localizer's Morse identification code (1020 Hz)
What you measure	In the audio spectrum, two lines at 90 Hz and 150 Hz. Their amplitude ratio varies with position in the beam

Reception is only possible near an airport and mostly along the approach axis: the localizer radiates towards the front of the runway and little to the sides. The glide is even more directional.

ILS

Instrument landing system: localizer, glide, DDM, markers

3 · Aviation

Good to know. The ILS is a safety-critical navigation aid. Never use it to navigate and **never transmit on these frequencies**: interference can endanger lives and is heavily penalised.

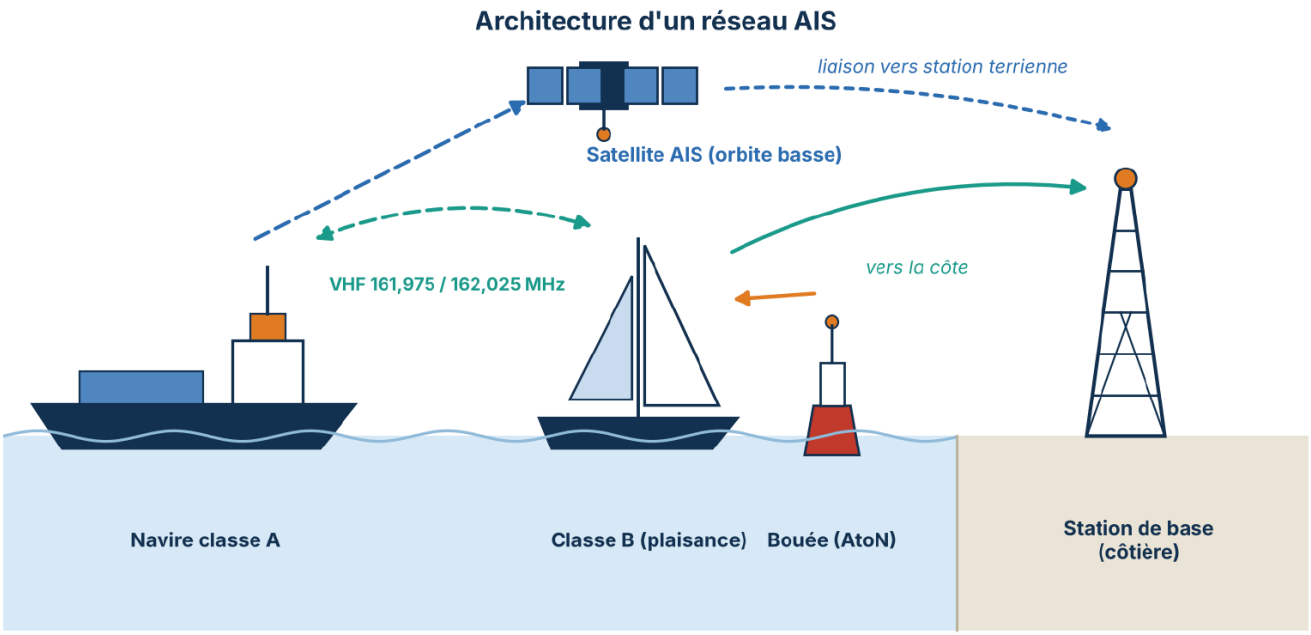
AIS and ATIS

Identifying ships by VHF radio: AIS and ATIS

In brief. AIS is a system for identifying and locating ships by VHF radio. Each transmitter automatically broadcasts, in clear, its identity, position, course and speed. The aim: avoid collisions, help the authorities monitor traffic and make rescue at sea easier. Carriage has been mandatory under the SOLAS convention (IMO) for the merchant ships concerned since 2002, with a phased introduction up to 2004.

2 channels	9600 bit/s	2250	9 digits
161.975 and 162.025 MHz	GMSK, 25 kHz channels	slots per minute per channel	MMSI identifier (sheet 12)

How it works



Chaque émetteur diffuse en clair, sans destinataire précis : tout récepteur à portée peut décoder.

There is no addressee: all transmitters (ships, base stations, buoys) broadcast and every receiver in range decodes. Coastal stations relay the data to traffic centres; low-orbit satellites pick up the same signals over the oceans.

Equipment classes

Equipment	Who it is for	Power	Access	Reporting rate
Class A	Ships subject to SOLAS: passenger ships, cargo ships of 300 GT and over on international voyages, 500 GT and over otherwise	12.5 W	SOTDMA	2 s to 3 min depending on speed and manoeuvres
Class B (CS)	Pleasure craft, fishing, small vessels	2 W	CSTDMA	about 30 s (3 min when stationary)
Class B (SO)	Same, higher-performance version	5 W	SOTDMA	more frequent than CS

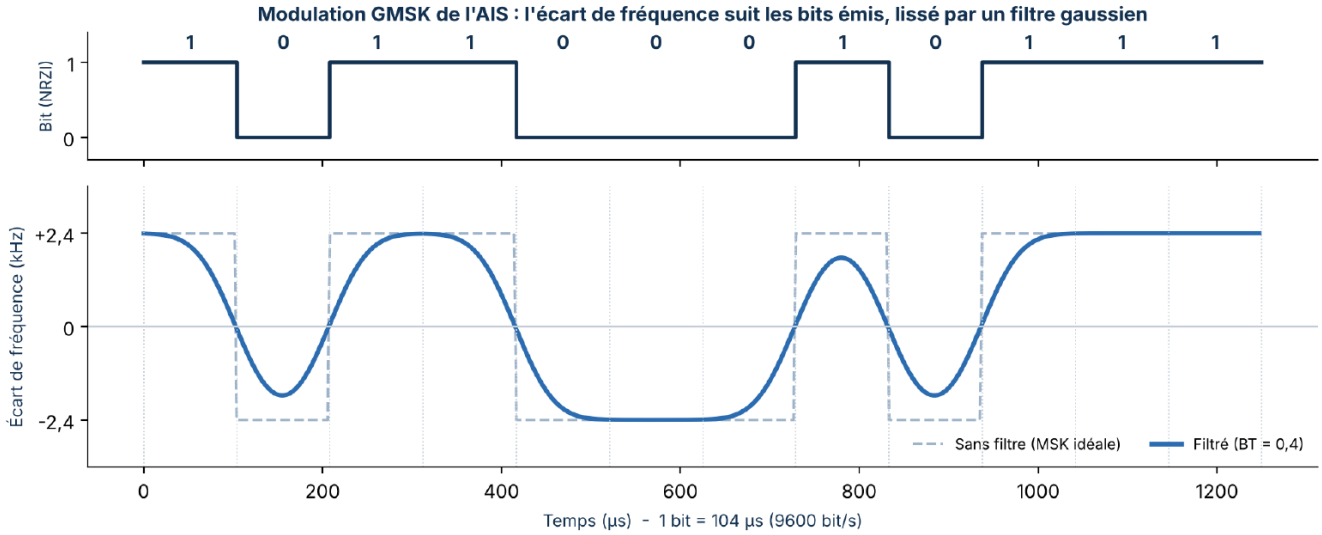
Class A reporting rate: about 3 min at anchor or moored, 10 s up to 14 knots, 6 s from 14 to 23 knots, 2 s above that, and faster still while changing course.

The radio layer

Parameter	Value
Channels	AIS 1: 161.975 MHz (channel 87B) / AIS 2: 162.025 MHz (channel 88B), the transmitter alternates between the two
Spacing	25 kHz
Modulation	GMSK, BT = 0.4, frequency deviation of ±2.4 kHz
Data rate	9600 bit/s
Coding	HDLC frame, NRZI coding, bit stuffing
Range	About 20 to 40 nautical miles between ships (line of sight, depending on antenna height)

Satellites pick up the same signals (satellite AIS). In very dense areas, transmissions overlap within their field of view, which limits detection. Message 27 was defined for long range.

The modulation: GMSK

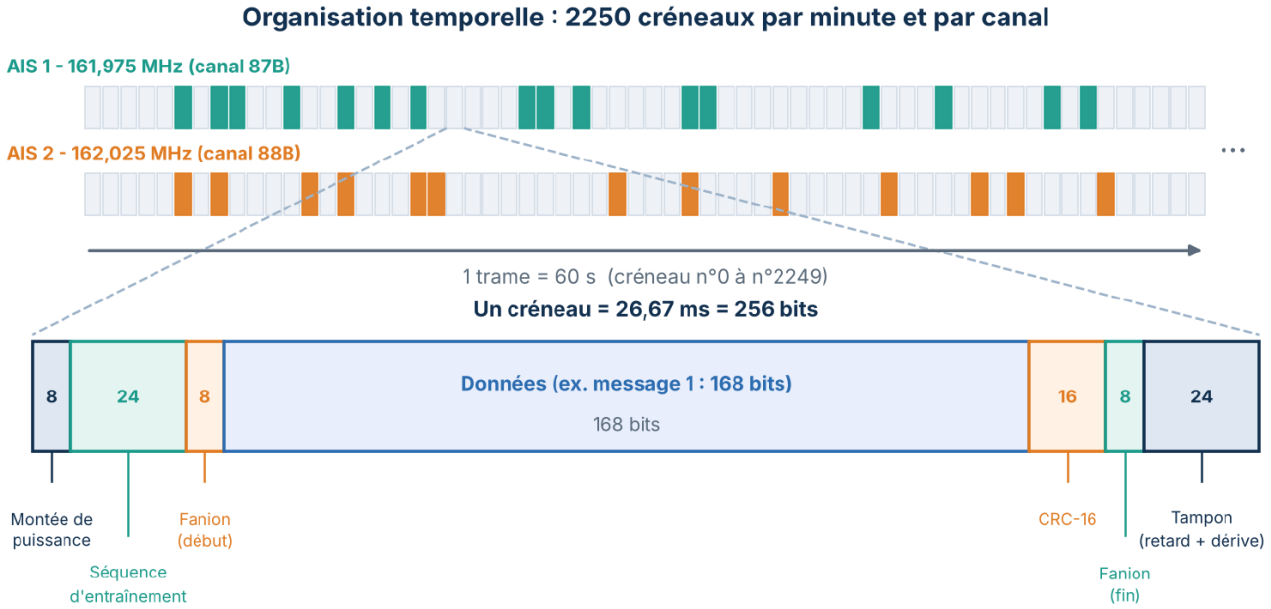


- GMSK is a continuous-phase frequency modulation: the frequency deviation switches between +2.4 and -2.4 kHz depending on the bit.
- A Gaussian filter (BT = 0.4) rounds off the transitions: the spectrum stays narrow enough to fit in a 25 kHz channel, at the cost of slight overlap between consecutive bits.
- NRZI coding means that a 0 bit causes a frequency change, whereas a 1 bit does not.
- Bit stuffing inserts a 0 after five consecutive 1s, so that the frame flag (01111110, i.e. 7E in hexadecimal) never appears by chance in the data.

Identifying a transmitter: the MMSI

The MMSI has 9 digits; the first three (MID) give the country for a ship. The full formats (ship, coast station, aid to navigation, AIS-SART beacons) and the list of MIDs are in sheet 12.

Sharing time: TDMA

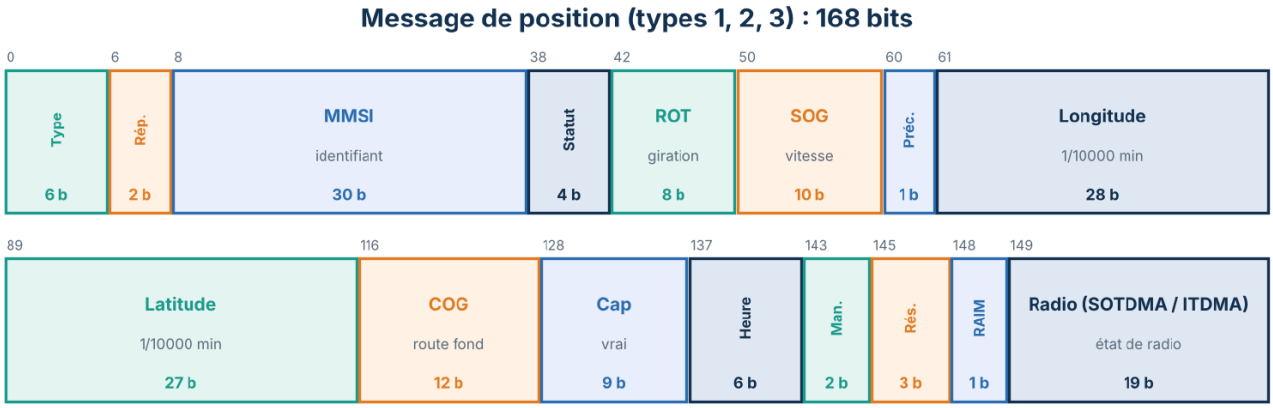


- **SOTDMA** (class A): each station picks a free slot and announces its future slots in the radio status field. Stations are synchronised to UTC time, usually via GPS.
- **CSTDMA** (class B CS): the station listens before transmitting and only uses a slot if it is free.
- **ITDMA**: one-off reservation, for example when entering the network or for longer messages.
- **FATDMA**: fixed slots reserved by base stations.

AIS and ATIS

Identifying ships by VHF radio: AIS and ATIS

A position message



Position en 1/10000 de minute d'arc, vitesse (SOG) en 1/10 de nœud, route (COG) en 1/10 de degré. Petit chiffre au-dessus d'un champ : rang du premier bit.

A position message fits in a single slot (168 data bits). Longer messages, such as type 5 static data (424 bits), take several slots.

Common messages

Type	Content
1, 2, 3	Class A position report
4	Base station report (UTC time and position)
5	Static and voyage data: name, callsign, ship type, dimensions, draught, destination, ETA (2 slots)
9	Position of a search and rescue aircraft
14	Free-text safety message (also used by AIS-SART beacons)
18, 24	Class B position and static data
21	Aid to navigation (AtoN): buoy, beacon, including virtual ones
27	Long-range position report (satellite detection)

Decoded example

```
!AIVDM,1,1,,B,177KQJ5000G?tO`K>RA1wUbN0TKH,0*5C
```

Element	Meaning
!AIVDM,1,1,,B	NMEA AIS sentence received, fragment 1 of 1, no sequence number, channel B
177KQJ5000G?...	Data in 6-bit ASCII (168 bits)
0*5C	0 padding bits, checksum (XOR) 5C
Decoded	Type 1, MMSI 477553000 (MID 477, Hong Kong), status 5 (moored), speed 0.0 knots, position 47.58283 N and 122.34583 W, course over ground 51.0 degrees, heading 181 degrees, UTC time: second 15

Receiving them

Need	Detail
Receiver	RTL-SDR dongle (or another SDR covering 162 MHz), sheet 2b
Antenna	Quarter-wave vertical (size: sheet 1, AIS row)
AIS-catcher	Decoder with web interface, for RTL-SDR
rtl_ais	Command-line decoder, NMEA output
OpenCPN	Nautical charting software that displays AIS targets
MarineTraffic, VesselFinder, AISHub	Community aggregators where you can share the data you receive

AIS and ATIS

Identifying ships by VHF radio: AIS and ATIS

Good to know. Transmitting is a different matter from receiving: you need approved equipment and an assigned MMSI, and broadcasting false positions or false beacons is illegal and dangerous. The data is entered by crews: it can be wrong or missing (AIS switched off), so do not rely on it alone for navigation.

Bonus: ATIS on inland waterways

On European rivers and canals, another identification system coexists with AIS: ATIS (Automatic Transmitter Identification System). Instead of transmitting continuously like AIS, the boat sends a short digital burst each time the push-to-talk (PTT) button is released at the end of a VHF voice call, on the same channel. It makes it possible to identify who has just spoken. Do not confuse it with aeronautical ATIS, the voice information message broadcast by airports (sheet 7).

	AIS	ATIS
Use	Worldwide, seagoing ships	European inland waterways
Transmission	Continuous, periodic beacons	One burst after each voice message
Frequency	161.975 and 162.025 MHz	The working channel used for voice
Modulation	GMSK, 9600 bit/s	FSK 1300 / 2100 Hz, 1200 baud, about 285 ms
Content	Position, course, speed, ship data	Identity only (callsign or coded MMSI)

The burst is built on DSC signalling, but on the voice channel. The modulation figures come from Wikipedia; they do not appear in the application's documentation.

ATIS RX on UV-K5 / UV-K1 radio (Labs edition)

The ATIS RX application in the experimental Labs firmware decodes this burst directly on the radio. Install the version matching your firmware with UV Studio (sheet 2b), then launch it with **F + 7**. First choose a working channel of the inland waterways, in FM.

Element	Detail
Decoding	Ten-digit identity; callsign and country reconstructed when possible. The decoded identity is authoritative, the reconstruction is a convenience
List	Eight stations in RAM, the most recently received goes to the top. The list is lost when you exit the application
Display	Callsign or identity, country, reception level, number of receptions, age of the last burst, frequency, recognised marine channel number
Speaker	Off by default and not needed for decoding; the setting is saved

Key	Action
UP / LEFT or DOWN / RIGHT	More recent or older station (UP / DOWN on UV-K5, LEFT / RIGHT on UV-K1, depending on SetNav)
1	Turns the speaker on or off, opened by the squelch
2	Clears the station list and the counters
EXIT	Exits the application

DSC, MMSI and NAVTEX

Digital selective calling (ITU-R M.493) and navigational warnings: signal, codes, settings

DSC (Digital Selective Calling): signals by band

Band	Frequency(ies)	Modulation	Audio tones	Data rate	Remarks
VHF channel 70	156.525 MHz	FM, AFSK	1300 Hz / 2100 Hz	1200 baud	Distress, urgency, safety, individual or group calls. No voice on this channel.
MF	2187.5 kHz (distress); 2177 kHz (ship to ship)	FSK, SSB (J2B)	1615 Hz / 1785 Hz (170 Hz shift)	100 baud	Range of a few hundred km.
HF	4207.5 · 6312 · 8414.5 · 12 577 · 16 804.5 kHz (distress)	FSK, SSB (J2B)	1615 Hz / 1785 Hz (170 Hz shift)	100 baud	Worldwide range. Alerts repeated every 3.5 to 4.5 min until acknowledged.

Structure of a distress call (in order)

Call element	Content	Details
Dot pattern	Sequence of alternating 1s and 0s	Bit synchronisation: 200 bits on MF/HF, 20 bits on VHF
Phasing sequence	7 symbols (DX / RX)	Frame synchronisation; the DX and RX channels carry the same message with time diversity
Format specifier (×2)	112 = distress	Others: 116 all ships, 120 individual call, 123 individual semi-automatic, 102 geographic area, 114 group
Address (if individual)	MMSI of the addressee, 9 digits	Distress: "all ships", no specific address
Category	112 = distress	Others: 100 routine, 108 safety, 110 urgency
Caller identity	MMSI, 9 digits	The first three digits (MID) give the country
Nature of distress	3-digit code	100 fire, 101 flooding, 102 collision, 103 grounding, 104 listing, 105 sinking, 106 disabled and adrift, 107 undesignated, 108 abandoning ship, 109 piracy, 110 man overboard, 112 EPIRB
Position	Quadrant + latitude and longitude	5 symbols, in degrees and minutes (e.g. 48°52' N, 002°20' E)
Time	4 UTC digits	HHMM; 8888 = time not available
Subsequent communication	Mode code	100 = telephony (156.800 MHz, 2182 kHz); 109 = SSB (J3E); 126 = unspecified
End of sequence (EOS)	117, 122 or 127	117: acknowledgement requested; 122: acknowledgement; 127: end without acknowledgement
ECC	Check character	Error check over the whole call

A DSC symbol is a 10-bit word: 7 information bits and 3 check bits (number of zeros in the 7 bits). A distress alert is sent 5 times in a row, then repeated until a coast station or a ship acknowledges it. Voice follows on channel 16 or on 2182 kHz.

Listening to and decoding DSC

- **VHF (channel 70)**: UV-K1 on 156.525 MHz in FM. You hear a short "bzzz" lasting a few seconds (1200 baud, two fast tones). A decoder on a PC or smartphone, fed with the audio, reads the message.
- **HF**: USB or LSB, tones at 1615 and 1785 Hz (centre around 1700 Hz). The exact dial frequency is set 1700 Hz below the stated carrier, in USB.
- **Decoders**: YADD (Yet Another DSC Decoder) and MultiPSK on Windows; for a home-made decoder, same approach as for NAVTEX: FSK detection, sync on the dot pattern, DX / RX voting, checking the 3 check bits. To be verified: availability of this software.
- If you decode a real distress call (format 112, category 112), note the MMSI, position and time, then alert CROSS (196) or 112 if no acknowledgement is heard (sheet 15).

MMSI and MID: reading a maritime identifier

Identifier type	Format	Example / remarks
Ship	MID + 6 digits	226 123456: MID 226 = France
Coast station	00 + MID + 4 digits	00 226 1234 (France)
Group of ships	0 + MID + 5 digits	Group call
SAR aircraft	111 + MID + 3 digits	Aeronautical search and rescue
Aid to navigation (AtoN)	99 + MID + 4 digits	Buoy, beacon (sheet 11)
AIS-SART / AIS MOB / AIS EPIRB	970 / 972 / 974 + 6 digits	AIS distress devices (sheet 16)

DSC, MMSI and NAVTEX

Digital selective calling (ITU-R M.493) and navigational warnings: signal, codes, settings

Common MIDs	
France	226, 227, 228
United Kingdom	232 to 235
Germany	211, 218
Spain	224, 225
Italy	247
Netherlands	244 to 246
Belgium	205
Ireland	250
United States	338, 366 to 369
Canada	316

The MID is the 3-digit country code.

NAVTEX (navigational warnings)

Parameter	Value
Frequencies	518 kHz (English, international); 490 kHz (national language); 4209.5 kHz (HF, supplementary transmissions)
Modulation	FSK 100 baud, 170 Hz shift, in LSB or USB depending on the decoder (SITOR-B, FEC mode)
Coding	7-bit code (CCIR 476): each character contains 4 bits set to 1 and 3 bits set to 0, which detects errors. Each character is sent twice: direct (DX), then repeated (RX) 4 characters later (time diversity)
Message format	ZCZC B1 B2 B3 B4 (text) NNNN: B1 = station letter; B2 = type (A navigational warnings, B weather - gale warnings, C ice, D SAR and piracy, E weather forecasts, F pilot service, L additional warnings, Z nothing to broadcast); B3B4 = serial number 00 to 99 (00 = urgent)
Schedule	Each station transmits in a 10-minute slot, every 4 hours. The slot is shifted by 10 minutes per letter (A at 00:00, B at 00:10, C at 00:20...) so that two stations do not interfere. The France zone is NAVAREA II
Decoding	MultiPSK, YAND (Windows), fldigi (NAVTEX / SITOR-B mode in recent versions, to be verified); or a home-made 100 baud FSK decoder with phase detection then DX / RX voting

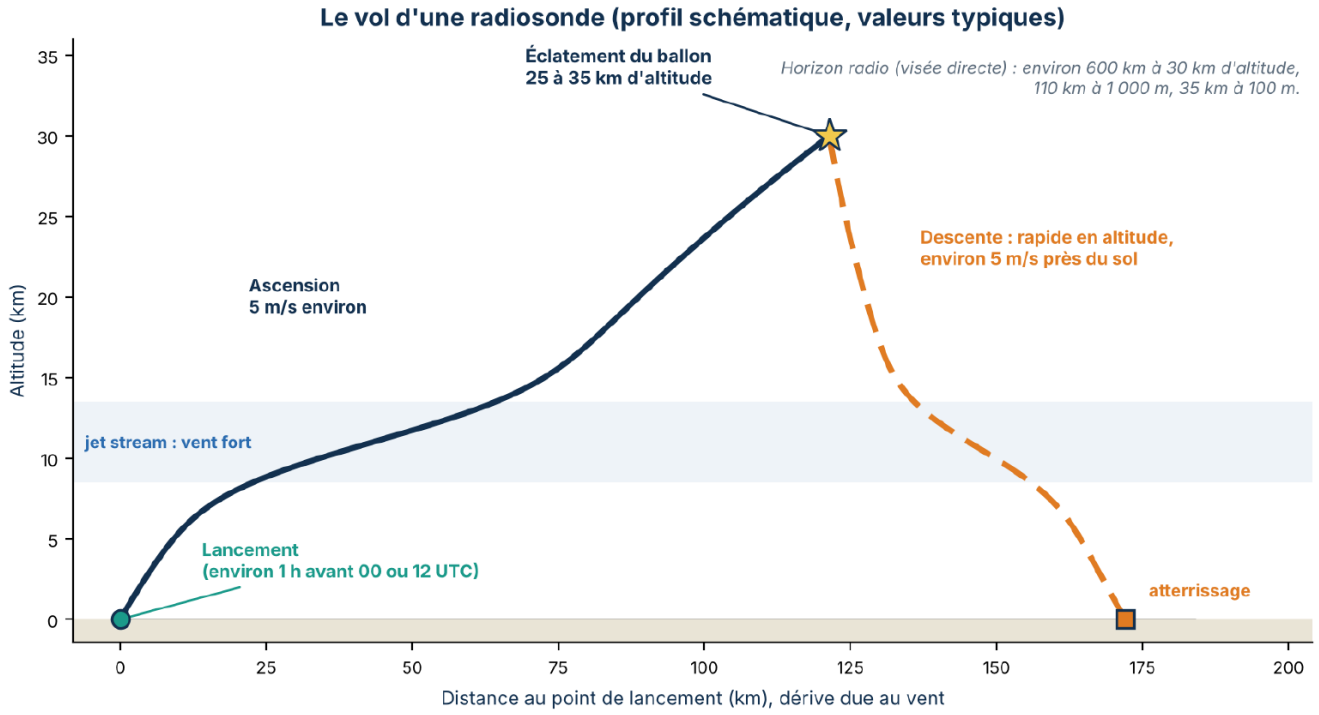
Radiosondes

Weather balloons: reception and tracking at 400 - 406 MHz

In brief. A radiosonde is a small battery-powered instrument carried by a weather balloon that measures the atmosphere and sends its readings by radio. About 1,300 sites worldwide launch balloons, mainly for the 00 and 12 UTC observations (launch about an hour earlier). The sonde measures temperature, humidity and, depending on the model, pressure; its GPS gives the position and, from its movements, the wind. It transmits in UHF at around 400 to 406 MHz (some models at 1680 MHz). The free software radiosonde_auto_rx scans the band, identifies the sondes, decodes the telemetry and publishes positions to SondeHub.

400 - 406 MHz	~1,300	00 and 12 UTC	~250 g
UHF radiosonde band	launch sites	observation times	typical mass

The flight of a radiosonde



Typical values: ascent of about 1 h 30 at 5 m/s, burst at 25 to 35 km altitude, then a descent of about forty minutes under parachute. Wind makes the sonde drift from several tens of km to over 100 km.

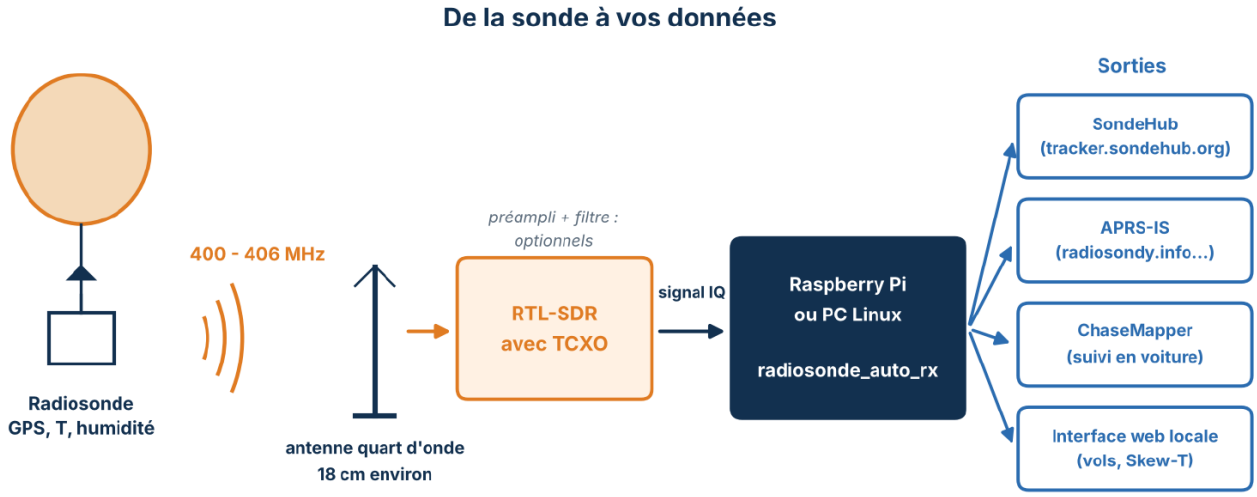
What the sonde measures

Quantity	Source
Temperature, relative humidity	Onboard sensors
Pressure	Onboard sensor, but some models do not transmit it
Position	GPS: latitude, longitude, altitude
Wind	Derived from GPS movement

Radiosondes

Weather balloons: reception and tracking at 400 - 406 MHz

From the sonde to your data

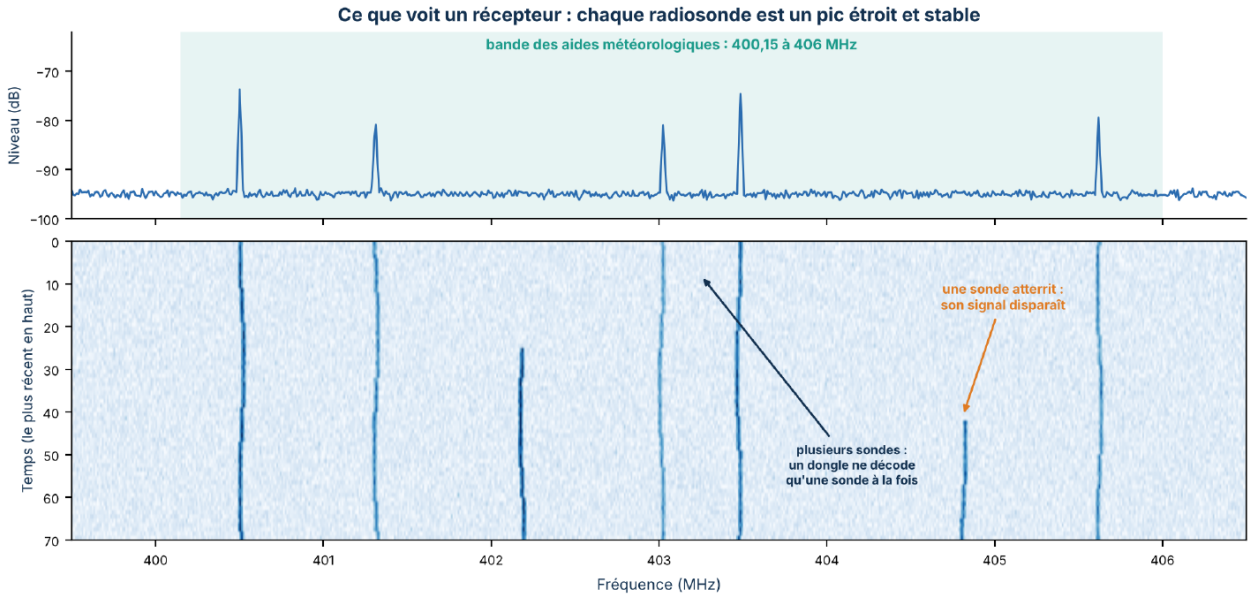


Item	Detail
Computer	Raspberry Pi 3 or newer, or any Linux PC
Receiver	RTL-SDR dongle with TCXO oscillator, ppm drift correction properly set (sheet 2b)
Several sondes at once	One dongle decodes only one; an Airspy (via KA9Q-Radio), a HydraSDR or an SDRPlay covers the whole band
Antenna	Quarter-wave for 400 - 406 MHz (sizes: sheet 1); an amateur 70 cm antenna probably works
More range	Preamplifier and bandpass filter recommended; good results are possible without

radiosonde_auto_rx installation	Detail
Docker	Probably the simplest
Raspberry Pi image	Preconfigured images, listed in the project wiki
Native	Raspbian Bullseye or newer, Python 3.6 or later
Configuration	station.cfg file: station position, frequency range, upload to SondeHub (enabled by default)

Data sent to SondeHub is licensed CC BY-SA 2.0. The local web interface shows the station status, past flights and Skew-T diagrams.

The spectrum: one peak per sonde



A sonde stays on the same frequency throughout the flight, with a slight drift: this is what lets auto_rx follow it. Several sondes are often active together.

Radiosondes

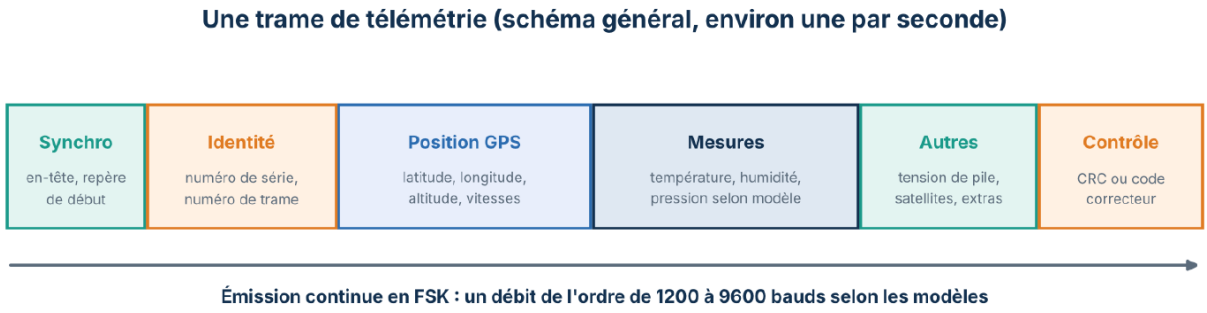
Weather balloons: reception and tracking at 400 - 406 MHz

Which sondes?

Manufacturer	Supported models
Vaisala	RS41-SG, RS41-SGP, RS41-SGM; RS92-SGP, RS92-NGP
Graw	DFM-06, DFM-09, DFM-17
Meteomodem	M10, M20
Intermet Systems	iMet-4, iMet-54
Lockheed Martin	LMS6-400, LMS6-1680
Meisei	iMS-100, RS11G
Others	MRZ-H1 (400 MHz), Meteosis MTS01, Changfeng CF-06, Changwang GTH-6, Meteolabor SRS-C50

The level of support varies by model and by measurement (some do not send pressure).

Telemetry



Le décodeur repère l'en-tête, relit les bits, vérifie le contrôle d'erreur, puis convertit en grandeurs physiques.

```
{ "type":"RS41", "serial":"S1234567", "frame":3120, "datetime":"2026-10-15T11:02:14Z",
  "freq":403.5, "lat":48.1234, "lon":2.3456, "alt":15234, "vel_v":5.1, "temp":-55.2, "humidity":4.3 }
```

Illustrative example, made-up values: altitude in metres, vertical speed in m/s, temperature in °C.

Tracking and recovery

Tool	Use
SondeHub Tracker	Live map of all received sondes: tracker.sondehub.org
APRS-IS	Positions viewable on radiosondy.info and wettersonde.net
ChaseMapper	Field tracking, by car, to go and fetch the sonde
auto_rx web interface	Station status, past flights, Skew-T

Troubleshooting

Symptom	Things to try
No signal	Wait for a nearby launch (00 and 12 UTC), antenna outdoors, check TCXO and ppm correction
Intermittent decoding	Preamplifier and filter, clear antenna, dongle away from the computer
Several sondes	Add dongles or use a wideband SDR
Pressure missing	Normal for some models

Good to know. Receiving is free, but in the field do not enter private property and avoid power lines, tall trees, water and roads. Sondes belong to weather services and often carry a label saying what to do if you find one. Never transmit on these frequencies.

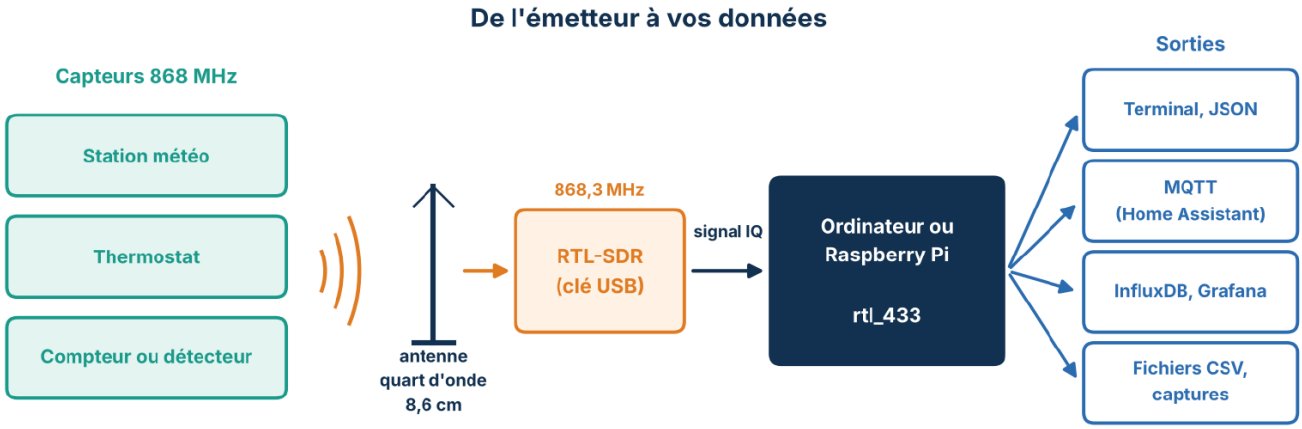
Wireless sensors: rtl_433

Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz

In brief. rtl_433 is a free decoder for wireless sensors in the ISM bands: mainly 433.92 MHz, but also 868 MHz (SRD in Europe), 315, 345 and 915 MHz. It works with an RTL-SDR dongle (or any SDR supported by SoapySDR). Weather stations, thermostats, detectors and meters send short unencrypted messages at regular intervals, without pairing; rtl_433 identifies them among its decoders (numbered up to 386).

868.3 MHz	250 kHz	433.92 MHz	386
very common channel	default sample rate	default frequency	highest decoder number

How it flows

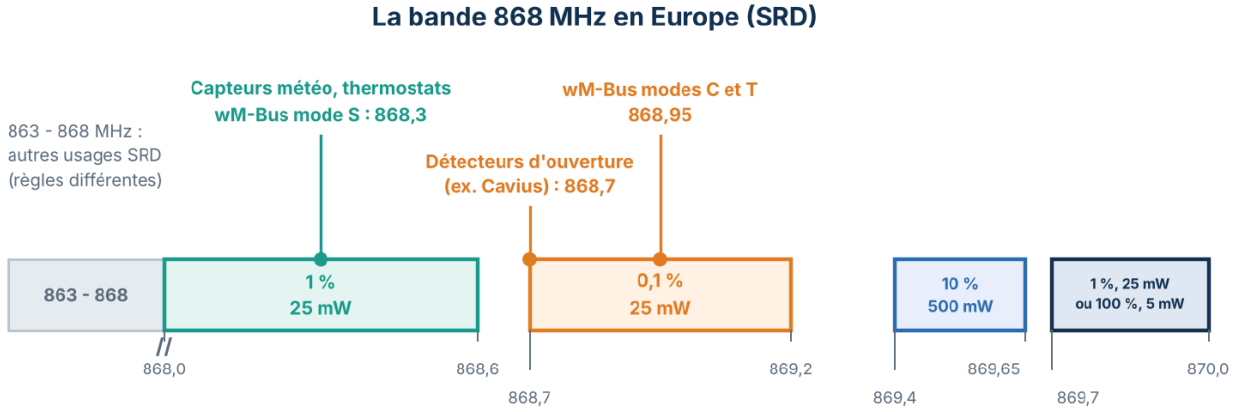


The dongle digitises the band and sends the raw signal (IQ) to rtl_433, which detects pulses, demodulates, splits frames and tries them against each decoder. Most check a checksum before publishing. Quarter-wave antenna: sizes in sheet 1.

Install and run

System	Command or method
Debian, Ubuntu	sudo apt install rtl-433
macOS	brew install rtl_433
FreeBSD	pkg install rtl-433
Docker	Community images (hertzg's rtl_433_docker project)
Source	docs/BUILDING.md in the merbanan/rtl_433 GitHub repository
First try	rtl_433 -f 868.3M: let it run for a few minutes, sensors do not transmit continuously

The 868 MHz band



Rapport cyclique maximal et puissance apparente rayonnée par sous-bande (valeurs indicatives, recommandation ERC 70-03).

LoRa (868,1 / 868,3 / 868,5) et Sigfox (868,13) utilisent aussi cette bande, mais rtl_433 ne les décode pas.

Wireless sensors: rtl_433

Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz

Each sub-band imposes a maximum duty cycle: at 1 %, a transmitter does not occupy the channel for more than 36 s per hour. Sensors therefore send very short messages, often every 30 to 60 s or more: be patient.

What you find there

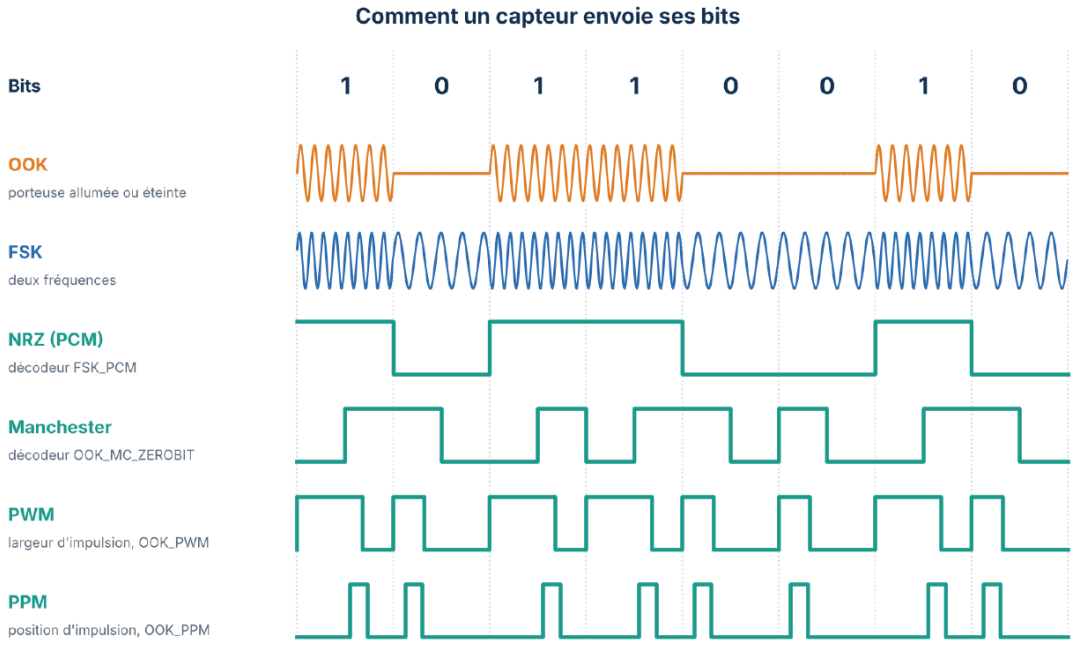
Family	Examples (decoder no.)	Note
Weather stations	Bresser 5-in-1, 6-in-1, 7-in-1 (119, 172, 173); Fine Offset WS80 and WS90 (213, 244); Ecowitt, LaCrosse, TFA	868 MHz version often sold in Europe: check the model
Thermostats	Eberle Instat 868r1 (358), Watts Vision (366)	Watts Vision: 868.3 MHz
Detectors	Cavius, door and window (368)	868.7 MHz
Meters	Wireless M-Bus modes S, T, C, R, F (104 to 107, 238)	High data rate: sample rate 1000 to 1200 kHz; data sometimes encrypted
Home automation	Somfy io-homecontrol (189)	Bidirectional, 868 MHz

Tuning the reception

Option	Purpose
-f 868.3M	Listening frequency (default 433.92 MHz)
-s 250k	Sample rate (default 250 kHz); 1000k or 1200k for wM-Bus
-g 0	Automatic gain; if overloaded or silent, try fixed values
-p 20	Dongle drift correction, in ppm (example value)
-f 433.92M -f 868.3M -H 30	Alternating listening, switching every 30 s (default 600 s)

With frequency hopping, frames on the other band go unnoticed: one dongle per band is more reliable.

How sensors encode their bits



Les conventions de lecture varient selon les fabricants (bit 1 ou 0 inversé, front montant ou descendant).

- The **modulation** (OOK, FSK) says what varies on the carrier; the **encoding** (NRZ, Manchester, PWM, PPM) says how the bits are represented.
- At 868 MHz, FSK is often preferred to OOK, which is more common at 433 MHz: better resistance to noise and distance.
- A frame generally contains a preamble, identifier, measurements and a checksum or CRC, which rtl_433 verifies and reports in the mic field.

No.	868 MHz decoder	Suggested setting
104	Wireless M-Bus, modes C and T, 100 kbit/s	-f 868.95M -s 1200k
105	Wireless M-Bus, mode S	-f 868.3M -s 1000k
106, 107	Wireless M-Bus, modes R and F	Disabled by default

Wireless sensors: rtl_433

Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz

No.	868 MHz decoder	Suggested setting
238	Wireless M-Bus, mode T	-f 868.3M -s 1000k
358	Eberle Instat 868r1	-
366	Watts Vision (thermostat)	-f 868.3M
368	Cavius (door and window detector)	-f 868.7M

```
{ "time":"2026-10-14 07:12:03", "model":"Bresser-6in1", "id":12345678, "channel":0,
  "battery_ok":1, "temperature_C":14.2, "humidity":71, "mic":"CRC" }
```

Illustrative example, made-up values. Fields vary by sensor; -M level adds the signal level.

Useful commands

Command	Effect
rtl_433 -f 868.3M	Listens on 868.3 MHz and displays decoded messages
rtl_433 -f 868.3M -F json -M level -M protocol	JSON output with level and protocol number
rtl_433 -f 868.3M -s 1000k -R 105	Decoder 105 only (wM-Bus mode S)
rtl_433 -f 868.3M -F mqtt://localhost:1883	Publishes to MQTT (Home Assistant, Node-RED)
rtl_433 -f 868.3M -A	Pulse analyser for an unknown signal
rtl_433 -f 868.3M -S unknown	Saves undecoded signals
rtl_433 -r capture.cu8	Replays a capture without hardware
rtl_433 -f 868.3M -T 300	Stops after 300 s

Other outputs: csv, influx, kv, syslog, http. The generic decoder -X lets you describe a simple protocol yourself.

Troubleshooting

Symptom	Things to try
No messages	Wait several minutes, move the antenna closer, check the sensor's actual frequency
Noise without decoding	Gain too high or too low: -g 0 then fixed values
Frequency offset	Correct with -p or use a TCXO dongle
Interference	Move the dongle away from the computer (USB extension), mind the power supply
Unknown signal	-A and -S unknown, then ask on the GitHub repository with the capture

Good to know. Receiving is generally permitted, but data from your neighbours' sensors (meters, alarms, thermostats) remains private: do not publish or exploit it. Never transmit to replay a command (shutter, door, alarm): that is intrusion. To go further: merbanan/rtl_433 repository, Home Assistant, Node-RED, InfluxDB and Grafana.

Meshtastic and MeshCore

LoRa mesh networks without Internet: principle, EU 868 MHz settings, listening

In brief. Meshtastic and MeshCore are two free firmwares that turn a small LoRa board (Heltec, LilyGO T-Beam or T-Deck, RAK, Seeed XIAO) into a text messenger that needs no Internet or cell tower, controlled from a phone. They use the same LoRa modulation but cannot talk to each other: all nodes need the same system, and the same board can run either one by reflashing. In Europe, both live in the SRD band 869.4 - 869.65 MHz, right next to the 868 MHz sensors of sheet 14.

869.4 - 869.65 MHz	10 %	500 mW	LoRa
EU sub-band (SRD)	max. transmit time	27 dBm e.r.p. maximum	spread-spectrum modulation

Two philosophies

	Meshtastic	MeshCore
Routing	Managed flooding: each node rebroadcasts a new packet unless it has already seen it	Companions do not rebroadcast; dedicated repeaters relay. The first message is flooded, then the path is remembered
Roles	Every device sends and relays	Companion (personal device), repeater (infrastructure), room server (keeps messages for absent users)
Max hops	7 (3 by default)	64
Scale	Small and medium networks, risk of congestion with many nodes	Large planned networks with repeaters; detailed delivery acknowledgements
App	Free Android and iOS apps, Bluetooth or Wi-Fi, position map	Mobile app, topology and delivery confirmations
Encryption	AES, channel protected by a key	Own encryption, not compatible
Choose it for	Getting started, small group, lots of documentation	Larger network, messages needing confirmation, ability to place repeaters

Settings in Europe

	Value	Note
Meshtastic, region	EU_868: 869.4 - 869.65 MHz	10 % transmit time over a rolling hour, 27 dBm max. The node stops when the limit is reached
Meshtastic, preset	LONG_FAST by default, channel centred on 869.525 MHz	No "Turbo" presets: 500 kHz bandwidth is not allowed here. LONG_MODERATE: maximum range
Meshtastic, other region	EU_N_868 (narrow band): NARROW_FAST / SLOW, 62.5 kHz	Same sub-band, narrower channel
MeshCore, example (Essex Ham)	869.618 MHz, 62.5 kHz, SF 8, CR 8, 22 dBm	Community settings vary: a Czech group publishes 869.525 MHz, SF 7, CR 5. Ask for those of your local network
MeshCore, duty cycle	Adjustable from 1 to 100 %, 50 % by default	50 % exceeds the 10 % limit: lower it to 10 %

Nodes must share exactly the same frequency, bandwidth, spreading factor (SF) and coding rate to talk to each other. Check the figures against the official documentation before configuring.

Getting started

- **Meshtastic**: flash the board with the web tool or the installer from meshtastic.org, set the region (until it is set, the node does not transmit), then pair the app over Bluetooth.
- **MeshCore**: flash the "companion", "repeater" or "room server" firmware with the project's web flasher from a WebSerial-compatible browser. The role is set by the firmware: changing role means reflashing. Bluetooth pairing uses the code shown on the board's screen.
- Never power a board without its antenna: the transmit stage can be damaged.
- Boards with an nRF52 processor (RAK4631) are the most power-efficient. Antennas are calculated as in sheet 1 (quarter-wave at about 869 MHz).

Listening with the UV-K1 and the SDR dongle

- **UV-K1**: useless. LoRa modulation (chirps) cannot be demodulated as FM or AM; the radio only shows activity. GOGUFW Messenger (sheet 2b) is the narrowband equivalent for these radios.
- **RTL-SDR dongle**: a waterfall centred on 869.5 MHz (SDR++ or GQRX) shows packets as small slanted lines, a sign of a nearby mesh network. They cannot be decoded with rtl_433 (sheet 14), which does not handle LoRa; read them with a Meshtastic or MeshCore node, or a dedicated LoRa receiver.

- A burst every few seconds on 869.525 MHz is typically Meshtastic; the sensors of sheet 14 are shorter and elsewhere (868.3 MHz).

Legality. These are transmissions: the 869.4 - 869.65 MHz sub-band imposes a 10 % duty cycle and 500 mW e.r.p. (27 dBm, about 823 mW e.i.r.p.); antenna gain and cable loss count. An amateur radio callsign gives no extra rights on this band; encrypted messages are not allowed on amateur bands. The power shown by the firmware is a software ceiling, not your country's legal limit. Sources: Meshtastic documentation and community guides; check your regulator's rules.

Emergency frequencies

Listenable from 1 kHz to 20 GHz: frequency, modulation, coding, listening with the UV-K1

Reading the table. Direct: voice can be heard as is (AM, FM, SSB). **External decoding**: you hear data tones that must be fed to a decoder (PC, smartphone, SDR). **Detection only**: a burst or a pulse, with no readable content. **Not listenable**: encrypted or out of reach of an ordinary receiver.

Frequency	Modulation	Coding / format	Use	Listening
MF AND HF (1 kHz to 30 MHz): nothing listenable below 457 kHz				
457 kHz	OOK	Short pulse about every second	Avalanche transceivers: victim detectors	Detection only
500 kHz	A1A (CW)	Morse	Former distress frequency, abandoned in 1999	Historical
518 · 490 · 4209.5 kHz	FSK 100 bd, 170 Hz	SITOR-B, 7 elements	NAVTEX: notices, distress and SAR relays (sheet 12)	External decoding
2174.5 kHz	FSK 100 bd	NBDP (radio telex)	MF printer distress	External decoding
2182 kHz	SSB, USB	Voice	MF distress and calling	Direct
2187.5 kHz	FSK 100 bd	DSC (ITU-R M.493)	MF digital alert (sheet 12)	External decoding
3023 · 5680 kHz	SSB, USB	Voice	On-scene SAR, aircraft and ships	Direct
4125 · 6215 · 8291 · 12290 · 16420 kHz	SSB, USB	Voice	Maritime HF voice distress (4125 also SAR)	Direct
4177.5 · 6268 · 8376.5 · 12520 · 16695 kHz	FSK 100 bd	NBDP	HF printer distress	External decoding
4207.5 · 6312 · 8414.5 · 12577 · 16804.5 kHz	FSK 100 bd	DSC	HF DSC alerts, worldwide range (sheet 12)	External decoding
8364 kHz	SSB, USB	Voice	Survival craft and life rafts	Direct
3760 · 7110 (LSB) · 14,300 kHz (USB)	SSB	Voice	IARU Region 1 amateur emergency networks; 18,160 and 21,360 kHz in USB	Direct
27.065 MHz (CB 9)				
VHF (30 - 300 MHz)				
121.500 MHz	AM	Voice; ELT beacons: siren 1600 → 300 Hz	Aviation distress, final homing of beacons	Direct
123.100 MHz	AM	Voice	On-scene SAR, aviation	Direct
156.800 MHz (channel 16)	FM	Voice	Maritime distress, urgency, safety; CROSS keeps watch	Direct
156.525 MHz (channel 70)	FM, AFSK 1200 bd	DSC	Maritime digital alert (sheet 12)	External decoding
156.300 MHz (channel 6)	FM	Voice	Ship to ship; SAR coordination	Direct
161.975 · 162.025 MHz	GMSK 9600 bd	AIS (sheet 11)	AIS-SART, AIS MOB, AIS EPIRB in distress	External decoding
144.800 MHz	AFSK 1200 bd	AX.25 (APRS)	Positions and messages used in emergencies	External decoding
145.500 MHz	FM	Voice	VHF amateur calling, emergency networks	Direct
162.400 - 162.550 MHz	FM	SAME: AFSK 520.83 bd	NOAA Weather Radio, alerts (United States, not in France)	External decoding
UHF (300 MHz - 3 GHz)				
243.000 MHz	AM	Voice	Military distress	Direct
225 - 400 MHz	AM; HAVE QUICK	Voice, sometimes encrypted	Aviation and military emergencies	Direct (partial)
380 - 400 MHz	TETRAPOL / TETRA	Encrypted digital	French emergency services (ANTARES, INPT): SDIS, SAMU, police	Not listenable
406.0 - 406.1 MHz	BPSK ±1.1 rad, 400 bit/s	Manchester, 112 or 144 bits, BCH	EPIRB, PLB, ELT: Cospas-Sarsat (sheet 16)	Detection only
433.500 MHz	FM	Voice	UHF amateur calling, emergency networks	Direct

Emergency frequencies

Listenable from 1 kHz to 20 GHz: frequency, modulation, coding, listening with the UV-K1

Frequency	Modulation	Coding / format	Use	Listening
1090 MHz	PPM, 1 Mbit/s	Mode S, 112 bits, CRC-24	ADS-B: emergency (squawk 7700, sheet 8)	External decoding
1537 - 1545 MHz	BPSK 600 bd	Inmarsat-C, SafetyNET	Maritime satellite alerts	Dedicated antenna
1544.1 · 1544.5 MHz	Satellite link	406 MHz translation	Cospas-Sarsat downlink	Not listenable
SHF (3 - 20 GHz): nothing listenable with an ordinary receiver				
9.2 - 9.5 GHz	X-band radar	Response: 12 dots on the radar screen	Radar SART	Not listenable

With the UV-K1

VHF and UHF in AM and FM depending on the firmware's range; HF with the SI4732 in AM and SSB. FM can only be heard on the VHF/UHF part. Data tones go by audio cable or microphone to a decoder (sheet 2b).

The three levels of voice call (said three times)

Word	Meaning
MAYDAY	Distress: life or vessel in grave and immediate danger
PAN-PAN	Urgency: serious situation without immediate danger
SÉCURITÉ	Important safety notice (weather, obstacle)

If you hear a real distress call

- Note the time, the position, the vessel's name or MMSI and the nature of the problem.
- Listen to see whether a CROSS or a vessel acknowledges.
- If there is no acknowledgement: call the CROSS on 196 (at sea) or 112, by phone.
- Respect the confidentiality of other exchanges: do not disclose their content.

Reminder. Never transmit on a distress frequency without real need: false alerts are penalised and tie up rescue services.

Distress beacons

EPIRB, PLB, ELT, AIS-SART, radar SART: signals, alert and listening

Beacon families

Type	For whom	Activation	Signal	Frequency
EPIRB	Ships, boats	Automatic (hydrostatic) or manual	Digital burst + homing	406 MHz; homing 121.5 MHz
PLB	People (mountain, hiking, boat)	Manual	Digital burst + homing	406 MHz; 121.5 MHz
ELT	Aircraft	Automatic on impact or manual	Digital burst + homing	406 MHz; 121.5 MHz (older: 243 MHz)
AIS-SART	Life rafts and survival craft	Manual	AIS message "SART ACTIVE" (MMSI 970xxxxxx)	161.975 and 162.025 MHz
AIS MOB / AIS EPIRB	Man overboard, boat beacon	Manual or automatic	AIS message (MMSI 972 / 974xxxxxx)	161.975 and 162.025 MHz
Radar SART	Rafts, boats	Manual	Answers a radar: 12 aligned dots	9.2 - 9.5 GHz (X band)

The 406 MHz signal

Parameter	Value
Band	406.0 - 406.1 MHz (beacon channels around 406.025 - 406.076 MHz)
Modulation	BPSK, phase ±1.1 rad, 400 bit/s; Manchester; no usable FM
Burst	About 0.5 s, repeated every ~50 s (interval slightly random)
Frame	112 or 144 bits, BCH correction: unique identifier, country (MID), beacon type, position if GPS
Position	With GPS: a few tens of metres. Without GPS: Doppler effect, a few km
Final homing	121.5 MHz in AM: descending two-tone sound, continuous, for rescuers with a direction finder

The alert path

Step	Role
1. Beacon	Sends the burst on 406 MHz
2. Cospas-Sarsat satellites	Low orbit (Doppler), geostationary (immediate alert), medium orbit (precise position)
3. Ground station and control centre	Compute the position, identify the owner in the registration database
4. CROSS (in France)	Contacts the owner to rule out a false alert, then launches the rescue

Listening with the UV-K1 and the SDR dongle

- **406 MHz**: detection only. In FM on 406.025 - 406.076 MHz, at best you hear a brief crackle and the S-meter jumps. BPSK can only be decoded with an SDR and a decoder such as dec406. A burst detector (squelch + 50 s timer) fits in a small app (sheet 25).
- **121.5 MHz**: AM mode (AIR); continuous descending two-tone sound.
- **AIS-SART** (161.975 / 162.025 MHz): cannot be decoded with the UV-K1; SDR + AIS decoder (sheet 11).
- Registration is mandatory (in France, with the ANFR); battery of about 5 years; test only with the "self-test" function, never by triggering the beacon.
- Never transmit on 406 MHz. If you pick up a real alert, it is not a test: warn the CROSS (196) or 112.

HF DATA modes

FT8, FT4, PSK31, Olivia, Feld Hell: signal, settings and decoding

· DATA mode

Mode	Modulation	Speed / duration	Width	Coding	Reference (kHz, USB)	Decoders
FT8	8-GFSK, 8 tones spaced 6.25 Hz apart	6.25 bd; 15 s slots (12.6 s of transmission)	50 Hz	77 bits + CRC 14, LDPC (174,91), 79 symbols including 21 sync	14 074 (20 m)	WSJT-X, JTDX; Android: FT8CN
FT4	4-GFSK, 4 tones spaced 20.83 Hz apart	20.83 bd; 7.5 s slots (4.5 s)	~90 Hz	Same coding as FT8, 103 symbols	14 080 (20 m)	WSJT-X, JTDX
PSK31	BPSK 31.25 bd (QPSK31 with FEC)	~50 words/min	~31 Hz	Varicode, no FEC	14 070 (20 m)	fldigi, MultiPSK
Olivia	MFSK 4 to 64 tones with FEC	8/250, 16/500, 32/1000 (tones/Hz)	125 to 2000 Hz	Block FEC, Walsh-Hadamard	Digital segments (e.g. 14 070 - 14 099)	fldigi, MultiPSK
Feld Hell	OOK, ~980 Hz tone	122.5 bd, ~2.5 char./s	~250 Hz	None: text drawn as pixels	Digital segments (e.g. 14 063)	fldigi, MultiPSK

By ear and on screen

Mode	What you hear	Threshold
FT8	Regular chirping, starting at seconds 00 / 15 / 30 / 45; clock within ±1 s (NTP or GPS)	-21 dB (2500 Hz bandwidth)
FT4	Faster than FT8, designed for contests	-17.5 dB
PSK31	Soft purring, very thin trace in the waterfall; variants PSK63, PSK125, PSKxxxR (with FEC)	
Olivia	Burst of gliding tones; very robust (fading, QRM)	~-14 dB
Feld Hell	Fast Morse-like rhythm; the text appears as an image (double line) that you read by eye	

Settings and UV-K1

- USB, moderate volume without clipping, 2.4 to 3 kHz filter. For FT8 and FT4, the PC or phone clock must be synchronised (NTP or GPS).
- UV-K1: HF with the SI4732 in USB (LSB below 10 MHz), audio to a decoder on a PC or smartphone by cable or microphone. None of these modes can be decoded inside the radio.
- Decoding FT8 or FT4 with a home-made FSK decoder is possible in principle, but it needs clock synchronisation and LDPC decoding: much heavier than RTTY.
- Frequencies for all amateur radio bands: sheet 4. Transmitting as a ham requires a callsign.

RTTY

Radioteletype: listening and decoding sheet

· DATA mode

What is it? Text transmission in FSK: two tones alternate, one for bit 1 (mark), the other for bit 0 (space). It is the oldest digital mode still active, used by amateur radio (mostly in contests), for weather and by commercial or military stations.

Parameter	Value
Coding	Baudot (ITA2): 5 bits, two character sets (letters LTRS and figures FIGS)
Frame	1 start bit (space) + 5 bits (least significant first) + 1.5 stop bits (mark): 7.5 bits per character
Speed	45.45 bd (amateur radio, ~60 words/min); 50 bd (weather, Europe); 75 and 100 bd
Shift	170 Hz (amateur radio); 425 / 450 Hz; 850 Hz (commercial, military)
Usual tones	2125 Hz (mark) / 2295 Hz (space) for 170 Hz shift
Width	About 250 to 300 Hz for 170 Hz / 45.45 bd
Sideband	LSB by convention. In USB, mark and space are swapped: enable "Reverse" in the decoder

Where to listen

Frequency	Parameters	Note
3.580 - 3.600 MHz (80 m)	45.45 bd, 170 Hz, LSB	Amateur radio
7.035 - 7.045 MHz (40 m)	45.45 bd, 170 Hz, LSB	Amateur radio
14.080 - 14.100 MHz (20 m)	45.45 bd, 170 Hz, LSB	The most active, crowded during contests
21.080 - 21.100 / 28.080 - 28.100 MHz	45.45 bd, 170 Hz, LSB	15 m and 10 m depending on propagation
4583 · 7646 · 10 100.8 kHz	50 bd, 450 Hz, LSB	DWD weather; some transmissions may have been discontinued
518 · 490 kHz	100 bd, 170 Hz (SITOR-B)	NAVTEX: sheet 12

Baudot ITA2 alphabet (LTRS set)

A	B	C	D	E	F	G	H	I	J	K	L	M
00011	11001	01110	01001	00001	01101	11010	10100	00110	01011	01111	10010	11100
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
01100	11000	10110	10111	01010	00101	10000	00111	11110	10011	11101	10101	10001

Commands: LTRS 11111 · FIGS 11011 · space 00100 · CR 01000 · LF 00010. In the FIGS set, the keys Q W E R T Y U I O P give 1 2 3 4 5 6 7 8 9 0. Classic test: RYRYRY.

Listening and decoding

- By ear: two notes alternating non-stop, about 6 characters per second at 45.45 bd, with no pause between packets (unlike APRS).
- UV-K1 (SI4732): LSB on 14.080 - 14.100 MHz, audio to fldigi (PC), MMTTY or 2Tone (Windows). See sheet 2b.
- Garbled text: try "Reverse", then check the speed (45.45 or 50 bd) and the shift (170 or 450 Hz). Typical CQ: "CQ CQ DE F4ABC F4ABC K".

CCIR Selcall

Selective calling by sequential tones: listening and decoding sheet

· DATA mode

What is it? Selcall calls a specific station with a series of audio tones: one digit = one frequency. The receiver only rings if it recognises its number (4 or 5 digits). The CCIR set is used on HF and VHF: commercial networks, NGOs, remote medical services, fleets. Not to be confused with DTMF (two simultaneous tones): here only one tone at a time, several in a row. The signal goes through the audio channel, in FM (VHF/UHF) or SSB (HF); the frequency depends on the network.

0	1	2	3	4	5	6	7
1981	1124	1197	1275	1358	1446	1540	1640
8	9	A	B	C	D	E (repeat)	
1747	1860	2400	930	2247	991	2110	

CCIR tones in Hz. E marks a repeated digit; A, B, C, D are service or group tones depending on the network. From one tone to the next the spacing is 70 to 120 Hz: a good signal-to-noise ratio is needed.

The signal in brief

Parameter	Value
Tone duration	100 ms (CCIR-1, the most common) or 70 ms
Call length	4 to 6 tones, i.e. ~0.5 to 0.6 s for 5 tones at 100 ms
Repeat	Two identical consecutive digits: the second becomes E (2110 Hz)
Transmission	Audio over FM or SSB, before or after the voice message

Number	Sequence (Hz)	Note
12345	1124 - 1197 - 1275 - 1358 - 1446	Five different digits
11234	1124 - 2110 - 1197 - 1275 - 1358	The second "1" becomes E
55021	1446 - 2110 - 1981 - 1197 - 1124	Repeated 5, then 0, 2, 1

Other set	Tone duration	Reference
CCIR	100 or 70 ms	Digits 1124 to 1981 Hz, repeat 2110 Hz
ZVEI-1	70 ms	Digits 1060 to 2400 Hz, repeat 2600 Hz (Germany, fire brigades, industry)
EEA	40 ms	Same range as CCIR, shorter tones (United Kingdom)
EIA	33 ms	Digits from 600 to 1869 Hz (United States)

To identify the set: measure the duration of a tone and the frequency of the highest tone.

Listening and decoding

- By ear: a burst of 5 short notes of different pitches in a little over half a second, like a short melody, often at the start of an exchange.
- UV-K1: FM (or SSB on HF with the SI4732) on the network frequency, squelch low so as not to miss the call. The radio does not decode selcall.
- Decoding: sequential-tone decoder on a PC or smartphone; Goertzel detection on the 15 frequencies, 100 ms window, is enough. Simple output format: "11234".
- Idea for SigID (sheet 25): recognise 4 to 6 tones of 100 ms among these 15 frequencies and display "CCIR Selcall".

APRS

Automatic Packet Reporting System: listening and decoding sheet

· DATA mode

What is it? Amateur radio network for positions, messages, weather and telemetry, created by Bob Bruninga (WB4APR). Stations send short packets, relayed by digipeaters and iGates to the Internet (APRS-IS, visible on aprs.fi). Packets are AX.25 UI in 1200 baud AFSK (Bell 202) on a VHF FM carrier. Transmitting requires an amateur radio callsign.

Region	Frequency	Note
Europe / France	144.800 MHz	FM, 1200 bd: the main one to monitor
ISS (digipeater)	145.825 MHz	FM, 1200 bd, when active; Doppler ±3.5 kHz
North America	144.390 MHz	FM, 1200 bd
Australia / Japan	145.175 / 144.640 MHz	FM, 1200 bd
HF (30 m)	10.1476 MHz	USB, 300 bd, long distance

Parameter	Value
Modulation	AFSK: 1200 Hz (mark) / 2200 Hz (space), 1200 bd, on FM
Coding	NRZI, bit-stuffing, 0x7E flags, CRC-16 (FCS)
Frame	AX.25 UI: source, destination, path (digipeaters), data
Timing	One beacon every 1 to 10 min, faster when moving (smart beaconing)

A decoded packet

```
F4ABC-9>APRS,WIDE1-1,WIDE2-1:!4852.00N/00220.00E>Test mobile
```

Element	Meaning
F4ABC-9	Source callsign (fictitious), SSID 9 = mobile
APRS	Generic destination (identifies the software, or encodes the position in Mic-E)
WIDE1-1,WIDE2-1	Requested path: up to 2 hops
!	Position without timestamp or messaging
4852.00N / 00220.00E	48°52.00' N, 002°20.00' E
/ ... >	Symbol: primary table + code > = car
Test mobile	Free comment

SSID	Use	Starts with	Type
-1	Digipeater / fixed	! or =	Position
-5	Smartphone	/ or @	Position + time
-7	Handheld	` or '	Mic-E position
-9	Mobile	:	Message
-10	Internet iGate	;	Object
-11	Balloon / aircraft	_	Weather
-15	Generic	>	Status

Listening with the UV-K1

- 144.800 MHz, narrow FM, squelch open or low: short bursts of "screeching" lasting 0.5 to 1 s.
- The radio does not decode APRS: audio to Direwolf, APRSdroid or Xastir by cable or microphone (TNC link: sheet 20b). To see stations received by others: aprs.fi.

KISS TNC

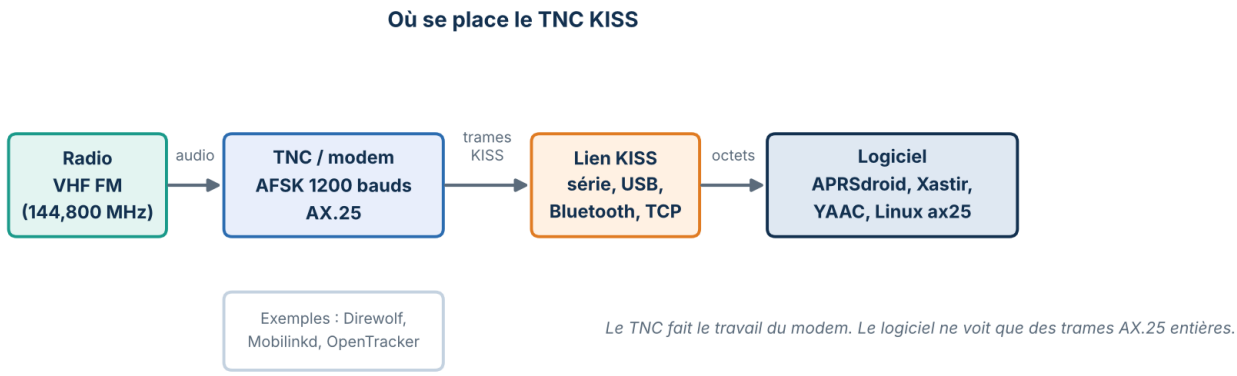
KISS protocol between a TNC and software: frame, commands, connection, APRS

· DATA mode

In brief. KISS (Keep It Simple, Stupid) is a very simple protocol between a TNC (Terminal Node Controller, the packet modem) and a computer or phone. The TNC handles the radio side: 1200 baud AFSK modulation, detection, AX.25 frames. KISS only carries whole frames, byte by byte, over a serial, USB, Bluetooth or TCP link. All the intelligence (APRS, connected packet, BBS) stays in the software. Created by Mike Chepponis and Phil Karn in 1987, it is the most widespread interface between a TNC and APRS software (sheet 20).

0xC0	1200 bd	8001	1987
FEND: start and end of frame	AFSK on VHF, APRS	Direwolf KISS TCP port	year of the specification

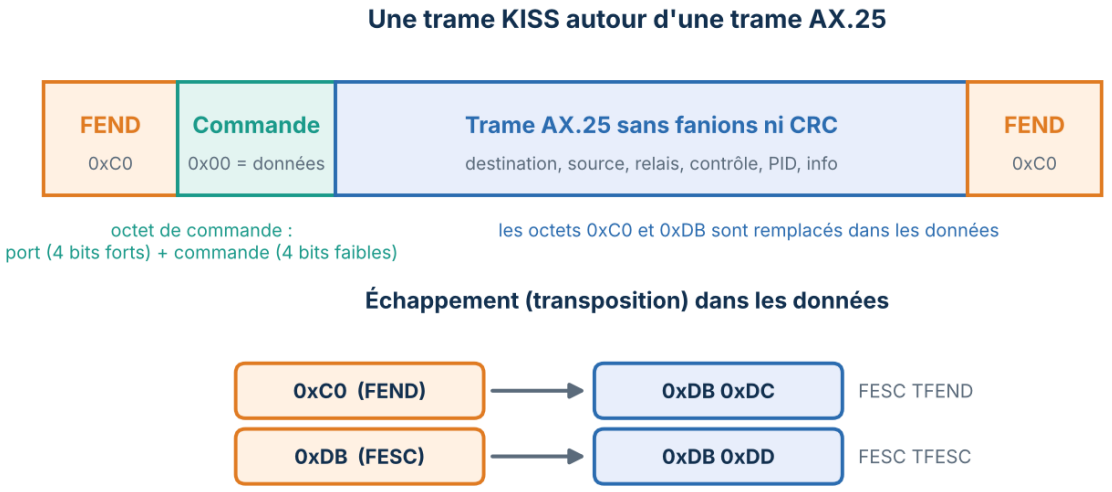
Where the TNC fits



A TNC can be a box (Mobilinkd, OpenTracker, Kantronics), a board (TNC-Pi) or software: Direwolf turns a computer's sound card into a TNC and exposes a KISS interface.

Layer	Role	Handled by
APRS	Meaning of the data: position, message, weather	Software (APRSdroid, Xastir, YAAC)
AX.25	Addresses, digipeaters, frame type, CRC	TNC (the software reads the addresses in the frame)
KISS	Transport of frames between TNC and software	Both ends of the link
AFSK 1200	1200 / 2200 Hz tones on FM, channel access	TNC and radio

The KISS frame



Sans cet échappement, un octet 0xC0 dans les données serait pris pour la fin de la trame.

- FEND (0xC0) opens and closes each frame. Two consecutive FENDs are ignored: you can send an extra one to resynchronise.
- The command byte carries the port number (high 4 bits, for a TNC with several radios) and the command (low 4 bits).
- The data is the AX.25 frame without flags or CRC: the TNC adds them when transmitting and removes them when receiving.

- In the data, 0xC0 becomes 0xDB 0xDC and 0xDB becomes 0xDB 0xDD (escaping).

The commands

Code	Name	Role	Usual value
0x00	Data	AX.25 frame to transmit, or received	-
0x01	TXDELAY	Delay between transmitter key-up and the first bit, in units of 10 ms	50 (500 ms)
0x02	P	Persistence: probability of transmitting when the channel is free, (P + 1) / 256	63 (25 %)
0x03	SlotTime	Length of a wait slot, in units of 10 ms	10 (100 ms)
0x04	TXtail	Delay after the end of the frame (obsolete)	0
0x05	FullDuplex	0 = half duplex, any other value = full duplex	0
0x06	SetHardware	Hardware-specific setting of the TNC	model-dependent
0xFF	Return	Leaves KISS mode (depending on the TNC)	-

Default values from the specification; Direwolf sets these parameters in its configuration file.

A decoded example

```
C0 00 82 A0 A4 A6 40 40 60 8C 68 82 84 86 40 72 AE 92 88 8A 62 40 62
AE 92 88 8A 64 40 63 03 F0 21 34 38 35 32 2E 30 30 4E 2F 30 30 32 32
30 2E 30 30 45 3E 54 65 73 74 C0
```

Bytes	Content
C0 00	FEND, command 0: data frame on port 0
82 A0 A4 A6 40 40 60	Destination APRS, SSID 0: each letter is shifted one bit to the left (0x41 << 1 = 0x82)
8C 68 82 84 86 40 72	Source F4ABC-9 (fictitious): SSID 9 is in byte 72
AE 92 88 8A 62 40 62	Digipeater WIDE1-1
AE 92 88 8A 64 40 63	Digipeater WIDE2-1: the least significant bit set to 1 marks the end of the addresses
03 F0	UI frame (03), protocol "no layer 3" (F0)
21 34 38 … 54 65 73 74	ASCII text: !4852.00N/00220.00E>Test, APRS position without timestamp
C0	FEND: end of frame

Illustrative example, made-up callsign, worked out by hand. No byte equals 0xC0 or 0xDB: nothing to escape.

Connecting and using

Link	Detail
Serial or USB	Often 9600 baud, 8N1; some TNCs accept 115200. Same speed on both sides
Bluetooth	Mobilinkd and others: serial profile. On Android, APRSdroid picks the TNC from the paired list
TCP	Direwolf listens on port 8001 by default; the software connects to the computer's address
Linux	kattach or kissattach (ax25-tools package) creates a network interface ax0 from the KISS link

Item	Use
Direwolf	Software TNC: sound card to KISS (TCP, pseudo-terminal, serial), APRS iGate and digipeater
APRSdroid	Android: KISS TNC over Bluetooth, USB or TCP
Xastir, YAAC	Computer: APRS maps, via the KISS port
Mobilinkd TNC	Bluetooth box, on the audio output of a handheld radio
OpenTracker, TNC-Pi	Boards and boxes, with or without GPS
Linux ax25	Kernel AX.25 stacks: connected packet, BBS, network applications

With the UV-K1

- The UV-K1 has no built-in TNC (apart from Labs applications). For APRS, connect its audio output to a TNC: a box, or Direwolf on a computer (sheet 2b).
- Setting: 144.800 MHz in narrow FM, squelch low, medium volume. Too high a volume overloads the TNC input.
- To transmit, the TNC must control the PTT: a suitable cable or VOX, and an amateur radio callsign. For listening only, a phone with APRSdroid and a Mobilinkd on the headphone output is enough.

KISS TNC

KISS protocol between a TNC and software: frame, commands, connection, APRS

· DATA mode

Troubleshooting

Symptom	Things to check
No frames received	Audio level, 144.800 MHz frequency, squelch, same serial speed
Garbled characters	Wrong serial speed, or TNC not in KISS mode (text command mode)
Many frames rejected	Audio level too high or too low, power supply noise, frequency offset
Transmission has no effect	PTT wired wrongly, TXDELAY too short, channel busy (P and SlotTime)
Port 8001 refused	Direwolf not running, or firewall between the computer and the phone

Good to know. Default values, ports and speeds are from memory, based on the KISS specification: confirm them with the documentation of your TNC or Direwolf. APRS positions are public, but do not share those of people who did not want it.

SSTV

Slow-Scan Television: listening and decoding sheet

· DATA mode

What is it? Still images sent as sound, line by line, in a few tens of seconds to a few minutes. The brightness of a pixel is an audio frequency: 1500 Hz = black, 2300 Hz = white; line syncs at 1200 Hz (same principle as fax, sheet 22). The signal fits in 3 kHz: SSB on HF, FM on VHF. Transmitting requires an amateur radio callsign.

Frequency	Mode	Note
145.800 MHz	FM	ISS (ARISS) during SSTV events; ~10 min pass, Doppler ±3.5 kHz
14.230 MHz	USB	International calling frequency (20 m), the most active
28.680 / 21.340 MHz	USB	10 m and 15 m depending on propagation
7.171 MHz	LSB	40 m, mainly Europe
3.730 MHz	LSB	80 m, regional traffic

Common modes

Mode	Duration	Resolution	VIS	Note
Robot 36	36 s	320 x 240	8	Fast, very common on VHF
Robot 72	72 s	320 x 240	12	
Martin M1 / M2	114 s / 58 s	320 x 256	44 / 40	Very common in Europe
Scottie S1 / S2	110 s / 71 s	320 x 256	60 / 56	Very common in the Americas
Scottie DX	269 s	320 x 256	76	Long distance, slow
PD50 / PD90	50 s / 90 s	320 x 256	93 / 99	
PD120	126 s	640 x 496	95	Usual ISS mode
PD180	187 s	640 x 496	96	High quality

Start of a transmission (VIS header)

Element	Frequency	Duration
Calling tone (leader)	1900 Hz	300 ms
Break	1200 Hz	10 ms
Calling tone	1900 Hz	300 ms
Start bit	1200 Hz	30 ms
7 VIS code bits (least significant first)	1100 Hz = 1 / 1300 Hz = 0	30 ms each
Parity (even) then stop bit	1100 or 1300 Hz, then 1200 Hz	30 ms each

Listening and decoding

- ISS: UV-K1 on 145.800 MHz, narrow FM, moderate volume; correct for Doppler in steps of 1 to 2 kHz during the pass.
- HF: USB above 10 MHz, LSB below; a frequency drift slants the image, so tune finely.
- The radio does not decode the image: audio to Robot36 (Android), Black Cat SSTV (iOS), MMSSTV (Windows) or QSSTV (Linux), in a quiet room.

Fault	Probable cause
Slanted image	Frequency drift or inaccurate audio clock
Shifted colours, torn image	Wrong mode, or sync lost (weak signal)
Streaks, noise	Weak signal, QSB, interference or volume too high
Image cut off at the top	Start missed (VIS header not received)

Weather fax

HF radiofax: weather charts by radio

· DATA mode

What is it? Official weather stations broadcast charts on HF (analyses, forecasts, waves, ice, satellite images), widely used by offshore sailors. Same principle as SSTV (1500 Hz black, 2300 Hz white), but much slower. Reception is free and unrestricted.

Parameter	Value
Modulation	FM of an audio subcarrier, transmitted in USB
Audio frequencies	Centre 1900 Hz; black 1500 Hz; white 2300 Hz (±400 Hz)
Speed	120 lines/min (2 per second), the most common; 60, 90 and 240 also exist
IOC	576 (standard, ~1810 dots per line) or 288
Start	~5 s tone (300 Hz for IOC 576, 675 Hz for IOC 288), then ~30 s phasing
End	450 Hz tone for ~5 s, then a black line
Tuning	In USB, 1.9 kHz below the published frequency (assigned carrier)

Station	Frequencies (carriers, kHz)	Note
DWD (Germany)	3855 · 7880 · 13 882.5	Europe and North Atlantic: the easiest from France
Northwood (United Kingdom)	2618.5 · 4610 · 8040 · 11 086.5	North Atlantic
Boston NMF (United States)	4235 · 6340.5 · 9110 · 12 750	North-West Atlantic
Point Reyes NMC (United States)	4346 · 8682 · 12 730 · 17 151.2 · 22 527	Pacific

Frequencies and schedules are from memory and may change or be discontinued: check the official schedules (DWD, NOAA, UK Met Office). Lower frequencies mostly at night, higher ones by day.

What you receive

- Surface analyses (isobars, fronts), forecasts at 24, 48, 72 and 96 h.
- Wind, wave and swell charts; sea temperature; ice; satellite images; broadcast schedule.
- APT from satellites at 137 MHz is a cousin of fax (2400 Hz subcarrier in AM, 2 lines/s): sheet 24.

Listening and decoding

- By ear: a low, regular "tick-tick", 2 lines per second, with tone variations depending on light and dark areas.
- UV-K1 (SI4732): USB on the published frequency minus 1.9 kHz; audio to fldigi (WEFAX 576 and 288), MultiPSK, Black Cat HF Weather Fax (iOS) or HF Weather Fax (Android).

Fault	Probable cause
Slanted image	Wrong speed (60/90/120/240) or inaccurate audio clock
Shifted image	Phasing missed: realign or wait for the next chart
Negative image	Wrong sideband (LSB instead of USB) or "invert" option
Noise, streaks	Weak signal, fading: change frequency (day / night)

Images in the waterfall

Drawing an image in the spectrum: principle, calculations, reception

· DATA mode

Principle. You build a sound whose spectrogram draws an image. Each line of the image is a piece of sound of duration T; each column is an audio frequency; the pixel brightness becomes the amplitude of the corresponding sine wave. In a waterfall (SDR, WSJT-X, fldigi), the image appears. There is no decoding: no coding, no error correction, no fixed frequency; the image is read by eye, within the SSB audio band (about 300 to 2700 Hz). Feld Hell (sheet 17) already does this for text.

Calculation	Formula	Example
Column width	df = B / N (B audio bandwidth, N columns)	B = 2400 Hz, N = 100: df = 24 Hz
Minimum line duration	T ≥ 1 / df; take 1.5 to 2 times more for a sharp image	df = 24 Hz: T ≥ 42 ms, i.e. ~80 ms
Image duration	Timage = number of lines x T	60 lines x 80 ms = 4.8 s
Column frequency	fk = f0 + k x df	f0 = 300 Hz, k = 50: 1500 Hz
Signal of one line	s(t) = sum Ak sin(2π fk t + φk), Ak = brightness	Random phases φk to avoid a power peak
Trade-off	More columns: more detail but longer lines	200 columns at 12 Hz: T ≥ 83 ms

Reception and good practice

- Reception: USB, moderate volume, waterfall set to 0 to 3 kHz. In LSB the image is reversed (mirrored). Adjust the FFT size: too fine, it stretches the image; too coarse, it blurs it.
- Test without transmitting: generate the sound on your computer and view it in Spectrum Lab or Audacity (spectrogram). No radio needed.
- Transmitting: only if your callsign and the band allow it. The signal is wide: stay clear of active frequencies, FT8, distress frequencies and beacons.

Satcom

Weather, amateur radio, Inmarsat, Iridium and military UHF: frequencies, formats and equipment

· DATA mode

Service	Frequency	Format	Content	Equipment / decoding
Civil and amateur radio satellites (downlink)				
NOAA APT	137.100 · 137.620 · 137.9125 MHz	FM, 2400 Hz subcarrier (AM), 2 lines/s	Polar weather image (NOAA 15, 18, 19 decommissioned or near end of life: check)	52 cm V-dipole; RTL-SDR; satdump, WXtoImg
Meteor-M2 LRPT	137.1 · 137.9 MHz	QPSK 72 kbit/s	High-resolution digital images	SDR, SatDump
Orbcomm	137 - 138 MHz	SDPSK 4800 bit/s	Fleet data	SDR, dedicated decoder
GOES HRIT / EMWIN	1694.1 MHz	DVB-S2 (QPSK)	Images, charts, notices (Americas)	Dish, SDR, SatDump; not visible from Europe
Meteosat (MSG)	1695.15 MHz (HRIT)	BPSK + FEC	Geostationary images	Dish + SDR; also EUMETCast (Ku)
FM amateur radio satellites	Downlink 435 - 438 MHz; uplink 145 MHz	FM, sometimes CTCSS 67 Hz	Voice (SO-50, AO-91...), beacons	UV-K1 + handheld Yagi; Doppler ±10 kHz
QO-100 (Es'hail-2)				
Commercial and military satcom (SDR)				
Inmarsat-C (STD-C)	1537 - 1545 MHz	BPSK 600 bit/s, TDM	SafetyNET, EGC: maritime notices, distress	1.5 GHz patch, SDR, STD-C decoder
Inmarsat AERO	1545 - 1555 MHz	BPSK / QPSK 600 to 1200 bit/s	ACARS by satellite	Patch, SDR; partly encrypted
Iridium	1616 - 1626.5 MHz	QPSK 25 kbaud, TDMA	Voice, SMS, ACARS, signalling	SDR, gr-iridium, iridium-toolkit
Military UHF	243 - 270 MHz	AM, FM, 25 kHz DAMA, encryption	Military voice and data	SDR + 250 MHz Yagi; partial reception

Reading and limits

- UV-K1: FM amateur radio satellites (435 - 438 MHz, ISS voice on 145.800 MHz) with a suitable antenna. 137 MHz needs a wider bandwidth than narrow FM: use an SDR. Inmarsat, Iridium, GOES and QO-100 are out of reach of the radio.
- 406 MHz beacons: sheet 16. NOAA images: close to weather fax (sheet 22) through the 2400 Hz subcarrier.
- Listening is free; do not transmit without a callsign; commercial and military satcom are restricted; do not disclose the content of private communications.

SigID

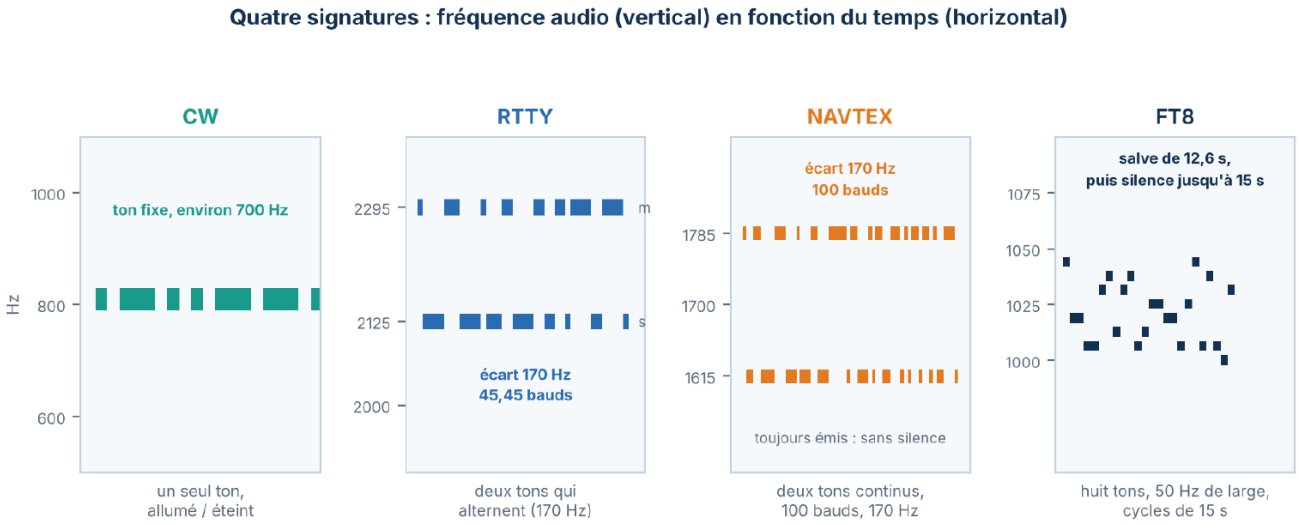
Identifying a radio signal from its tones (UV-K1)

· DATA mode

In brief. SigID is a small UV-K1 application that identifies the type of signal received (CW, RTTY, NAVTEX, FT8) and displays it. It does not decode the content: once the mode is known, decoding is done on a smartphone or PC. It is built on the SelCall CCIR application (Labs F4HWN v6.0.0 overlay), which measures the energy of a few tones with the Goertzel algorithm on audio sampled at 8 kHz. Main constraint: 4 KB maximum once compiled.

4 KB	8 kHz	4 modes	Goertzel
maximum size	audio sampling	CW, RTTY, NAVTEX, FT8	analysis of a few tones

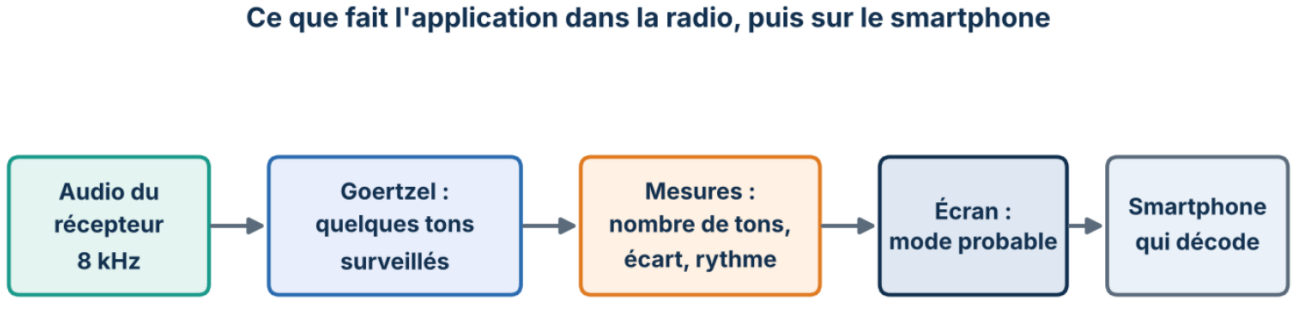
Four signatures to recognise



Illustrative diagram: each mode leaves a different fingerprint in the audio time / frequency plane.

Mode	Tones	Baud rate and shift	Timing cue	Where to listen
CW	1	Tone on / off, variable speed	A dash lasts about three dots	Amateur radio bands
RTTY	2	45.45 Bd, 170 Hz shift (amateur); 50 Bd, 425 or 450 Hz (commercial, weather)	Continuous, 5-bit Baudot	Amateur radio bands, HF utility (sheet 18)
NAVTEX	2	100 Bd, 170 Hz shift, CCIR 476 (SITOR-B)	10-minute slot per station, letter A at 00:00 UTC, every 4 h; phasing of at least 10 s, then ZCZC	518 kHz, 490 kHz, 4209.5 kHz (sheet 12)
FT8	8	~6.25 Bd, tones spaced ~6.25 Hz apart	12.64 s burst every 15 s (00, 15, 30, 45)	7.074, 14.074, 21.074, 28.074 MHz (sheet 17)

The processing chain in the radio



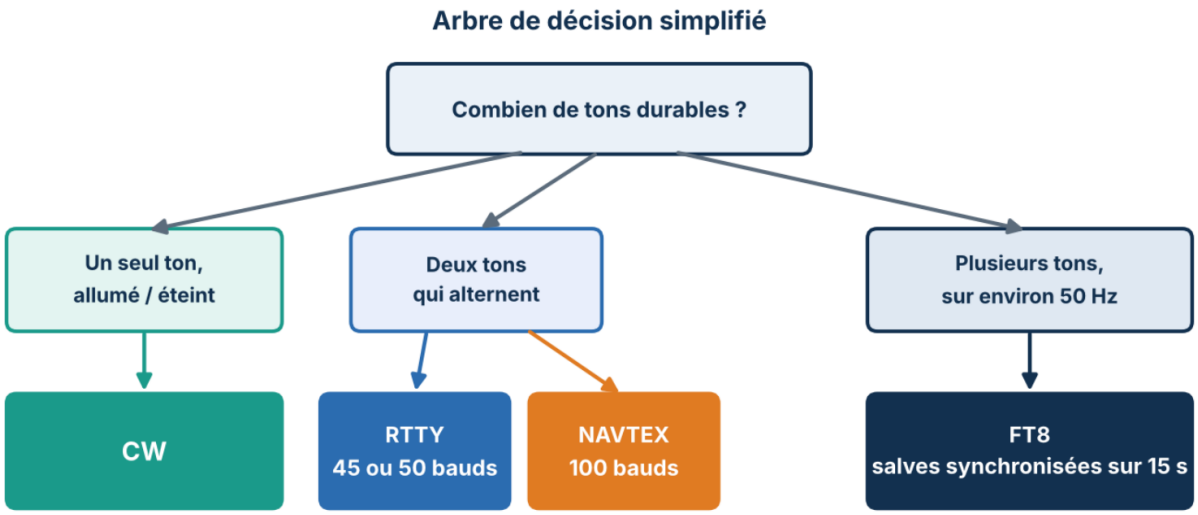
Le décodage complet (texte NAVTEX, messages FT8) reste sur le smartphone : la radio se contente d'identifier le signal.

Goertzel: measuring a tone cheaply

A full FFT needs too much memory for 4 KB. The Goertzel algorithm computes the energy at a single frequency with two state variables and one multiplication per sample; a few are run in parallel, one per monitored tone.

Concept	Key point
Channel width	fs / N. At 8 kHz with N = 160: 50 Hz per channel, one measurement every 20 ms
Trade-off	The larger N is, the more precise but slower it is. NAVTEX (10 ms bit) requires a short N, FT8 (6.25 Hz tones) a long N
Cost	One coefficient 2cos(2πk/N) per tone, then s = x + c·s1 - s2 for each sample: possible in integer arithmetic
Decision	Compare the tone energies with each other and with the noise, using an adaptive threshold

Decision tree



The number of tones sorts first, then the rhythm separates RTTY from NAVTEX, and the 15 s synchronisation confirms FT8.

Limits and difficulties

Difficulty	Approach
RTTY and NAVTEX look alike	Same 170 Hz shift: only the rhythm (45.45 or 50 Bd versus 100 Bd) tells them apart; measure over several seconds
FT8 is very narrow	Tones spaced ~6.25 Hz apart: long N, slow analysis; rely on the 15 s structure
Weak signal	Require several consistent measurements, display "unknown" when in doubt
Tuning offset	Measure the spacing between tones rather than their absolute value
Receiver	NAVTEX at 518 kHz and FT8 on HF require a medium- and shortwave receiver in a suitable mode (SI4732)
Size	Few monitored tones, no large tables, integer arithmetic

Identified mode	Decoding on smartphone or PC
CW	Morse decoder working from the audio
RTTY	RTTY decoder (shift and baud rate according to the identification): sheet 18
NAVTEX	NAVTEX FEC decoder, 100 Bd, text between ZCZC and NNNN: sheet 12
FT8	FT8 application with a well-synchronised clock: sheet 17

Going further. Add other modes (SelCall sheet 19, DTMF, SSTV, AIS, ATIS), display the level, duration and measured baud rate, and test on known references (NAVTEX 518 kHz, beacons). This sheet gives the principle, not the code: the values of N, the thresholds and the keys must be adapted to your version.

Sig Finder

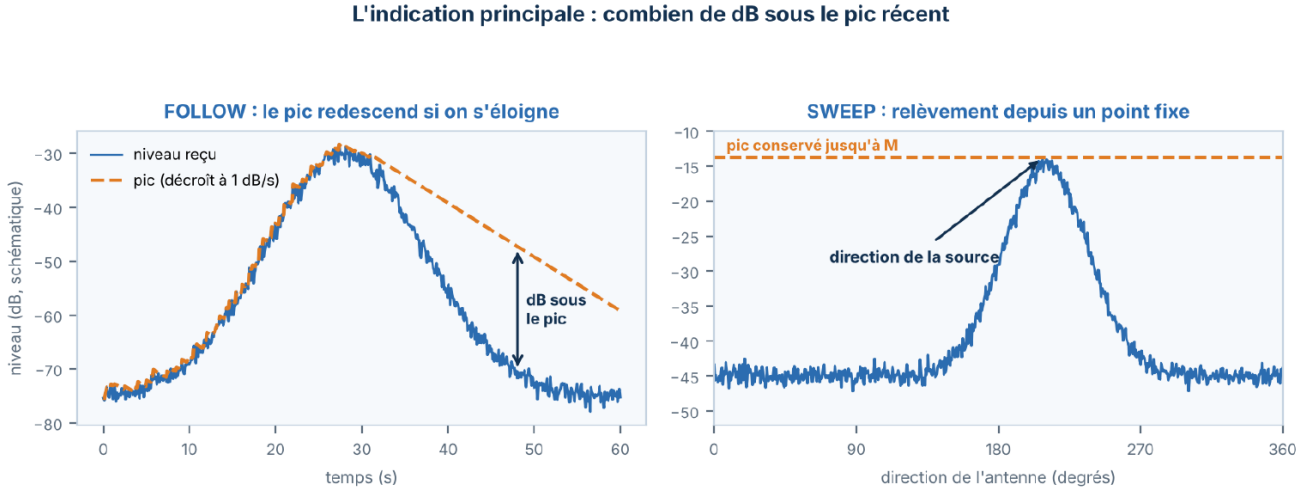
Direction finding on UV-K5 / UV-K1 (Labs edition)

· DATA mode

In brief. Sig Finder is a Labs edition application that helps find the direction of a carrier: a fox in a fox hunt, a jammer, or the 121.5 / 243 MHz homing signal of a distress beacon. Its main indication is not the absolute level but the number of dB below the strongest recent level. It also recognises the downward-swept tone of a homing signal and displays ELT. Receive only.

dB below peak	2 modes	121.5 / 243	F + 7
main indication	FOLLOW and SWEEP	MHz, homing signal	launch

Principle: getting closer by comparison



In FOLLOW, the stored peak falls back at a chosen rate: if the gap in dB grows, you are moving away. In SWEEP, the peak stays frozen until you press M.

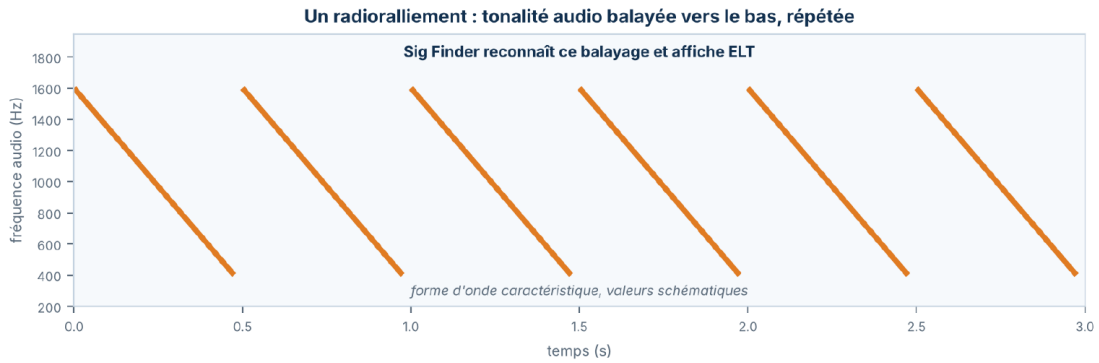
Setting up and launching

Step	Detail
Install	With UV Studio, the application version matching the firmware (sheet 2b)
Set the VFO	AM for 121.5 / 243 MHz, otherwise the appropriate modulation
Launch	F + 7
Navigation	Depends on SetNav: UP / DOWN on UV-K5, LEFT / RIGHT on UV-K1 (UP/LEFT and DOWN/RIGHT mean the equivalent keys)

FOLLOW and SWEEP

Mode	Behaviour	Use
FOLLOW	The peak decays at an adjustable rate: 0.5, 1, 2 or 4 dB/s	You move towards the source
SWEEP	The peak is kept; M resets it to zero for a new bearing	Bearing from a fixed point

Automatic attenuator. Close to the source, the signal can overload the receiver. The attenuator adds attenuation before overload and removes it when the signal weakens; it can also be set by hand (key 3 or UP/LEFT and DOWN/RIGHT).
Recognising a homing signal: ELT. Sig Finder detects the tone that sweeps downward and displays ELT. The audio-off and listen modes detect it; the proximity beeps mode does not, because it cuts the sound between beeps.



Sig Finder

Direction finding on UV-K5 / UV-K1 (Labs edition)

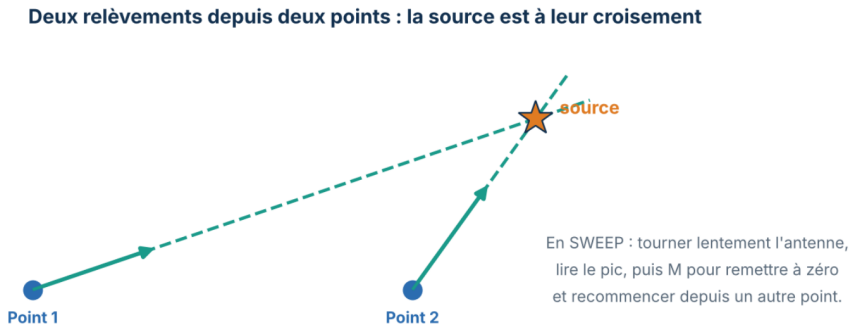
· DATA mode

The keys

Key	Action
1	Toggles FOLLOW / SWEEP and resets the peak to zero
2 or F then 2	Next or previous audio mode: off, proximity beeps, normal listening, USB listening
3, UP/LEFT, DOWN/RIGHT	Changes the attenuation step and switches to manual; F then 3: previous step
4	Turns automatic attenuation on or off
5 or F then 5	Peak decay rate in FOLLOW (0.5, 1, 2, 4 dB/s)
6	Backlight held on, or normal timeout
M	Resets the peak to zero; in SWEEP, starts a new sweep
Hold F	Locks or unlocks the application keypad
EXIT	Exits the application (keypad unlocked)

Saved settings: mode, audio, attenuation, auto attenuation, decay rate, backlight.

Field tips



- SWEEP to take a bearing, then FOLLOW to move towards the source.
- Reduce the attenuation by hand if the signal is weak; leave it on automatic near the source.
- Take several bearings from different points: where they cross gives the position.
- Lock the keypad (hold F) to avoid accidental key presses while walking.

Warning. Never transmit a test homing signal on 121.5, 243 or 406 MHz: these are monitored distress frequencies. Sig Finder is a receive-only tool.

Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for the radio amateur and the listener

Alphabet, Morse and spelling

	Morse	ICAO		Morse	ICAO
A	•—	Alfa	N	—•	November
B	—•••	Bravo	O	———	Oscar
C	—•—•	Charlie	P	•——•	Papa
D	—••	Delta	Q	——•—	Quebec
E	•	Echo	R	•—•	Romeo
F	••—•	Foxtrot	S	•••	Sierra
G	——•	Golf	T	—	Tango
H	••••	Hotel	U	••—	Uniform
I	••	India	V	•••—	Victor
J	•———	Juliett	W	•——	Whiskey
K	—•—	Kilo	X	—••—	X-ray
L	•—••	Lima	Y	—•——	Yankee
M	——	Mike	Z	——••	Zulu

0	1	2	3	4	5	6	7	8	9
—————	•————	••———	•••——	••••—	•••••	—••••	——•••	———••	————•

International digits: Zero, Wun, Too, Tree, Fower, Fife, Six, Seven, Ait, Niner.

Punctuation and procedural signals

Sign	Morse	Signal	Meaning	Morse
.	•—•—•—	AR	End of message	•—•—•
,	——••——	SK	End of contact	•••—•—
?	••——••	BT	Pause (=)	—•••—
/	—••—•	KN	Over to you only	—•——•
=	—•••—	AS	Wait	•—•••
+	•—•—•	CT	Start of message	—•—•—
-	—••••—	SOS	Distress	•••———•••
( )	—•——• / —•——•—	Error	Correction	••••••••
: " ' @	———••• / •—••—• / •————• / •——•—•			

Timing: dot = 1 unit, dash = 3, gap between elements = 1, between letters = 3, between words = 7. Dot duration (ms) = 1200 / speed in words per minute: 60 ms at 20 wpm.

RST reports and CW abbreviations

- RST: R = readability (1 to 5), S = strength (1 to 9), T = tone (1 to 9, CW only). 599 = perfect in CW; 59 in phone.
- Typical call: CQ CQ CQ DE F4ABC F4ABC K; reply: F4ABC DE DL1XYZ K.

Abbr.	Meaning	Abbr.	Meaning	Abbr.	Meaning
CQ	General call	73	Best regards	UR	Your
DE	From ("this is")	88	Love and kisses	FB	Fine business (excellent)
K	Over to you	GM / GA / GE	Good morning / afternoon / evening	DX	Long distance
KN	Over to you only	YL / XYL	Young lady operator / wife	ES	And
AR	End of message	RIG / ANT	Radio / antenna	BK	Break (back to you)
SK	End of contact	PWR	Power	R	Received
PSE	Please	WX	Weather	OM	Old man (friend)
TNX / TKS / TU	Thanks	AGN	Again, repeat		

Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for the radio amateur and the listener

Q code (the most useful)

Code	Meaning	Code	Meaning
QRA	Name of my station	QRU	I have nothing for you
QRG	Exact frequency	QRV	I am ready
QRH	My frequency varies	QRX	Stand by, I will call again
QRI	Quality of the tone	QRZ	Who is calling me?
QRK	Readability of signals	QSA	Strength of signals (1 to 5)
QRL	Frequency busy	QSB	Fading
QRM	Interference from other stations	QSL	Received; QSL card
QRN	Atmospheric noise (static)	QSO	Radio contact
QRO	Increase power	QSP	Relay a message
QRP	Reduce power / low power	QSY	Change frequency
QRQ	Send faster	QTC	Messages to send
QRS	Send more slowly	QTH	Location
QRT	Stop transmitting, end of traffic	QTR	Exact time

A Q code followed by '?' is a question (QRZ? = who is calling me?). QNH, QFE, QFU, QDM and QDR are aeronautical Q codes (sheet 7).

CB code (10-codes and slang)

On CB (27 MHz, channels in sheet 6), the "10-codes" that came from the United States are used, often mixed with slang. Usage varies by region and group: they are guidelines, not a standard.

Code	Meaning	Code	Meaning
10-1	Poor reception	10-20	Location (your location?)
10-2	Good reception	10-21	Call me by phone
10-3	Stop transmitting	10-22	Meet me in person
10-4	Received, OK	10-23	Stand by
10-5	Relay the message	10-24	Assignment completed
10-6	Busy, wait	10-26	Disregard last information
10-7	Out of service, leaving the air	10-27	Moving to channel ...
10-8	In service, listening	10-28	Give your identification
10-9	Repeat	10-32	Reception test (radio check)
10-10	End of transmission, still listening	10-33	Emergency traffic
10-13	Road or weather conditions	10-36	Exact time
10-16	Pick up	10-42	Traffic accident
10-18	Anything for us?	10-62	Unable to copy, call me by phone
10-19	Nothing for you, return to base	10-70	Fire

Slang term	Meaning
Break (break 19)	I want to speak (on channel 19, for example)
Over / out	Over: end of message, it is your turn; out: end of contact
Copy, roger	Received
Affirmative, negative	Yes, no
Radio check	Request for a reception test
Mobile, base	Station in a vehicle, fixed station

Values from memory, to be verified: several codes differ from one group to another. Channel 9 is the emergency channel by convention (sheet 15). CW punctuation and abbreviations (73, 88, QSL...) are given above.

Amateur radio callsigns

Structure: prefix (country) + digit + suffix. F4ABC = F (France), 4, ABC. Suffixes: /P portable, /M mobile, /MM ship at sea, /AM aircraft.

Prefix	Territory	Prefix	Territory
F	Metropolitan France	TK	Corsica

Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for the radio amateur and the listener

Prefix	Territory	Prefix	Territory
FM	Martinique	FG	Guadeloupe
FS	Saint Martin	FY	French Guiana
FR	Réunion	FH	Mayotte
FP	Saint Pierre and Miquelon	FK	New Caledonia
FO	French Polynesia	FW	Wallis and Futuna
FT	French Southern Territories (TAAF)	TM / TO	French special callsigns

Country	Pref.	Country	Pref.	Country	Pref.
Germany	DL	Czech Rep.	OK	South Africa	ZS
United Kingdom	G/M	Hungary	HA	United States	K/W/N
Spain	EA	Romania	YO	Canada	VE
Italy	I	Croatia	9A	Mexico	XE
Belgium	ON	Slovenia	S5	Brazil	PY
Netherlands	PA	Bulgaria	LZ	Argentina	LU
Switzerland	HB9	Russia	UA	Chile	CE
Luxembourg	LX	Ukraine	UR	Japan	JA
Portugal	CT	Turkey	TA	China	BY
Ireland	EI	Monaco	3A	India	VU
Austria	OE	Andorra	C3	South Korea	HL
Poland	SP	Morocco	CN	Thailand	HS
Sweden	SM	Algeria	7X	Indonesia	YB
Norway	LA	Tunisia	3V	Israel	4X
Denmark	OZ	Senegal	6W	Australia	VK
Finland	OH	Ivory Coast	TU	New Zealand	ZL
Greece	SV	Madagascar	5R		

UV-K1: keys and overlay applications

Key functions by firmware, then the list and keys of the overlay applications (games excluded)

UV-K1 keys by firmware

How to read. Long = long press; **F +** = F key (or F #) then the key. Fusion is the base: Labs is an experimental edition of it, GOGUFW and NR7Y CW derive from it, and Falcon follows its logic with its own additions. The actions of the side keys SK1 and SK2 and of "M Long" can be programmed in the menu (F1Shrt, F1Long, F2Shrt, F2Long, M Long): the values below are those given in the documentation. An UP / DOWN arrow is called LEFT / RIGHT on the UV-K1 depending on the SetNav setting (hidden menu 76 / 77).

Keypad

Key	Fusion (F4HWN): long press, then F +	Falcon 2.3.0: long press, then F +
1 BAND	Long: same as F +. F +: changes band 1 to 7; in memory mode, copies the channel to the VFO	Long: copies the displayed channel to the VFO at the same frequency. F +: moves the channel to the next list
2 A/B	F +: switches the main VFO (top / bottom)	Long: switches VFO A/B in DUAL. F + SK1 / SK2: changes the frequency step
3 VFO/MR	F +: toggles frequency / memory	Long: toggles frequency / memory
4 FC	F +: copies frequency and CTCSS; save with M	Long: changes the frequency step. F +: adds or removes the frequency from the watch list (50 max.)
5 NOAA	Long: in memory mode, cycles the scan list (OFF, 1 to 24, ALL); in frequency mode, scan range. F +: spectrum analyser	Long: wide / narrow bandwidth. F +: applications menu (OA version)
6 H/M/L	F +: channel power	Long: power. F +: Sentinel (LIST / BAND / RANGE)
7 VOX	Long: VOX on/off. F +: Labs: application launcher	Long: Falcon quiet mode. F +: LIST spectrum
8 R	Long: reverse (TX on the RX frequency). F +: manual backlight on/off	Long: squelch 1 to 9. F +: BAND spectrum
9 Call	Long: call channel. F +: ends manual backlight	Long: backlight. F +: RANGE spectrum
0 FM	F +: FM broadcast radio	Long: modulation FM, AM, USB. F +: FM broadcast radio
*** SCAN**	Short: DTMF entry. Long: channel or frequency scan. F +: DCS / CTCSS scan	Long: scan of the list (or of the range between two free VFOs). F +: fast scan of all memories

Control keys

Key	Fusion	Falcon (additional)
MENU (M)	Short: menu. Long: "M Long" action; during a scan, excludes the channel. During a scan, keeps the last channel found	Long: torch or configured shortcut. F + MENU: new channel, or EDIT / NEW / DEL
EXIT	Short: exits, deletes a digit. Long: clears the entry, exits DTMF, monitor and ScnRng	Long: new frequency entry
UP / DOWN	Menus and lists. F +: squelch	F + left / right: squelch threshold in dBm
F #	Short: activates F +. Long: locks the keypad (SetLck selects what is locked)	Same
PTT	Transmits (SetPTT mode: CLASSIC or ONEPUSH). PTT + SK2: 1750 Hz tone. PTT + key: DTMF	SK1 / SK2: PTT 2 (transmit on the unselected VFO)
SK1 / SK2	F1Shrt, F1Long, F2Shrt, F2Long programmable. F + SK1 / SK2: changes the frequency step. F + long press: selects the action	Same + RXMOD, ScanLst Up, ScanLst Dw

SetLck: KEYS locks the keypad; KEYS + ACTIONS also locks the side shortcuts and M Long; KEYS + PTT locks transmission; KEYS + ACTIONS + PTT locks everything.

In the spectrum and Sentinel (Falcon)

Key	Action
1 / SK2	Skips the current reception / blacklists the frequency
2 · 3	Backlight · normal / fast speed
5 · 6 · 7	Frequency details · LIST / BAND / RANGE · saves the settings
9 · 0	History (200 detections) · session modulation
*** / F**	Relative squelch up or down
SK1	NORMAL, FREQ LOCK, MONITOR
MENU · long MENU	List or band selections · settings page

UV-K1: keys and overlay applications

Key functions by firmware, then the list and keys of the overlay applications (games excluded)

Sonic V71: different keys

Main screen (L = long press)	Action
L0 · L1 · L2	MR: copies the channel to the VFO · stronger squelch · bandwidth
L3 · L4 · L5	Modulation AM / FM / USB · power L / M / H · frequency step
L6 · L7	Toggles MR / VFO · MR: channel list number
L8 · L9	Torch indication on reception · constant backlight
F2	MR: deletes the channel; VFO: keeps the frequency
F6 · F7 · F8	Spectrum by range (RANGE) · by lists · by bands
F0	FM broadcast radio

In the spectrum / ULTRA WATCH	Action
1 · 2 · 3	Skips the frequency · backlight · fast or slow scan
4 · 5 · 6	Scanned lists or bands · options menu (values with 1 and 3) · switches between lists / bands / range mode
7 · 8 · 9	Saves the settings · raw or smoothed spectrum and ULTRA WATCH · modulation
0	History (200 entries)
*** · #**	DYNSQL threshold up or down; long press: DYNSQL off
SK1 · SK2	Locks the frequency (FL), then monitor · blacklist (# in the history)
MENU	Low-level registers: LNA, PGA, MIX... (navigate with 2 and 8)
In the history	1: scans the history · 2: saves to a free channel · 3: deletes · 5: menu · 6: close call · 8: saves to flash

Sonic: full reset by powering on with PTT + SK2 held (warning screen, hold for 4 seconds). LED red when transmitting, green when receiving, flashing yellow during a scan or a close call.

Labs: applications (F + 7)

Application	Main keys
Sig Finder (sheet 26)	1: FOLLOW / SWEEP · 2: audio mode · 3: attenuation · 4: auto attenuation · 5: decay rate · M: reset to zero · hold F: lock · EXIT: exit
ATIS RX (sheet 11)	UP / LEFT and DOWN / RIGHT: newer / older station · 1: speaker · 2: clear · EXIT: exit

NR7Y CW (Apex?): CW keys

Key (long press)	Action
0 FM	Modulation FM, AM, USB, CW (squelch opens in USB and CW)
4 FC	CW filter width (on K1: F then long 4)
5 NOAA	CW practice oscillator (VFO in CW; on K1: F then long 5)
7 VOX	CW break-in
8 R	SSB cross-mode: transmission offset by the tone (CWfreq)

To be completed. GOGUFW: no key table found online; open the radio menu or the project notes. Fusion: the detailed spectrum keys are not covered here. Labs keys depend on the installed version of the application. Sources: F4HWN wiki (Button functions), Falcon v2.3.0 user guide, Sonic V71 manual, NR7Y CW documentation.

UV-K1: keys and overlay applications

Key functions by firmware, then the list and keys of the overlay applications (games excluded)

Overlay applications (Labs)

Reading. F4HWN's Labs edition is launched with **F + 7** and opens applications installed by UV Studio (sheet 2b). First set the receive VFO (modulation, frequency), then launch the application. They generally require firmware 6.3.0 or later and the matching application version. **NAV** = UP / DOWN on the UV-K5, LEFT / RIGHT on the UV-K1. **F then X** = F then the key, usually the opposite action. **EXIT** quits all applications. Applications marked "transmit" send a signal: an amateur radio callsign or authorised use is required. Only documented keys are listed. Games are not included.

Finding a source or beacon

Application	Function	Keys
Sig Finder	Direction-finding aid: relative signal feedback and detection of a distress homing signal (sheet 26)	1: FOLLOW / SWEEP · 2 (F then 2): audio mode · 3 (F then 3) or NAV: attenuation · 4: auto attenuation · 5: decay speed · 6: backlight · M: reset · F held: lock
FoxHunt	Fox-hunting aid: signal strength, history, attenuation and audio guidance (sheet 26)	1: bar or history · 2: audio (silence, beeps, tone) · 3: attenuation 0 / 6 / 15 / 27 dB · F then 2 or 3: backwards · M: resets PK, MN and trend · side key: Beacon
Beacon (transmit)	Repeating ARDF-style Morse beacon on the chosen transmit VFO	1: transmit time 5 to 60 s · 2: silence 5 to 240 s · 3: ID MOE / MOI / MOS / MOH / MO5 / MO / CALL · M: stops, new silence · side key: FoxHunt
Beam (transmit)	Transfers a channel configuration between compatible radios over the air	NAV: transmit or receive · M: sends the channel or waits for a packet · EXIT: stops or quits. Only one channel received per launch
EPIRB 406	Receives and decodes messages from first-generation Cospas-Sarsat 406 MHz beacons (sheet 16)	NAV: previous or next message · 4 / 6: -5 / +5 kHz (±50 kHz) · 5: back to VFO · 1: speaker · *: compact view · M: clear · 7 then M: QR Map
QR Map	Shows a position sent by another application as a map QR code to scan with a phone	1 Google · 2 Apple · 3 geo: · 4 OpenStreetMap · 5 text · NAV: next format · EXIT: back to the application · EXIT held: normal screen

Aircraft, ships, messages

Application	Function	Keys
ACARS RX	Receives and scrolls through VHF ACARS messages from aircraft (sheet 9)	NAV: scroll, then newer or older message · *: compact view · 1: audio · F: decoded or raw message · 2: clear (AM, local ACARS frequency)
AIS RX	Receives AIS reports from ships on AIS 1 or AIS 2 (sheet 11)	NAV: scroll, then next ship (8 kept) · *: compact view · 1: raw audio · M: freezes the list · 2: clear (FM, 161.975 or 162.025 MHz)
ATIS RX	Decodes the identity of inland-waterway ATIS stations at the end of transmission (sheet 11)	NAV: newer or older station (8 kept) · 1: speaker · 2: clear
ATIS DSC RX	Receives VHF digital selective calling on marine channel 70 (sheet 12)	NAV: newer or older call (6 kept) · 1: audio · 2: clear. Distress and urgency: inverted screen, green LED, beep (channel 70, 156.525 MHz)
POCSAG RX	Receives and scrolls through POCSAG 512 and 1200 baud messages, optionally keeping tone-only pages	NAV: scroll, then next page · *: compact view · 1: raw audio · 2: clear · 3: details or statistics · 4: keep tone-only pages (wide FM, 10 pages)

Radiosondes

Application	Function	Keys
Sonde M10, M20, RS41	Decode telemetry from Meteomodem M10 and M20 and Vaisala RS41 radiosondes: position, movement and, depending on the model, temperature, humidity, battery (sheet 13)	1: speaker · 2 (F then 2): +5 / -5 kHz, ±50 kHz · 3 (F then 3): next / previous attenuation · 4: back to the VFO frequency · 5: decode beep · EXIT: restores the VFO. M10: standard 0x9F frames only; RS41: no temperature, humidity or pressure, and a clean signal is needed (no error correction)
GeoNav	Shows azimuth, distance, altitude and elevation between the APRS "home" position and a decoded radiosonde (sheet 13)	M in a sonde application (M10, M20 or RS41) opens GeoNav; EXIT returns to the sonde. "Home" position taken from APRS Edit. Reception is paused during GeoNav

UV-K1: keys and overlay applications

Key functions by firmware, then the list and keys of the overlay applications (games excluded)

APRS and KISS

Application	Function	Keys
APRS RX	Receives, decodes and scrolls through APRS packets on the receive VFO (sheet 20)	NAV: scroll, then next frame (5 kept) · *: compact view · 1: speaker · 2: clear
APRS TX (transmit)	Sends an APRS position packet from the transmit VFO (sheet 20)	PTT or M: sends a frame · 1 (F then 1): tone level · 2 (F then 2): 2200 Hz twist · 3: beacon interval 1 / 5 / 15 / 30 / 60 min (F then 3: off) · 0: APRS Edit · *: view · NAV: scroll
APRS Edit	Edits the callsign, position, path, symbol and comment used by APRS TX (sheet 20)	NAV: field · M: opens the editor, confirms · EXIT: cancels the field, or saves and returns to APRS TX · digits: input · *: text mode, hemisphere · F: clear · PTT: preset messages (comment)
APRS DIGI (transmit)	Repeats eligible WIDE1-1 or WIDE2-1 packets, as a fill-in digipeater (sheet 20)	1: W1 or W12 · EXIT: stops. Callsign of 1 to 6 characters in CHIRP (Message Line 1), non-zero squelch
KISS TNC	Connects APRSdroid or another KISS client over USB-C (sheet 20b)	1: audio · 2: test frame to the host, without radio · 3: receive-only mode · EXIT: stops (sheet 20b)

Tones, Morse, images

Application	Function	Keys
ToneCall (transmit)	Sends editable ZVEI or CCIR sequential selective calls (sheet 19)	List: NAV selects the preset (1 to 4), M or PTT sends, F edits. Editor: 0 to 9 digits (8 max.), F clears, * standard (ZVEI-1 / 2, CCIR-1 / 2), M keeps
CW Decode	Decodes carrier telegraphy (A1A) with crystal correction and USB listening (sheet 27)	NAV: scroll · 0: audio · *: font · 1 / F then 1: threshold margin · 2 / F then 2: speed · 3: AGC · 4: recalibrate · 5: clear · 6 / F then 6: crystal correction · M: help
CW Keyer (transmit)	Sends Morse by hand with the PTT or the side keys (sheet 27)	PTT: carrier · 1: STRAIGHT / PADDLE · 8: AUTO, IAMB A, IAMB B · 2 / 3 / 4 (F then: backwards): speed, tone, delay · 5: swaps dots and dashes · 0: listening · 6: crystal · NAV: tuning ±50 kHz · M: help
CW USB (transmit)	Sends Morse with a key or a passive paddle on the USB-C port (sheet 27)	PTT: carrier · SIDE1 / SIDE2 or USB-C: key · 1: STRAIGHT / PADDLE · 2 / 3 / 4 / 6: speed, tone, delay, crystal (F then: decrease) · 5: swap · 7: SIDE / USB port · NAV: tuning 100 Hz · 0: listening. Passive adapter only, never on a charger or computer
SSTV (transmit)	Sends the start-up logo and receives monochrome SSTV images in 14 modes (sheet 21)	M: sends the logo · PTT: ignored · 1 (F then 1): transmit mode · 2: 1-bit / dithered · 3: speaker · 4: image / information · EXIT: interrupts. VFO in FM

Measurement and inspection

Application	Function	Keys
AF Scope	Shows the demodulated audio against time, or tunes a steady tone (sheet 25)	M: SCOPE, TUNE, AF Spectro · NAV: time base · * held: freeze · 1: STD / RAW · 2: speaker · 3: gain · 4: edge · 5: persistence · 0: clear
AF Spectro	Shows the demodulated audio as a spectrum or a waterfall (sheet 25)	M: SPECTRO, SONO, AF Scope · NAV: range 80 / 60 / 40 dB · * held: freeze · 1: STD / RAW · 2: speaker · 3: sampling 9.6 / 19.2 / 38.4 kHz · 0: clear
Spectrum3D	Live radio spectrum as a scrolling 3D landscape	NAV: centre · 1 / 3 (F then: reverse direction): ±6.25 kHz, range · 4 / 6: rotation · 2 / 8: tilt · 5: reset · 0: listen to the peak · *: speed · M: freeze
SystemInfo	Information on the firmware, hardware, battery, memory and channels	NAV: scroll (seven sections: firmware, system, battery, memory, identity, code, wiki)
Flash Map	Layout of the external flash memory and names of the firmwares and application slots	1 to 0: regions (config, calibration, logo, log, firmware, apps, Sonic 4K...) · NAV: region (F: first or last) · M: read-only dump · *: HEX / TEXT
Resistor	Calculates resistance, tolerance and temperature coefficient from 3 to 6 colour bands	NAV: band · * (F then *): next / previous colour · 0 to 9: colour · M: 3, 4, 5, 6 bands
Cube3D, Plasma	Graphics demos: 3D solid in filled or wireframe rendering; nine animated scenes	Cube3D: NAV speed · 1 to 9 shape · * next · F then *: wireframe / filled · M pause. Plasma: NAV speed · 1 to 9 scene · * rendering · 0 invert · F: auto cycle · M pause

Source: list provided and F4HWN wiki (App-... pages), as at the time of reading.

Antennas to build

Eight designs: V dipole, vertical, J-pole, half-wave wire, then four satellite antennas

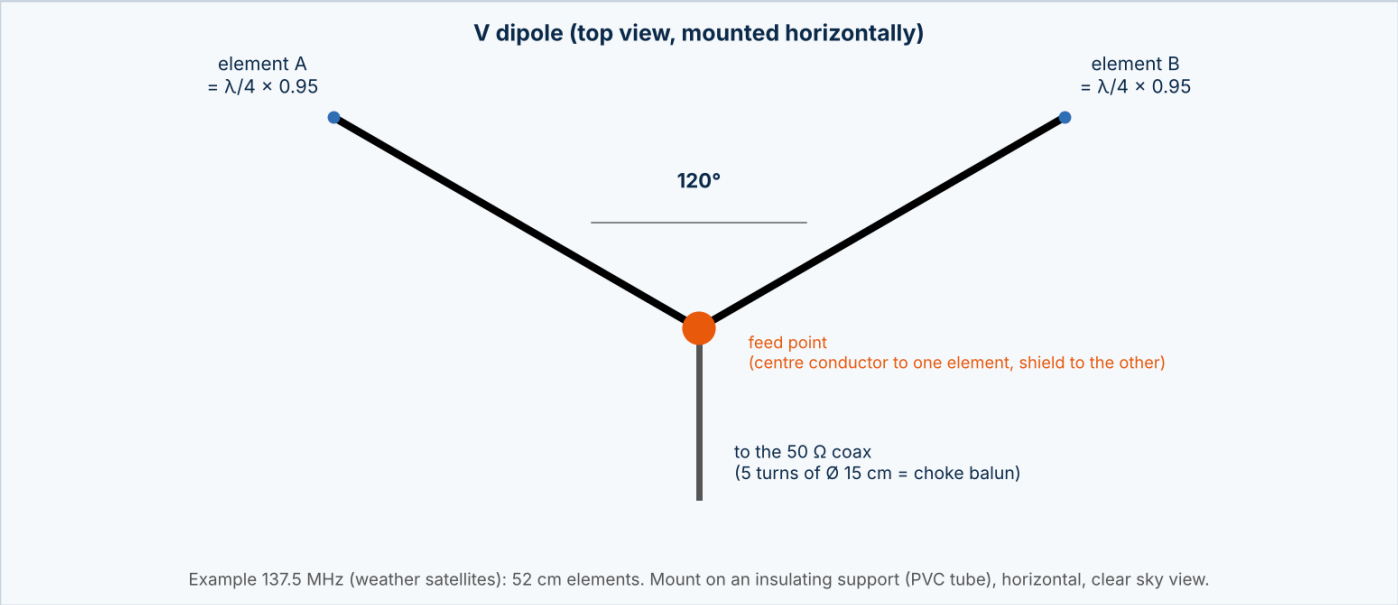
Before you start. The lengths come from the formula in sheet 1 (λ = 299.79 / f in m, with f in MHz) and the 0.95 factor for a conductor in air. They are **theoretical**: cut 3 to 5% too long, then shorten in small steps while watching the SWR on an analyser (such as a nanoVNA) or, without an instrument, by listening to a beacon or a known signal. These antennas are for **listening**; to transmit, you must be a licensed amateur on your band and hold the corresponding authorisation.

Which antenna to choose

Design	Frequency	For listening to	Difficulty
1. V dipole	137 MHz	Weather satellites (APT), VHF in general	Easy
2. λ/4 vertical	VHF to UHF	FM, aviation, ADS-B, PMR, radiosondes	Easy
3. J-pole	137 to 162 MHz	Marine VHF, AIS, 2 m	Medium
4. Half-wave wire (EFHW)	HF 3 to 30 MHz	80 / 40 / 20 m, broadcast, amateurs	Medium
5. CJU (loop + rod)	~435 MHz	UHF satellites, handheld	Medium
6. IOio (EA4CYQ)	145 and 435 MHz	FM amateur satellites, handheld	Hard
7. SATCOM EA1HBX	240 to 290 MHz	Military UHF satellites (listening only)	Medium
8. Turnstile I6IBE	255.55 MHz	Military UHF satellites, circular polarisation	Hard

A. Simple antennas

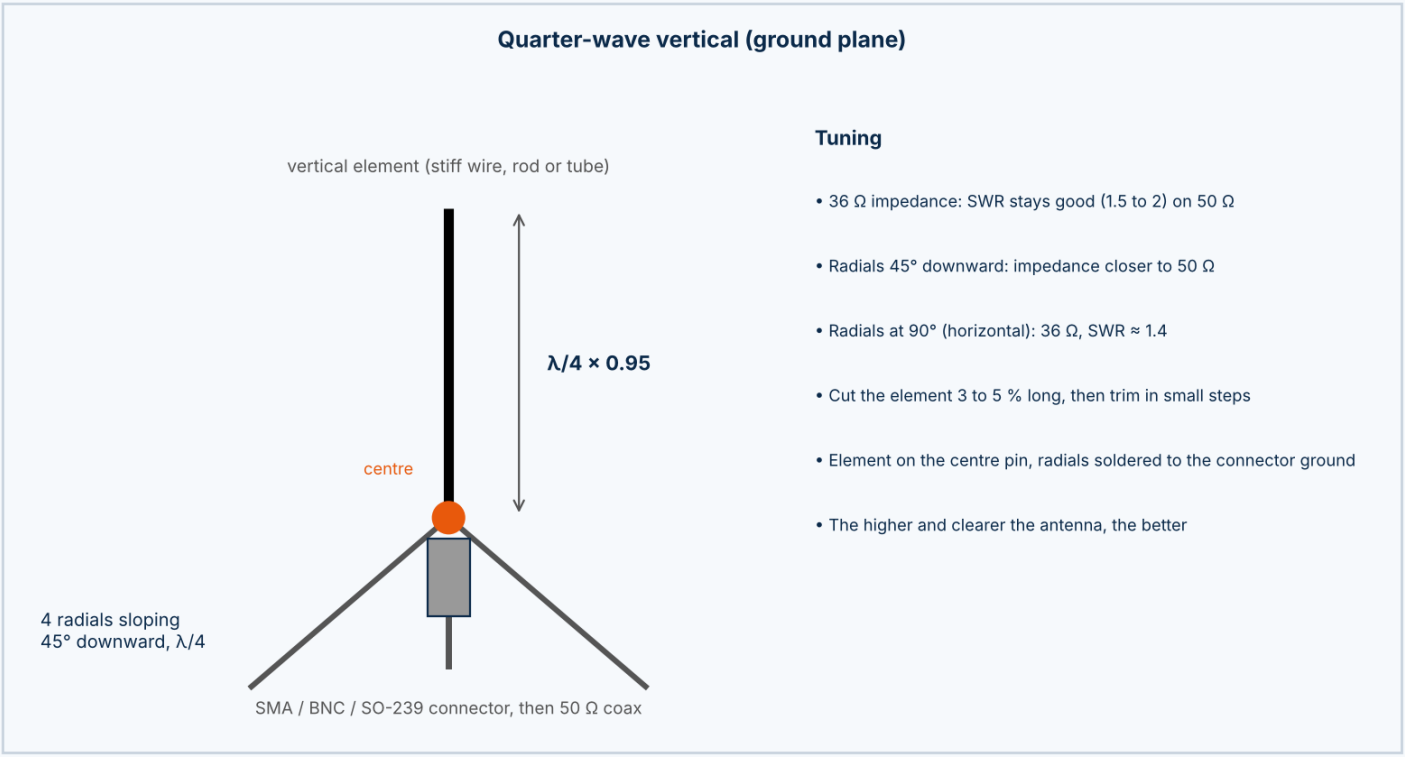
1. V dipole



Example 137.5 MHz (weather satellites): 52 cm elements. Mount on an insulating support (PVC tube), horizontal, clear sky view.

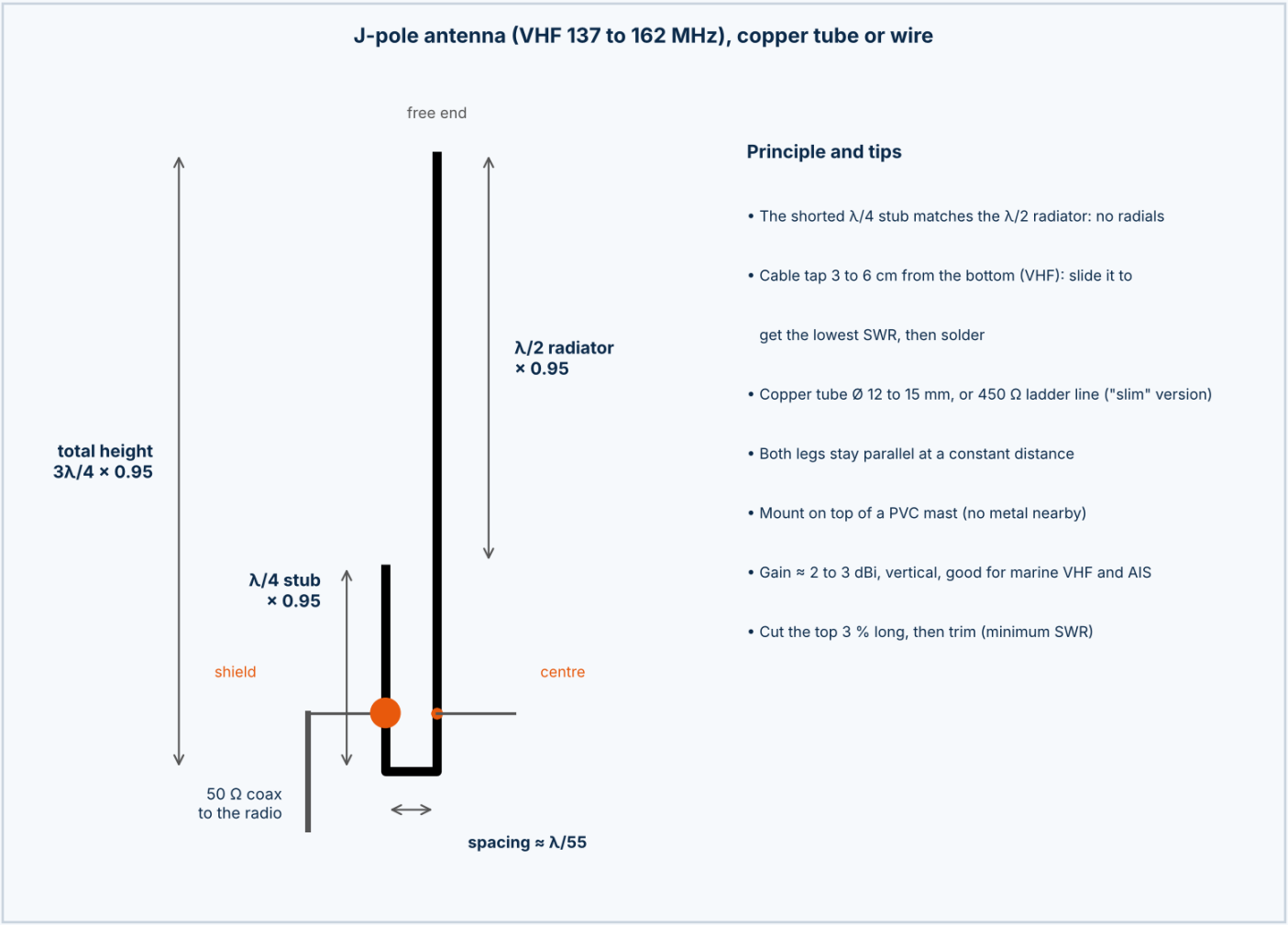
- Two λ/4 × 0.95 elements, spread 120° apart, horizontal, with a clear view of the sky. The feed point is at the centre, on an insulating support.
- The coax comes in from below. Make 5 turns of Ø 15 cm at the exit, to block currents travelling back up the cable (choke).
- For another frequency, take the element length from the table below.

2. Quarter-wave vertical (ground plane)



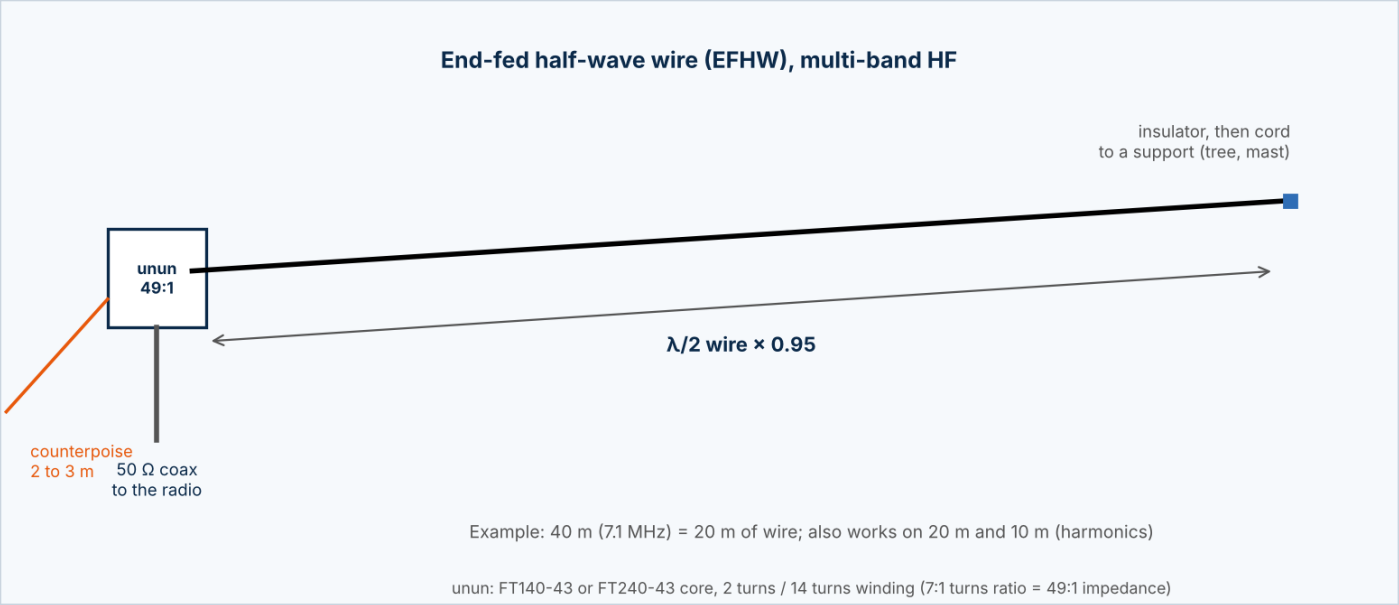
- One λ/4 × 0.95 element connected to the centre conductor of a connector base, four λ/4 radials connected to its ground, sloping down at 45°. The antenna is **omnidirectional** with vertical polarisation, like the whip antennas of handheld radios and repeaters.
- For the UV-K1, the element alone screws into the SMA (the radio's case acts as the ground plane); with radials, place the mount on a mast or roof and connect it to the radio with a short coax.
- ADS-B variant (1090 MHz): 6.5 cm element, 7 cm radials, on a piece of printed circuit board or a metal plate.

3. J-pole



- A 3λ/4 tube bent into a J: the short side (λ/4 stub) matches the impedance, the long side (λ/2) radiates. No radials, no balun: the antenna is mounted at the top of a mast, away from metal masses.
- The cable tap point is adjusted by sliding it (3 to 6 cm from the bottom on VHF), using an analyser. Solder once the SWR is at its minimum, then protect the joint (heat-shrink, silicone) and seal the top of the tube.
- Gain of 2 to 3 dBi, vertical polarisation: suited to marine, AIS and repeaters.

4. End-fed half-wave wire (EFHW)

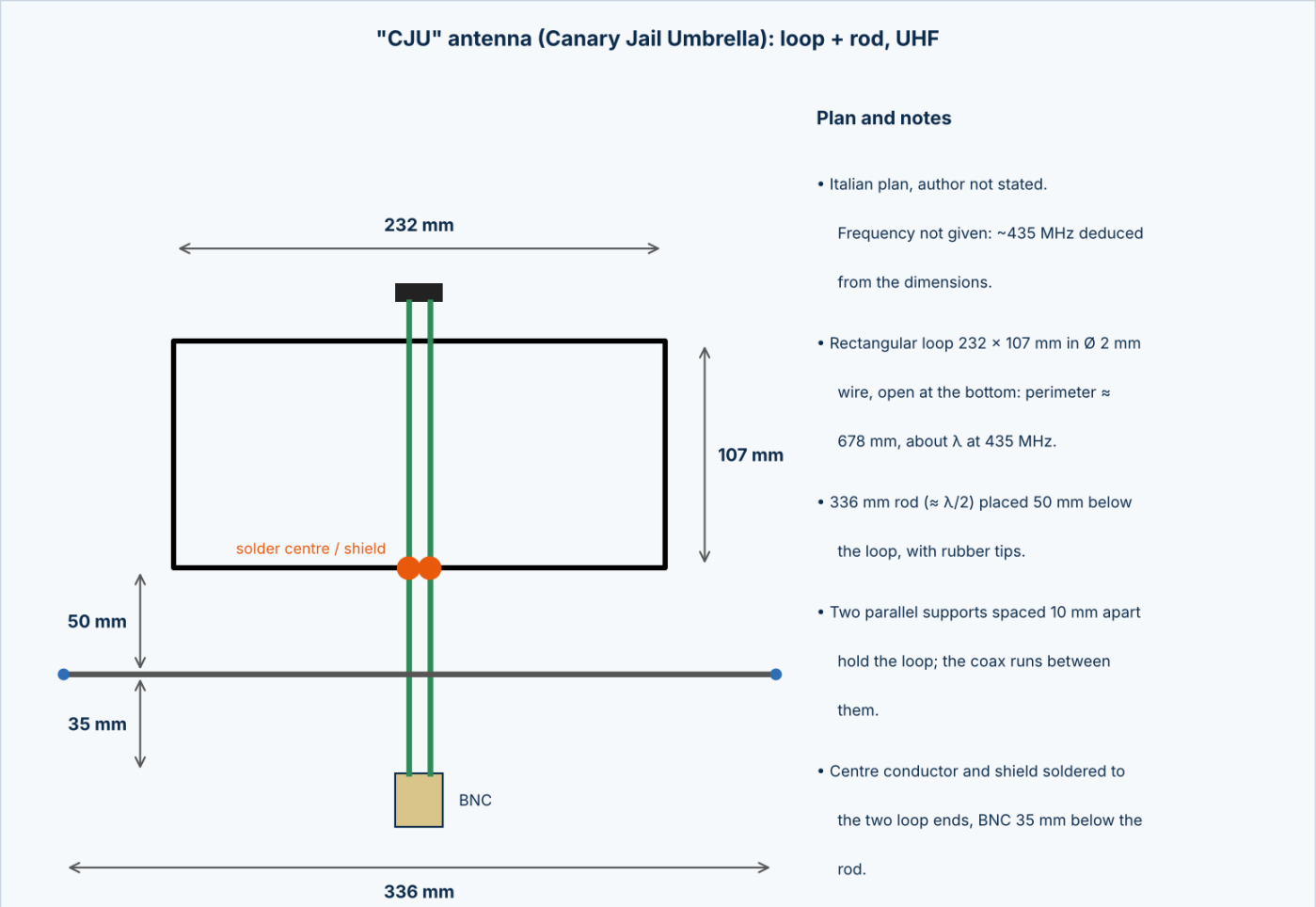


- A λ/2 × 0.95 wire connected to a 49:1 impedance transformer (unun): the end of a half-wave wire presents about 2500 Ω, which the unun brings down towards 50 Ω. Cut for 40 m, it also works on 20 m and 10 m (harmonics).
- A 2 to 3 m counterpoise runs from the unun; otherwise the coax acts as the counterpoise, with a few ferrites near the unun. Stretch the wire at 5 m height or more, sloping up to a tree or mast.
- It is the quickest to put up on HF: one wire, two ropes. The unun winding (2 turns / 14 turns on an FT140-43 or FT240-43 toroid) is given from memory: compare it with a manufacturer's diagram.

B. Satellite antennas (UHF)

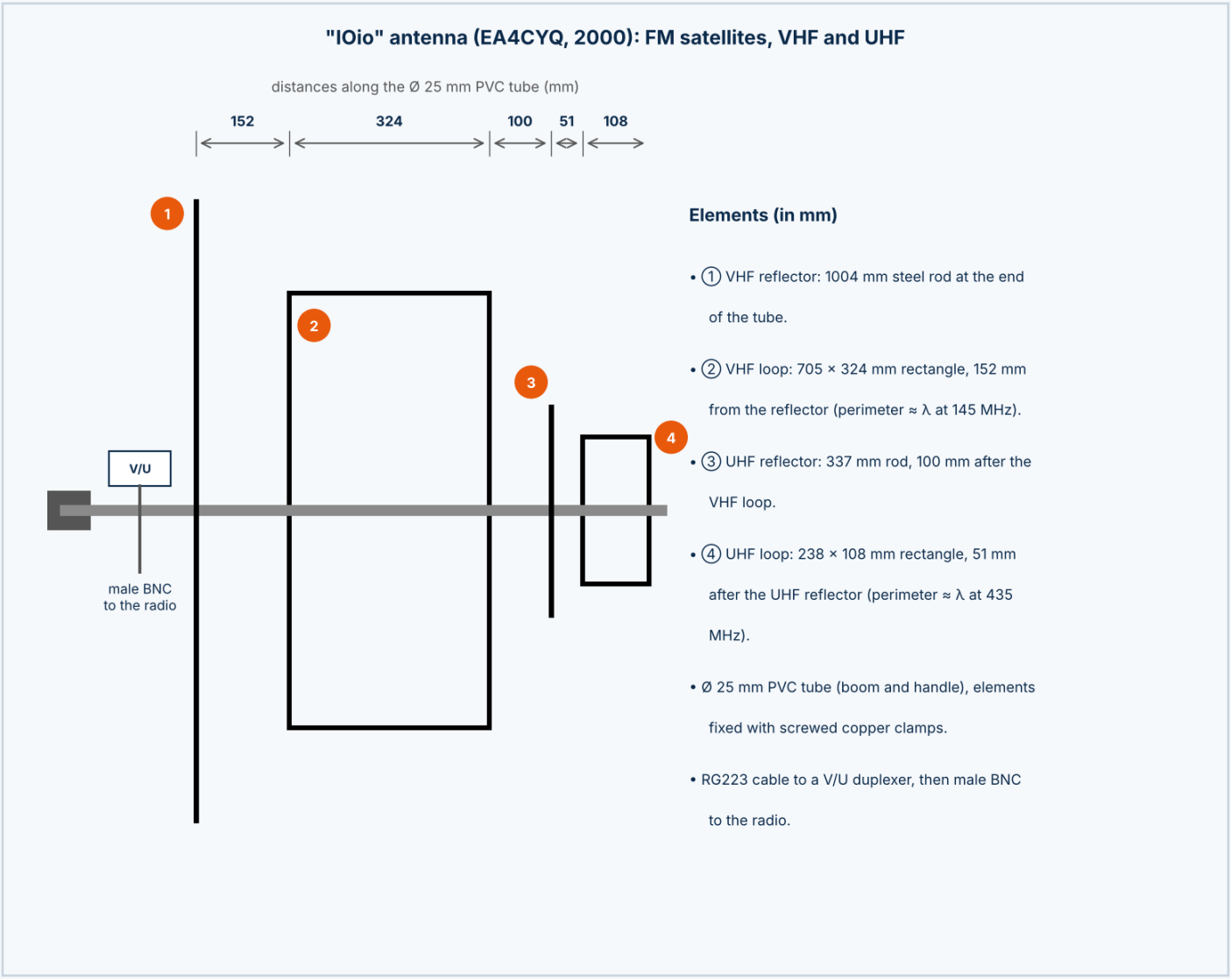
Satellites pass quickly and change direction: a directional antenna has to be tracked by hand. The antennas in this chapter are therefore **lightweight**, to hold in the hand or set on a tripod, and several have **circular polarisation** (turnstile), which avoids the fading caused by the satellite's rotation. The dimensions come from amateur designs that I redrew; I have not built them. For the frequencies and content of the links, see sheet 24. **Listening is free; never transmit** to military satellites.

5. "CJU" antenna: loop and rod



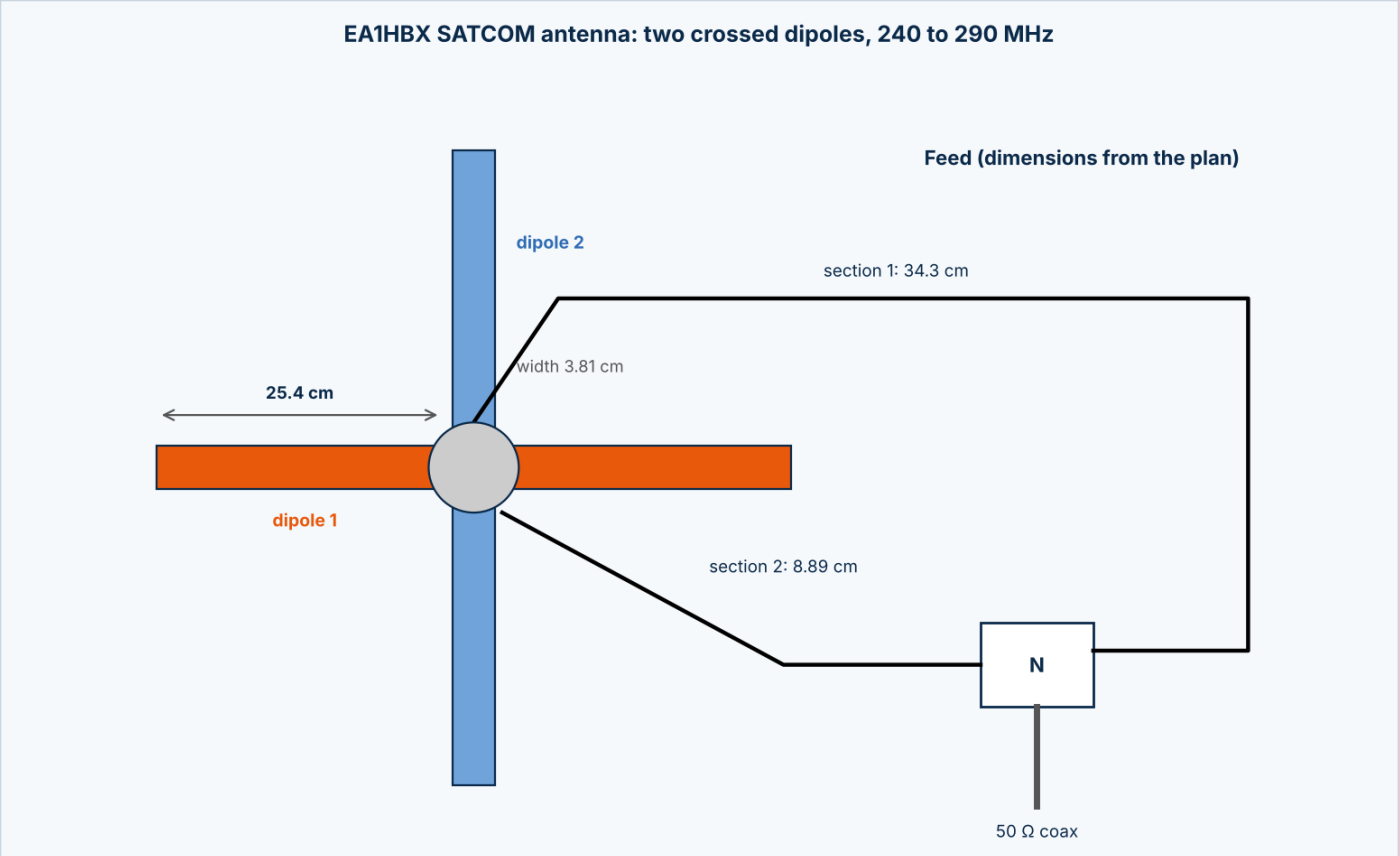
- A rectangular loop of 232 × 107 mm and a 336 mm rod, 50 mm below it, on two parallel supports: very compact, it fits in a bag.
- Italian design by an unnamed author. The frequency is not given: the dimensions correspond to about 435 MHz (loop perimeter ≈ λ, rod ≈ λ/2).
- The original design gives no tuning instructions: measure the SWR before using the antenna. Ø 2 mm wire, waterproof joints.

6. "IOio" antenna (EA4CYQ): FM satellites on VHF and UHF



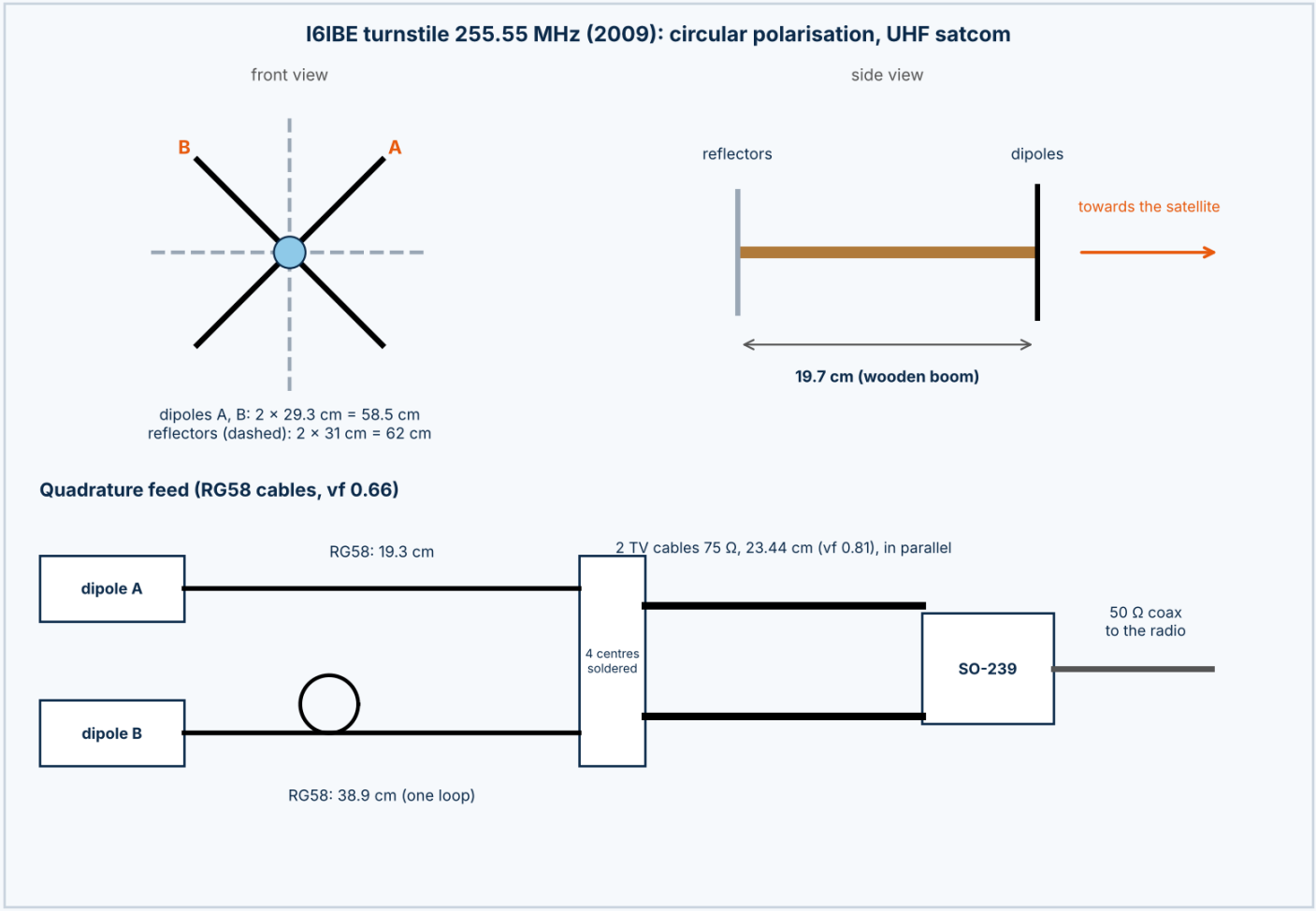
- Two rectangular loops, each with its reflector rod, on a hand-held Ø 25 mm PVC tube: one for 145 MHz, one for 435 MHz. A V/U duplexer combines the two onto a single cable.
- VHF loop 705 × 324 mm (perimeter ≈ λ at 145 MHz), UHF loop 238 × 108 mm (≈ λ at 435 MHz): the frequency is deduced from the dimensions. Design dated 31-10-2000.
- It is the most complex in the chapter: copper clamps, RG223 cable and a duplexer. Allow half a day.

7. EA1HBX SATCOM antenna: two crossed dipoles



- Four copper or aluminium plates of 25.4 × 3.81 cm form two crossed dipoles. They are fed by two lengths of RG59 (75 Ω, vf 0.66) of 34.3 and 8.89 cm, whose centre conductors and shields are joined at the N connector.
- Stated band: 240 to 290 MHz (military UHF satellites). Dimensions taken from the design; they have not been recalculated, so check the phase difference before relying on it.

8. I6IBE 255.55 MHz turnstile



- Two crossed 58.5 cm dipoles (Ø 2 mm brass on a printed-circuit hub), with two crossed 62 cm reflectors 19.7 cm behind, on a wooden boom. Right-hand circular polarisation.
- Feed: two RG58 cables of 19.3 and 38.9 cm. The 19.6 cm difference makes a quarter wave in the cable and shifts the dipoles by 90°. The four centre conductors are soldered together, then two 23.44 cm 75 Ω TV cables in parallel lead back to the SO-239 connector.
- Design by I6IBE (Ivo Brugnera, 2009), intended for receiving and transmitting to military UHF satellites. In this guide, it is for listening only.

Lengths to cut (factor 0.95)

Band / use	f (MHz)	V dipole: each element	Vertical: element; radials	J-pole: stub / radiator / total	J spacing
Weather satellite APT	137.5	51.8 cm	51.8 cm; 52 cm	51.8 / 103.6 / 155 cm	4.0 cm
2 m / APRS	145	49.1 cm	49.1 cm; 49 cm	49.1 / 98.2 / 147 cm	3.8 cm
Marine VHF	156.8	45.4 cm	45.4 cm; 45 cm	45.4 / 90.8 / 136 cm	3.5 cm
AIS	162	44.0 cm	44.0 cm; 44 cm	44.0 / 87.9 / 132 cm	3.4 cm
70 cm / PMR446	433	16.4 cm	16.4 cm; 16 cm	16.4 / 32.9 / 49 cm	1.3 cm
ADS-B	1090	6.5 cm	6.5 cm; 7 cm	not recommended	-

EFHW (λ/2 wire)	f (MHz)	Wire length	Counterpoise	Usable bands
80 m	3.65	39.0 m	3 to 5 m	80 m, 40 m (2nd harmonic), 20 m, 15 m
40 m	7.1	20.1 m	2 to 3 m	40 m, 20 m, 10 m
20 m	14.2	10.0 m	1.5 to 2 m	20 m, 10 m

Values calculated for elements in air, to be verified with an analyser: a thick tube or an insulated wire shortens the length by a few percent. These designs are amateur radio classics calculated from memory: check the unun and the J-pole tap point against a reliable diagram before any use for transmitting.

Antennas to build

Eight designs: V dipole, vertical, J-pole, half-wave wire, then four satellite antennas

Materials and safety

- **Supplies**: 50 Ω coax (RG58, or better RG213 beyond 10 m), SMA / BNC / SO-239 connector base, 2 mm copper wire, heat-shrink, stainless clamps, ferrites, PVC tube for the mast.
- **Soldering**: element to the centre conductor, radials to the connector base ground; then protect against water (varnish, silicone, heat-shrink).
- **Safety**: put them up away from any power line, in calm weather and with two people. A mast or wire that touches a line can be fatal. An outdoor antenna needs earthing and a lightning arrester.